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Kenta Imoto

Multifunctional Molecular Magnets Based on Octacyanidometalates

Doctoral Thesis accepted by
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Supervisor's Foreword

Multifunctional materials are intensively investigated in the field of chemistry, physics, and materials science due to the possibility of tuning a material's physical properties by external stimuli. The present thesis by Kenta Imoto deals with the demonstration of multifunctionalities in cyanido-bridged metal assemblies. One system presented in the thesis shows a conversion from the paramagnetic state to the ferromagnetically ordered state by light irradiation. This behavior is achieved by a rational chemical design and alignment of the photoactive spin-crossover sites with strong magnetic interactions. Other novel contributions assigned to this thesis are the demonstration of photo-induced magnetization in a two-step spin-crossover material and the realization of the coexistence between metamagnetism and super-ionic conductivity. These multifunctionalities are realized by the incorporation of a spin-crossover moiety or a hydrogen-bonding network to a magnetically coupled cyanido-bridged framework. In particular, in light-induced spin-crossover magnets, a magnetically non-ordered state can be altered to a magnetically ordered state by photo-irradiation, which is one of the attractive mechanisms for novel optical switching devices. The results in this work will be appealing to readers related to the field of materials science.

The studies in the thesis were performed between April 2011 and March 2014 at the Department of Chemistry, School of Science, The University of Tokyo, and have led to several publications and presentations at international conferences.

Tokyo, Japan
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Prof. Shin-ichi Ohkoshi

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(1) S. Ohkoshi, K. Imoto, Y. Tsunobuchi, S. Takano, H. Tokoro, ***Nature Chemistry*** 3, 564 (2011).
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(3) K. Imoto and S. Ohkoshi, ***Chem. Lett.***, 45, 359 (2016).

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Contents

1 Introduction 1
1.1 Molecule-Based Magnet 1
1.2 Molecular Field Theory 2
1.3 Van Vleck Paramagnetism 9
1.4 Cyanido-Bridged Metal Assemblies and Their Functionalities . . . 14
1.5 Octacyanidometalate-Based Magnets 16
1.6 Photo-Induced Magnetization in Cyanido-Bridged Metal Assemblies 17
1.7 Spin-Crossover Phenomenon 20
1.8 Light-Induced Excited Spin State Trapping (LIESST) 21
1.9 Ionic Conductivity 22
1.10 Objectives of This Work 23
References 23

2 Observation of Light-Induced Spin-Crossover Magnetism in a Fe-[Nb(CN)$_8$] Bimetal Assembly 29
2.1 Introduction 29
2.2 Experiments 30
2.2.1 Synthesis 30
2.2.2 Characterization 31
2.2.3 Photo-Irradiation Measurements 31
2.3 Results 31
2.3.1 Elemental Analysis 31
2.3.2 XRD Patterns and Crystal Structure 32
2.3.3 IR Spectra 33
2.3.4 Magnetic Properties 33
2.3.5 UV-Vis Spectra 36
2.3.6 ^{57}Fe Mössbauer Spectra 36
2.3.7 Photomagnetic Measurements 38
2.3.8 UV-Vis Spectra Under the Light Irradiation 39

2.4 Discussion 41
2.4.1 Electronic States of High-*T* and Low-*T* Forms 41
2.4.2 XRD Patterns and Crystal Structure 43
2.4.3 Magnetic Properties of Low-*T* Form 43
2.4.4 Photomagnetic Properties 44
2.5 Conclusion 45
References 45

3 Two-Step Spin-Crossover and Photo-Induced Spin-Crossover Ferromagnetism in $Fe_2^{II}[Nb^{IV}(CN)_8](4\text{-Methylpyridine})_8 \cdot 2H_2O$ 47
3.1 Introduction 47
3.2 Experiments 48
3.2.1 Synthesis 48
3.2.2 Characterization 48
3.2.3 Photo-Irradiation Measurements 49
3.3 Results 49
3.3.1 Elemental Analysis 49
3.3.2 Single Crystal X-Ray Analysis at Various Temperatures 49
3.3.3 IR Spectra 51
3.3.4 Magnetic Properties 53
3.3.5 Temperature Dependence of UV-Vis Spectra 54
3.3.6 Photomagnetic Measurements 54
3.3.7 UV-Vis Spectra Under the Light Irradiation at 3 K 54
3.4 Discussion 56
3.4.1 The Assignment of Spin-Transition in the Present System 56
3.4.2 The Analysis of Magnetic Susceptibility and the Electronic State 56
3.4.3 Temperature Dependence of Crystal Structure 61
3.4.4 Thermodynamical Properties 61
3.4.5 Magnetic Properties of Low Temperature State 65
3.4.6 Photomagnetic Properties 65
3.5 Conclusion 67
References 67

4 Observation of Coexistence of Super-Ionic Conductivity and Metamagnetism in $Mn_3[Nb^{IV}(CN)_8]_2(4\text{-Aminopyridine})_{10}(4\text{-Aminopyridinium})_2 \cdot 12H_2O$ 69
4.1 Introduction 69
4.2 Experiments 70
4.2.1 Synthesis 70
4.2.2 Characterization 70
4.2.3 Impedance Measurement 71

4.3 Results . . . 71
4.3.1 Elemental Analysis . . . 71
4.3.2 Single Crystal X-ray Analysis . . . 71
4.3.3 Powder X-ray Diffraction Pattern . . . 74
4.3.4 IR Spectra . . . 77
4.3.5 Magnetic Properties . . . 77
4.3.6 Impedance Measurements . . . 79
4.4 Discussion . . . 80
4.4.1 The Valence State, the Form of Organic Ligand, and the Crystal Structure . . . 80
4.4.2 Magnetic Properties . . . 82
4.4.3 Conducting Properties . . . 83
4.4.4 The Mechanism for Super-Ionic Conductivity . . . 84
4.5 Conclusion . . . 84
References . . . 85

5 Concluding Remarks . . . 87

About the Author . . . 89

Chapter 1
Introduction

Abstract This chapter firstly describes general introduction of molecule-based magnets, explaining advantages of molecule-based magnets compared to metal or metal oxide-based magnets. Secondary, as principal methods to treat magnetic properties, molecular-field theory and Van Vleck's theory are explained. Next, attractive points of cyanido-bridged metal assemblies, especially octacyanidometalate-based magnets, are described with examples of reported functional cyanido-bridged metal assemblies. Then, as the related functionalities in this thesis, spin-crossover phenomenon, light-induced spin-crossover phenomenon, photo-induced magnetization, and ionic-conductivity are explained. Finally, observation coupling effect between spin-crossover phenomenon and magnetic phase transition and demonstration of ionic conductive magnetic material in octacyanidometalate-based magnets are described as the objectives of this thesis.

Keywords Molecule-based magnet · Photo-induced magnetization · Spin-crossover · Ionic conductivity · Cyanido-bridged metal assemblies

1.1 Molecule-Based Magnet

Molecule-based magnets are magnetic materials composed of metal complex and/or organic radical [1–8]. Molecule-based magnets can take various colors in contrast with black or metallic colored metal oxide or magnetic metals. In addition, since molecule-based magnets can change their crystal structure and/or electronic state by various external stimuli such as solvent, humidity, pressure, or light, they are extensively studied as functional magnetic materials. For example, as functional magnetic materials, photo-induced magnetization in which a substance is optically switched between paramagnetic state and ferromagnetic state [9–13], humidity sensitive magnet which exhibit humidity-induced reversible change of magnetic properties and/or magnetic pole direction [14, 15], temperature-induced phase transition phenomenon with thermal hysteresis [16, 17] have been reported. In addition, unique magnetic properties such as mixed ferro-ferri magnetism [18, 19],

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inverted magnetic hysteresis loop [20], magnetic dimensional crossover phenomenon [21] have been realized. The researches about the magnetic properties of metal complexes have been studied from about the 1950s before the appearance of the term "molecule-based magnet" [22–31]. Most of these researches are devoted to make a theoretical treatment of magnetic properties of polynuclear metal complexes. On the other hand, the investigations of bulk magnetic properties of metal complexes started from late 1980s [32–34]. These pioneering works succeeded in observation of magnetic phase transition using designable advantage of metal complex or organic radicals. In addition, metal-free organic radical magnet has been realized in 1991 [35] and a variety of compounds are realized to exhibit ferromagnetic property. In such stream of research, Prussian blue analogs have been investigated as molecule-based magnet due to their unique properties. Prussian blue has been well-known for blue pigment from 17th century, which is said to be the first synthetic coordination compound. Prussian blue was realized as a ferromagnetic material with a Curie temperature of 5.6 K in 1968 [6] and the crystal structure of Prussian blue was solved in 1977 [7]. Chemical formula of Prussian blue is expressed as $Fe_4^{III}[Fe^{II}(CN)_6]_3$ and there are two types of metal ions, Fe^{II} and Fe^{III}. These two metal ions can be altered by other metal cations and metal-substituted compounds are called as Prussian blue analogs ($[M(CN)_6]$-based magnets). In Prussian blue analogues, various functions such as room-temperature magnetism [36–39], humidity-induced magnetism [15], photo-induced magnetism [9–13, 40–42] have been realized. Additionally, heptacyanidometalate-based magnets [43–46] or octacyanidometalate-based magnets [47–53] were proved to be useful building blocks for molecule-based magnets and various functionalities have been demonstrated according to the characteristics of the components. From the next section, I will introduce the background knowledge in the field of molecule-based magnet. First, I introduce two main analytical methods for characterizing magnetic properties of a material, that is, molecular-field theory and Van Vleck paramagnetism. Then, background and functionalities in cyanido-bridged metal assemblies, spin-crossover phenomenon, and ionic conductivity will be shown.

1.2 Molecular Field Theory

(i) Molecular field theory for unary metal system [54]

Molecular field theory is the first theory which succeeded in explaining ferromagnetic phase transition. Molecular field theory is mainly developed by P. Weiss and L. Néel. In this theory, the magnetic interaction is introduced as molecular field, which is proportional to the averaged magnetization value (M) of the nearest neighbor site, the number of nearest neighbor sites (z), and super exchange

interaction parameter (J). For unary metal system, when the magnetic moment is M, the molecular field H_{M} is expressed as

$$H_M = H_0 + nM, \text{where } n = \frac{2Z}{N_{\mathrm{A}}(g\mu_{\mathrm{B}})^2}J. \tag{1.1}$$

Here, H_0 is the external magnetic field, N_{A} is Abogadro constant, g is g factor, μ_{B} is Bohr magneton, and Z is the number of the nearest neighbor metal sites (coordination number of bridging ligand). The averaged sublattice magnetization ($\langle M \rangle$, where $M = N_{\mathrm{A}} g \mu_B \langle M \rangle$) is obtained by solving self-consistent equation,

$$\langle M \rangle = S_{\mathrm{M}} B_{S_{\mathrm{M}}}\left(\frac{g\mu_{\mathrm{B}} H_{\mathrm{M}} S_{\mathrm{M}}}{k_{\mathrm{B}}T}\right), \tag{1.2}$$

where S_{M} is spin quantum number of the metal and $B_{S_{\mathrm{M}}}$ is the Brillouin function which is given by the following formula,

$$B_{S_{\mathrm{M}}}(a) = \left(\frac{2S_{\mathrm{M}}+1}{2S_{\mathrm{M}}}\right)\tanh^{-1}\left(\frac{2S_{\mathrm{M}}+1}{2S_{\mathrm{M}}}a\right) - \left(\frac{1}{2S_{\mathrm{M}}}\right)\tanh^{-1}\left(\frac{1}{2S_{\mathrm{M}}}a\right). \tag{1.3}$$

To obtain spontaneous magnetization, we set $H_0 = 0$ and Eq. (1.3) is

$$\langle M \rangle = S_{\mathrm{M}} B_{S_{\mathrm{M}}}\left(\frac{g\mu_{\mathrm{B}} H_{\mathrm{M}} S_{\mathrm{M}}}{k_{\mathrm{B}}T}\right) = S_{\mathrm{M}} B_{S_{\mathrm{M}}}\left(\frac{2ZJS_{\mathrm{M}}}{k_{\mathrm{B}}T}\langle M \rangle\right). \tag{1.4}$$

When $x = \frac{2ZJS_{\mathrm{M}}}{k_{\mathrm{B}}T}\langle M \rangle$ is introduced, the Eq. (1.4) becomes

$$\langle M \rangle = \frac{k_{\mathrm{B}}T}{2ZJS_{\mathrm{M}}}x \tag{1.5}$$

and

$$\langle M \rangle = S_{\mathrm{M}} B_{S_{\mathrm{M}}}(x). \tag{1.6}$$

The solution $\langle M \rangle$ is obtained by the intersection of $\langle M \rangle$ versus x curves of Eqs. (1.5) and (1.6). As depicted in Fig. 1.1, M value takes non-zero value below specific temperature T_{C}, where tangent line of $M = S_{\mathrm{M}} B_{S_{\mathrm{M}}}(x)$ at $x = 0$ coincides with $\langle M \rangle = \frac{k_{\mathrm{B}}T}{2ZJS_{\mathrm{M}}}x$. The T_{C} value corresponds to the Curie temperature and is obtained as

$$T_{\mathrm{C}} = \frac{2ZJS_{\mathrm{M}}(S_{\mathrm{M}}+1)}{3k_{\mathrm{B}}}. \tag{1.7}$$

as the following way.

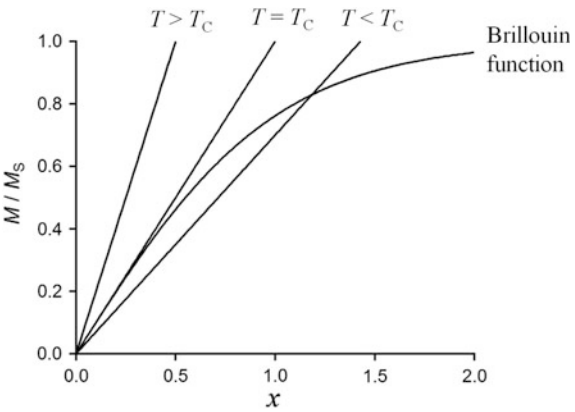


Fig. 1.1 Brillouin function and straight line at $T > T_C$, $T = T_C$, and $T < T_C$ obtained from molecular field theory. The cross point of *two curves* gives the magnetization value. Below T_C, the *two curves* cross at non-zero magnetization value, indicating the appearance of bulk magnetization

Around T_C, x value is almost zero. In this region we can apply the approximation of Brillouin function with $x \ll 1$. When $x \ll 1$, $B_{S_M}(x)$ is approximated as

$$B_{S_M}(x) \approx \frac{S_M + 1}{3S_M} x. \tag{1.8}$$

Therefore, the θ value is obtained as

$$\begin{aligned} S_M \frac{S_M + 1}{3S_M} x &= \frac{k_B T_C}{2ZJS_M} x \\ \Leftrightarrow \frac{S_M + 1}{3} &= \frac{k_B T_C}{2ZJS_M} \\ \Leftrightarrow T_C &= \frac{2ZJS_M(S_M + 1)}{3k_B}. \end{aligned} \tag{1.9}$$

Equation (1.9) enables us to estimate superexchange interaction parameter (J) by Curie temperature (T_C).

From here, we consider the magnetic susceptibility (χ) at high temperature region. In this case, we include external magnetic field (H_0) in Eq. (1.4) and the equation is

$$\langle M \rangle = S_M B_{S_M} \left\{ \frac{g\mu_B S_M}{k_B T} \left(H_0 + \frac{2ZJ}{g\mu_B} \langle M \rangle \right) \right\}. \tag{1.10}$$

At high temperature, $\frac{g\mu_B S_M}{k_B T} \left(H_0 + \frac{2ZJ}{g\mu_B} \langle M \rangle \right) \ll 1$, therefore,

$$\langle M \rangle = \frac{S_M + 1}{3} \frac{g\mu_B S_M}{k_B T} \left(H_0 + \frac{2ZJ}{g\mu_B} \langle M \rangle \right). \tag{1.11}$$

Introducing $\chi = M/H_0 = N_A g\mu_B \langle M \rangle / H_0$,

$$\begin{aligned}
\chi &= \frac{S_M + 1}{3} \frac{N_A (g\mu_B)^2 S_M}{k_B T} \left(1 + \frac{2ZJ}{N_A (g\mu_B)^2} \chi \right) \\
&\Leftrightarrow \left(T - \frac{2ZJS_M(S_M+1)}{3k_B} \right) \chi = \frac{N_A (g\mu_B)^2 S_M (S_M+1)}{3k_B} \\
&\Leftrightarrow \chi = \frac{C}{T-\theta} \\
&\Leftrightarrow \frac{1}{\chi} = \frac{T-\theta}{C}
\end{aligned} \tag{1.12}$$

This magnetic behavior is called Curie-Weiss law and $C = \frac{N_A (g\mu_B)^2 S_M (S_M+1)}{3k_B}$ is called Curie constant. θ is Weiss temperature, which takes positive value in the case of ferromagnetic interaction ($J > 0$) and takes negative value in the case of antiferromagnetic interaction ($J < 0$). In addition, Curie temperature (T_C) is related with θ by $T_C = |\theta|$ in this theory.

The magnetic interaction is clearly understood by plotting χ^{-1} versus T (Fig. 1.2).

(ii) Molecular field theory for more than binary metal system [18, 19]

Detail discussion about magnetic properties of cyanido-bridge metal assemblies was performed by Ohkoshi et al. in 1997. They focused on the magnetic interaction only between the nearest neighbor metal sites which is the feature of cyanido-bridged metal assemblies and applied molecular field theory to ternary metal Prussian blue analogs. When we consider the formula of $(M_A{}_x\ M_B{}_{1-x})_{y1}[M_C(CN)_6]_{y2} \cdot zH_2O$ composed of three metal ions of M_A, M_B, and M_C, the molecular field H_M are expressed by the following equation using molecular field coefficient (n_{pq}),

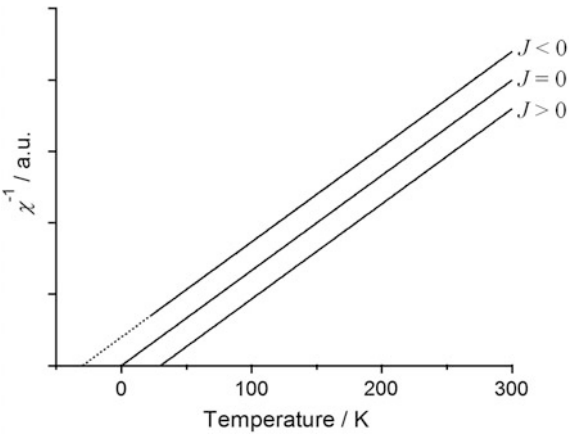


Fig. 1.2 The χ^{-1} versus temperature *curve* obtained by Curie-Weiss law. The *x*-intercept is called Weiss temperature and is negative for antiferromagnetic interaction (J < 0) and positive for ferromagnetic interaction (J > 0)

$$H_{\mathrm{A}} = H_0 + n_{\mathrm{AC}} M_{\mathrm{C}},\ H_{\mathrm{B}} = H_0 + n_{\mathrm{BC}} M_{\mathrm{C}},\ H_{\mathrm{C}} = H_0 + n_{\mathrm{CA}} M_{\mathrm{A}} + n_{\mathrm{CB}} M_{\mathrm{B}}, \quad (1.13)$$

because the nearest neighboring metal of $\mathrm{M_A}$ and $\mathrm{M_B}$ are always $\mathrm{M_C}$, and that of $\mathrm{M_C}$ is $\mathrm{M_A}$ or $\mathrm{M_B}$.

The molecular field coefficients (n_{pq}) are given by the following equations

$$\begin{aligned} n_{\mathrm{AC}} &= \frac{2Z_{\mathrm{AC}}}{y_2 N_{\mathrm{A}} (g\mu_{\mathrm{B}})^2} J_{\mathrm{AC}},\ n_{\mathrm{BC}} = \frac{2Z_{\mathrm{BC}}}{y_2 N_{\mathrm{A}} (g\mu_{\mathrm{B}})^2} J_{\mathrm{BC}}, \\ n_{\mathrm{CA}} &= \frac{2Z_{\mathrm{CA}}}{y_1 N_{\mathrm{A}} (g\mu_{\mathrm{B}})^2} J_{\mathrm{AC}},\ n_{\mathrm{CB}} = \frac{2Z_{\mathrm{CB}}}{y_1 N_{\mathrm{B}} (g\mu_{\mathrm{B}})^2} J_{\mathrm{BC}}. \end{aligned} \quad (1.14)$$

Here, Z_{pq} is the number of the nearest neighbor metal q sites around a p site (coordination number of bridging ligand to metal ion q around a metal ion p). The averaged sublattice magnetization ($\langle M \rangle$, where $M_{\mathrm{A}} = y_1 x N_{\mathrm{A}} g\mu_B \langle M \rangle$, $M_{\mathrm{B}} = y_1(1-x) N_{\mathrm{A}} g\mu_B \langle M \rangle$, $M_{\mathrm{C}} = y_2 N_{\mathrm{A}} g\mu_B \langle Mc \rangle$) is obtained by the following equations,

$$\langle M_i \rangle = S_i B_{S_i} \left(\frac{g\mu_{\mathrm{B}} H_i S_i}{k_{\mathrm{B}} T} \right) \quad (i = \mathrm{A}, \mathrm{B}, \mathrm{C}), \quad (1.15)$$

where S_i is spin quantum number of the metal and $B_{S_{\mathrm{M}}}$ is the Brillouin function.

To obtain spontaneous magnetization, we set $H_0 = 0$ and the equation is expressed as

$$\langle M_{\mathrm{A}} \rangle = S_{\mathrm{A}} B_{S_{\mathrm{A}}} \left(\frac{g\mu_{\mathrm{B}} n_{\mathrm{AC}} M_{\mathrm{C}} S_{\mathrm{A}}}{k_{\mathrm{B}} T} \right) = S_{\mathrm{A}} B_{S_{\mathrm{A}}} \left(\frac{2Z_{\mathrm{AC}} x J_{\mathrm{AC}} S_{\mathrm{A}}}{k_{\mathrm{B}} T} \langle M_{\mathrm{C}} \rangle \right), \quad (1.16a)$$

$$\langle M_{\mathrm{A}} \rangle = S_{\mathrm{A}} B_{S_{\mathrm{A}}} \left(\frac{g\mu_{\mathrm{B}} n_{\mathrm{AC}} M_{\mathrm{C}} S_{\mathrm{A}}}{k_{\mathrm{B}} T} \right) = S_{\mathrm{A}} B_{S_{\mathrm{A}}} \left(\frac{2Z_{\mathrm{AC}} x J_{\mathrm{AC}} S_{\mathrm{A}}}{k_{\mathrm{B}} T} \langle M_{\mathrm{C}} \rangle \right), \quad (1.16b)$$

$$\begin{aligned} \langle M_{\mathrm{C}} \rangle &= S_{\mathrm{C}} B_{S_{\mathrm{C}}} \left(\frac{g\mu_{\mathrm{B}} (n_{\mathrm{CA}} M_{\mathrm{A}} + n_{\mathrm{CB}} M_{\mathrm{B}}) S_{\mathrm{C}}}{k_{\mathrm{B}} T} \right) \\ &= S_{\mathrm{C}} B_{S_{\mathrm{C}}} \left(\frac{2Z_{\mathrm{CA}} x J_{\mathrm{AC}} S_{\mathrm{C}}}{k_{\mathrm{B}} T} \langle M_{\mathrm{A}} \rangle + \frac{2Z_{\mathrm{CB}} (1-x) J_{\mathrm{BC}} S_{\mathrm{C}}}{k_{\mathrm{B}} T} \langle M_{\mathrm{B}} \rangle \right). \end{aligned} \quad (1.16c)$$

By solving these self-consistent equations, we can obtain magnetization values of each metal ion and total magnetization value.

The T_{C} value is obtained by the approximating the Brillouin function as $B_{S_{\mathrm{M}}}(x) \approx \frac{S_{\mathrm{M}}+1}{3S_{\mathrm{M}}} x$, which is valid for $x \ll 1$. In this case, the equations are

$$\langle M_{\mathrm{A}} \rangle = S_{\mathrm{A}} \frac{S_{\mathrm{A}} + 1}{3S_{\mathrm{A}}} \frac{2Z_{\mathrm{AC}} x J_{\mathrm{AC}} S_{\mathrm{A}}}{k_{\mathrm{B}} T_{\mathrm{C}}} \langle M_{\mathrm{C}} \rangle, \quad (1.17a)$$

$$\langle M_{\mathrm{B}} \rangle = S_{\mathrm{B}} \frac{S_{\mathrm{B}} + 1}{3S_{\mathrm{B}}} \frac{2Z_{\mathrm{BC}} (1-x) J_{\mathrm{BC}} S_{\mathrm{B}}}{k_{\mathrm{B}} T_{\mathrm{C}}} \langle M_{\mathrm{C}} \rangle, \quad (1.17b)$$

$$\langle M_{\mathrm{C}}\rangle = S_{\mathrm{C}}\frac{S_{\mathrm{C}}+1}{3S_{\mathrm{C}}}\left(\frac{2Z_{\mathrm{CA}}xJ_{\mathrm{AC}}S_{\mathrm{C}}}{k_{\mathrm{B}}T_{\mathrm{C}}}\langle M_{\mathrm{A}}\rangle + \frac{2Z_{\mathrm{CB}}(1-x)J_{\mathrm{BC}}S_{\mathrm{C}}}{k_{\mathrm{B}}T_{\mathrm{C}}}\langle M_{\mathrm{B}}\rangle\right), \tag{1.17c}$$

therefore,

$$\begin{aligned}
\langle M_{\mathrm{C}}\rangle &= \frac{S_{\mathrm{C}}+1}{3}\left(\frac{2Z_{\mathrm{CA}}xJ_{\mathrm{AC}}S_{\mathrm{C}}}{k_{\mathrm{B}}T}\frac{S_{\mathrm{A}}+1}{3}\frac{2Z_{\mathrm{AC}}xJ_{\mathrm{AC}}S_{\mathrm{A}}}{k_{\mathrm{B}}T_{\mathrm{C}}}\langle M_{\mathrm{C}}\rangle + \frac{2Z_{\mathrm{CB}}(1-x)J_{\mathrm{BC}}S_{\mathrm{C}}}{k_{\mathrm{B}}T_{\mathrm{C}}}\frac{S_{\mathrm{B}}+1}{3}\frac{2Z_{\mathrm{BC}}(1-x)J_{\mathrm{BC}}S_{\mathrm{B}}}{k_{\mathrm{B}}T_{\mathrm{C}}}\langle M_{\mathrm{C}}\rangle\right)\\
&\Leftrightarrow \frac{S_{\mathrm{C}}+1}{3}\left(\frac{4Z_{\mathrm{CA}}Z_{\mathrm{AC}}x^2J_{\mathrm{AC}}^2S_{\mathrm{C}}}{3k_{\mathrm{B}}^2T_{\mathrm{C}}^2}S_{\mathrm{A}}(S_{\mathrm{A}}+1) + \frac{4Z_{\mathrm{CB}}Z_{\mathrm{BC}}(1-x)^2J_{\mathrm{BC}}^2S_{\mathrm{C}}}{3k_{\mathrm{B}}^2T_{\mathrm{C}}^2}S_{\mathrm{B}}(S_{\mathrm{B}}+1)\right) = 1\\
&\Leftrightarrow T_{\mathrm{C}}^2 = \frac{4S_{\mathrm{C}}(S_{\mathrm{C}}+1)}{(3k_{\mathrm{B}})^2}\left(Z_{\mathrm{CA}}Z_{\mathrm{AC}}x^2J_{\mathrm{AC}}^2S_{\mathrm{A}}(S_{\mathrm{A}}+1) + Z_{\mathrm{CB}}Z_{\mathrm{BC}}(1-x)^2J_{\mathrm{BC}}^2S_{\mathrm{B}}(S_{\mathrm{B}}+1)\right)
\end{aligned}$$

$$\begin{aligned}
\Leftrightarrow T_{\mathrm{C}} = \frac{2}{3k_{\mathrm{B}}}\Big[&S_{\mathrm{C}}(S_{\mathrm{C}}+1)\{Z_{\mathrm{CA}}Z_{\mathrm{AC}}x^2J_{\mathrm{AC}}^2S_{\mathrm{A}}(S_{\mathrm{A}}+1)\\
&+Z_{\mathrm{CB}}Z_{\mathrm{BC}}(1-x)^2J_{\mathrm{BC}}^2S_{\mathrm{B}}(S_{\mathrm{B}}+1)\}\Big]^{1/2}
\end{aligned} \tag{1.18}$$

This equation relates the superexchange interaction parameters (J) and Curie temperature (T_{C}).

As performed in the unary metal system, we consider the magnetic susceptibility (χ) at high temperature region. In this case, we include external magnetic field (H_0) and the equation is

$$\langle M_{\mathrm{A}}\rangle = S_{\mathrm{A}}B_{S_{\mathrm{A}}}\left(\frac{g\mu_{\mathrm{B}}H_{\mathrm{A}}S_{\mathrm{A}}}{k_{\mathrm{B}}T}\right) = S_{\mathrm{A}}B_{S_{\mathrm{A}}}\left(\frac{g\mu_{\mathrm{B}}(H_0+n_{\mathrm{AC}}M_{\mathrm{C}})S_{\mathrm{A}}}{k_{\mathrm{B}}T}\right), \tag{1.19a}$$

$$\langle M_{\mathrm{B}}\rangle = S_{\mathrm{B}}B_{S_{\mathrm{B}}}\left(\frac{g\mu_{\mathrm{B}}H_{\mathrm{B}}S_{\mathrm{B}}}{k_{\mathrm{B}}T}\right) = S_{\mathrm{B}}B_{S_{\mathrm{B}}}\left(\frac{g\mu_{\mathrm{B}}(H_0+n_{\mathrm{BC}}M_{\mathrm{C}})S_{\mathrm{B}}}{k_{\mathrm{B}}T}\right), \tag{1.19b}$$

$$\langle M_{\mathrm{C}}\rangle = S_{\mathrm{C}}B_{S_{\mathrm{C}}}\left(\frac{g\mu_{\mathrm{B}}H_{\mathrm{C}}S_{\mathrm{C}}}{k_{\mathrm{B}}T}\right) = S_{\mathrm{C}}B_{S_{\mathrm{C}}}\left(\frac{g\mu_{\mathrm{B}}(H_0+n_{\mathrm{CA}}M_{\mathrm{A}}+n_{\mathrm{CB}}M_{\mathrm{B}})S_{\mathrm{B}}}{k_{\mathrm{B}}T}\right). \tag{1.19c}$$

Because we consider the magnetization at high temperature, we introduce the approximation of the Brillouin function as $B_{S_{\mathrm{M}}}(x) \approx \frac{S_{\mathrm{M}}+1}{3S_{\mathrm{M}}}x$ again. Then the equation is

$$\frac{M_{\mathrm{A}}}{y_1xN_{\mathrm{A}}g\mu_{\mathrm{B}}} = \frac{S_{\mathrm{A}}+1}{3}\left(\frac{g\mu_{\mathrm{B}}(H_0+n_{\mathrm{AC}}M_{\mathrm{C}})S_{\mathrm{A}}}{k_{\mathrm{B}}T}\right), \tag{1.20a}$$

$$\frac{M_{\mathrm{B}}}{y_1(1-x)N_{\mathrm{A}}g\mu_{\mathrm{B}}} = \frac{S_{\mathrm{B}}+1}{3}\left(\frac{g\mu_{\mathrm{B}}(H_0+n_{\mathrm{BC}}M_{\mathrm{C}})S_{\mathrm{B}}}{k_{\mathrm{B}}T}\right), \tag{1.20b}$$

$$\frac{M_{\mathrm{C}}}{y_2N_{\mathrm{A}}g\mu_{\mathrm{B}}} = \frac{S_{\mathrm{C}}+1}{3}\left(\frac{g\mu_{\mathrm{B}}(H_0+n_{\mathrm{CA}}M_{\mathrm{A}}+n_{\mathrm{CB}}M_{\mathrm{B}})S_{\mathrm{C}}}{k_{\mathrm{B}}T}\right). \tag{1.20c}$$

Introducing $\chi_i = M_i/H_0$ (i = A, B, and C),

$$\chi_{\mathrm{A}} = y_1 x \frac{N_{\mathrm{A}}(g\mu_{\mathrm{B}})^2 S_{\mathrm{A}}(S_{\mathrm{A}}+1)}{3k_{\mathrm{B}}T}(1+n_{\mathrm{AC}}\chi_{\mathrm{C}}), \tag{1.21a}$$

$$\chi_{\mathrm{B}} = y_1 (1-x) \frac{N_{\mathrm{A}}(g\mu_{\mathrm{B}})^2 S_{\mathrm{B}}(S_{\mathrm{B}}+1)}{3k_{\mathrm{B}}T}(1+n_{\mathrm{BC}}\chi_{\mathrm{C}}), \tag{1.21b}$$

$$\chi_{\mathrm{B}} = y_2 (1-x) \frac{N_{\mathrm{A}}(g\mu_{\mathrm{B}})^2 S_{\mathrm{C}}(S_{\mathrm{C}}+1)}{3k_{\mathrm{B}}T}(1+n_{\mathrm{CA}}\chi_{\mathrm{A}}+n_{\mathrm{CB}}\chi_{\mathrm{B}}). \tag{1.21c}$$

Using Curie constants (C_i), $C_i = \frac{N_{\mathrm{A}}(g\mu_{\mathrm{B}})^2 S_i(S_i+1)}{3k_{\mathrm{B}}}$ (i = A, B, and C),

$$\chi_{\mathrm{A}} = y_1 x \frac{C_{\mathrm{A}}}{T}(1+n_{\mathrm{AC}}\chi_{\mathrm{C}}), \tag{1.22a}$$

$$\chi_{\mathrm{B}} = y_1 (1-x) \frac{C_{\mathrm{B}}}{T}(1+n_{\mathrm{BC}}\chi_{\mathrm{C}}), \tag{1.22b}$$

$$\chi_{\mathrm{C}} = y_2 \frac{C_{\mathrm{C}}}{T}(1+n_{\mathrm{CA}}\chi_{\mathrm{A}}+n_{\mathrm{CB}}\chi_{\mathrm{B}}). \tag{1.22c}$$

From these equations, χ_{A}, χ_{B}, χ_{C}, and $\chi = \chi_{\mathrm{A}} + \chi_{\mathrm{B}} + \chi_{\mathrm{C}}$ are obtained as

$$\begin{aligned}
\chi_{\mathrm{C}} &= y_2 \frac{C_{\mathrm{C}}}{T}\left\{1+n_{\mathrm{CA}}y_1 x\frac{C_{\mathrm{A}}}{T}(1+n_{\mathrm{AC}}\chi_{\mathrm{C}})+n_{\mathrm{CB}}y_1(1-x)\frac{C_{\mathrm{B}}}{T}(1+n_{\mathrm{BC}}\chi_{\mathrm{C}})\right\} \\
&\Leftrightarrow \chi_{\mathrm{C}}\left(1-n_{\mathrm{CA}}n_{\mathrm{AC}}y_1 x\frac{C_{\mathrm{A}}}{T}-n_{\mathrm{CB}}n_{\mathrm{BC}}y_1(1-x)\frac{C_{\mathrm{B}}}{T}\right) \\
&= y_2 \frac{C_{\mathrm{C}}}{T}\left\{1+n_{\mathrm{CA}}y_1 x\frac{C_{\mathrm{A}}}{T}+n_{\mathrm{CB}}y_1(1-x)\frac{C_{\mathrm{B}}}{T}\right\} \\
&\therefore \quad \chi_{\mathrm{C}} = \frac{y_2 C_{\mathrm{C}}\{T+n_{\mathrm{CA}}y_1 x C_{\mathrm{A}}+n_{\mathrm{CB}}y_1(1-x)C_{\mathrm{B}}\}}{T^2-\{n_{\mathrm{CA}}n_{\mathrm{AC}}y_1 x C_{\mathrm{A}}+n_{\mathrm{CB}}n_{\mathrm{BC}}y_1(1-x)C_{\mathrm{B}}\}\,T} \\
&= y_2 \frac{C_{\mathrm{C}}}{T}\frac{\{T^2+n_{\mathrm{CA}}y_1 x C_{\mathrm{A}}T+n_{\mathrm{CB}}y_1(1-x)C_{\mathrm{B}}T\}}{T^2-\{n_{\mathrm{CA}}n_{\mathrm{AC}}y_1 x C_{\mathrm{A}}+n_{\mathrm{CB}}n_{\mathrm{BC}}y_1(1-x)C_{\mathrm{B}}\}\,T} \\
&= y_2 \frac{C_{\mathrm{C}}}{T}[1+\frac{(1+n_{\mathrm{AC}})n_{\mathrm{CA}}y_1 x C_{\mathrm{A}}T+(1+n_{\mathrm{BC}})n_{\mathrm{CB}}y_1(1-x)C_{\mathrm{B}}T}{T^2-\{n_{\mathrm{CA}}n_{\mathrm{AC}}y_1 x C_{\mathrm{A}}+n_{\mathrm{CB}}n_{\mathrm{BC}}y_1(1-x)C_{\mathrm{B}}\}\,T},
\end{aligned} \tag{1.23a}$$

$$\begin{aligned}
\chi_{\mathrm{A}} &= y_1 x \frac{C_{\mathrm{A}}}{T}\left[1+n_{\mathrm{AC}}\frac{y_2 C_{\mathrm{C}}\{T+n_{\mathrm{CA}}y_1 x C_{\mathrm{A}}+n_{\mathrm{CB}}y_1(1-x)C_{\mathrm{B}}\}}{T^2-\{n_{\mathrm{CA}}n_{\mathrm{AC}}y_1 x C_{\mathrm{A}}+n_{\mathrm{CB}}n_{\mathrm{BC}}y_1(1-x)C_{\mathrm{B}}\}\,T}\right] \\
&= y_1 x \frac{C_{\mathrm{A}}}{T}\left[1+\frac{n_{\mathrm{AC}}y_2 C_{\mathrm{C}}T+n_{\mathrm{AC}}y_2 C_{\mathrm{C}}n_{\mathrm{CA}}y_1 x C_{\mathrm{A}}+n_{\mathrm{AC}}y_2 C_{\mathrm{C}}n_{\mathrm{CB}}y_1(1-x)C_{\mathrm{B}}}{T^2-\{n_{\mathrm{CA}}n_{\mathrm{AC}}y_1 x C_{\mathrm{A}}+n_{\mathrm{CB}}n_{\mathrm{BC}}y_1(1-x)C_{\mathrm{B}}\}\,T}\right],
\end{aligned} \tag{1.23b}$$

$$\chi_B = y_1(1-x)\frac{C_B}{T}\left[1+n_{BC}\frac{y_2C_C\{T+n_{CA}y_1xC_A+n_{CB}y_1(1-x)C_B\}}{T^2-\{n_{CA}n_{AC}y_1xC_A+n_{CB}n_{BC}y_1(1-x)C_B\}\ T}\right]$$
$$= y_1(1-x)\frac{C_B}{T}\left[1+\frac{n_{BC}y_2C_CT+n_{BC}y_2C_Cn_{CA}y_1xC_A+n_{BC}y_2C_Cn_{CB}y_1(1-x)C_B\}}{T^2-\{n_{CA}n_{AC}y_1xC_A+n_{CB}n_{BC}y_1(1-x)C_B\}\ T}\right], \tag{1.23c}$$

and

$$\chi = y_1x\frac{C_A}{T}\left[1+\frac{n_{AC}y_2C_CT+n_{AC}y_2C_Cn_{CA}y_1xC_A+n_{AC}y_2C_Cn_{CB}y_1(1-x)C_B}{T^2-\{n_{CA}n_{AC}y_1xC_A+n_{CB}n_{BC}y_1(1-x)C_B\}\ T}\right]$$
$$+y_1(1-x)\frac{C_B}{T}\left[1+\frac{n_{BC}y_2C_CT+n_{BC}y_2C_Cn_{CA}y_1xC_A+n_{BC}y_2C_Cn_{CB}y_1(1-x)C_B\}}{T^2-\{n_{CA}n_{AC}y_1xC_A+n_{CB}n_{BC}y_1(1-x)C_B\}\ T}\right]$$
$$+y_2\frac{C_C}{T}\left[1+\frac{(1+n_{AC})n_{CA}y_1xC_AT+(1+n_{BC})n_{CB}y_1(1-x)C_BT}{T^2-\{n_{CA}n_{AC}y_1xC_A+n_{CB}n_{BC}y_1(1-x)C_B\}\ T}\right] \tag{1.24}$$

where

$$n_{AC}=\frac{2Z_{AC}}{y_2N_A(g\mu_B)^2}J_{AC},\ n_{BC}=\frac{2Z_{BC}}{y_2N_A(g\mu_B)^2}J_{BC},$$
$$n_{CA}=\frac{2Z_{CA}}{y_1N_A(g\mu_B)^2}J_{AC},\ n_{CB}=\frac{2Z_{CB}}{y_1N_A(g\mu_B)^2}J_{CB}. \tag{1.25}$$

The lease square fitting of the experimental magnetic susceptibility allows us to estimate the superexchange interaction constant (J) and g factor.

1.3 Van Vleck Paramagnetism

(i) Van Vleck equation [22]

Van Vleck theory is developed to perform theoretical treatment of magnetic properties of mononuclear or polynuclear metal complexes. This theory is based on quantum chemistry and we calculate magnetic susceptibility by taking account of Boltzmann distribution of each quantum state which is split by Zeeman effect, ligand field, spin-orbit coupling, and exchange interaction (spin-spin coupling). The Van Vleck equation is obtained by considering Zeeman effect as follows.

The magnetization value (I) is related to the energy value of a material (W) as

$$I=-\frac{\partial W}{\partial H}, \tag{1.26}$$

where H is the external magnetic field.

When we consider that an atom is placed in an external magnetic field, the energy of level i (W_i) can be expanded as

$$W_i = W_i^{(0)} + W_i^{(1)}H + W_i^{(2)}H^2 + o(H^3), \tag{1.27}$$

where $W_i^{(1)}$, $W_i^{(2)}$, … are called as first, second, … order Zeeman coefficients.

Ignoring the third and higher order term,

$$I_i = -\frac{\partial W}{\partial H} = -W_i^{(1)} - 2W_i^{(2)}H. \tag{1.28}$$

Considering the Boltzmann distribution, the magnetic susceptibility ($\chi = I/H$) is

$$\chi = \frac{N}{H}\frac{\sum I_i \exp(W_i/kT)}{\sum \exp(W_i/kT)} = \frac{(N/H)\sum(-W_i^{(1)} - 2W_i^{(2)}H)\exp(W_i/kT)}{\sum \exp(W_i/kT)}, \tag{1.29}$$

where N is Avogadro constant. Considering the order of Zeeman energy is a few wavenumber and $k \sim 0.695\ \mathrm{cm}^{-1}$, at high temperature region, the approximation of $W_i^{(1)} \ll kT$ can be applied. In this condition,

$$\begin{aligned}
&\exp(W_i^{(1)}/kT) \sim 1 - W_i^{(1)}/kT \text{ and } \exp(W_i^{(2)}/kT) \sim 1 - W_i^{(2)}/kT, \text{ so} \\
&\qquad \exp(-W_i/kT) = \exp(-W_i^{(0)} - W_i^{(1)}H - W_i^{(2)}H^2) \\
&\qquad\qquad = \exp(-W_i^{(0)})(1 - W_i^{(1)}H/kT)(1 - W_i^{(2)}H^2/kT) \\
&\qquad\qquad \sim \exp(-W_i^{(0)})(1 - W_i^{(1)}H/kT)
\end{aligned}$$

Therefore,

$$\begin{aligned}
\chi &= \frac{(N/H)\sum(-W_i^{(1)} - 2W_i^{(2)}H)\exp(W_i/kT)}{\sum \exp(W_i/kT)} \\
&= \frac{(N/H)\sum(-W_i^{(1)} - 2W_i^{(2)}H)\exp(-W_i^{(0)}/kT)(1 - W_i^{(1)}H/kT)}{\sum \exp(-W_i^{(0)}/kT)(1 - W_i^{(1)}H/kT)}
\end{aligned}$$

$$\begin{aligned}
&= \frac{N\sum(-W_i^{(1)}/H)\exp(-W_i^{(0)}/kT) + N\sum\{(W_i^{(1)})^2 - 2W_i^{(2)}\}\exp(-W_i^{(0)}/kT)}{\sum \exp(-W_i^{(0)}/kT)(1 - W_i^{(1)}H/kT)} \\
&+ \frac{N\sum(2W_i^{(1)}W_i^{(2)}H/kT)\exp(-W_i^{(0)}/kT)}{\sum \exp(-W_i^{(0)}/kT)(1 - W_i^{(1)}H/kT)}.
\end{aligned} \tag{1.30}$$

Here, noting that Zeeman splitting does not change the average energy value, all terms which have odd order of $W_i^{(1)}$ become zero. Therefore,

$$\chi = \frac{N\sum(-W_i^{(1)}/H)\exp(-W_i^{(0)}/kT) + N\sum\{(W_i^{(1)})^2 - 2W_i^{(2)}\}\exp(-W_i^{(0)}/kT)}{\sum\exp(-W_i^{(0)}/kT)(1 - W_i^{(1)}H/kT)} + \frac{N\sum(2W_i^{(1)}W_i^{(2)}H/kT)\exp(-W_i^{(0)}/kT)}{\sum\exp(-W_i^{(0)}/kT)(1 - W_i^{(1)}H/kT)} = \frac{N\sum\{(W_i^{(1)})^2 - 2W_i^{(2)}\}\exp(-W_i^{(0)}/kT)}{\sum\exp(-W_i^{(0)}/kT)}. \quad (1.31)$$

When we calculate magnetic susceptibility, we obtain $W_i^{(1)}$, $W_i^{(2)}$ by considering spin-orbit coupling, and exchange interaction (spin-spin coupling), and Zeeman splitting, and then we assign the obtained values to the above equations. When Zeeman splitting is smaller enough than spin-orbit coupling and/or exchange interaction, we can treat Zeeman splitting as perturbation. The example calculation of magnetic susceptibility of octahedral Fe(II) ion (d^6) is given in the following.

(ii) Magnetic susceptibility of octahedral Fe(II) ion (d^6) with $^5T_2$ term[27]

As an example, I will introduce the calculation of magnetic susceptibility of octahedral Fe(II) ion as follows.

When we consider free ion, the five d orbitals are degenerated and the orbital magnetic quantum number (M_L) is a good quantum number. Here, we represent d orbitals by $|M_L\rangle$ (M_L = 2, 1, 0, −1, −2), which is a eigenstate corresponding to the eigenvalue of M_L. However, in octahedral field, the energy of d orbital splits into three t_{2g} orbitals and two $e_{.g.}$ orbitals and the orbital magnetic quantum number (M_L) is no longer a good quantum number. Taking the principle axis with 4-fold rotation axis, the ground states of octahedral Fe(II) ion (d^6) are triply degenerated and are expressed by the linear combination of $|M_L\rangle$ as

$$|1\rangle, |-1\rangle, \text{and } \sqrt{\frac{1}{2}}(|2\rangle - |-2\rangle).$$

Considering that the Fe(II) ion (d^6) has spin quantum number (S) of 2, the wavefunctions of $^5T_2$ term is described as the following 15 wavefunctions shown as $|m_L\rangle[m_S]$.

$$|1\rangle[+2], \quad |1\rangle[+1], \quad |1\rangle[0], \quad |1\rangle[-1], \quad |1\rangle[-2],$$
$$-|1\rangle[+2], \quad |-1\rangle[+1], \quad |-1\rangle[0], \quad |-1\rangle[-1], \quad |-1\rangle[-2]$$
$$\sqrt{1/2}(|+2\rangle - |-2\rangle)[+2], \sqrt{1/2}(|+2\rangle - |-2\rangle)[+1],$$
$$\sqrt{1/2}(|+2\rangle - |-2\rangle)[0], \sqrt{1/2}(|+2\rangle - |-2\rangle)[-1], \sqrt{1/2}(|+2\rangle - |-2\rangle)[-2]$$

From here, we are going to consider the spin-orbit coupling (λ) as the following Hamiltonian.

$$H = \lambda LS.$$

The interaction among the above 15 wavefunctions can be obtained by the following relation,

$$\begin{aligned}
\boldsymbol{L}_{\mathbf{z}}|L, M_L, S, M_S\rangle &= M_L|L, M_L, S, M_S\rangle \\
\boldsymbol{L}_{+}|L, M_L, S, M_S\rangle &= \sqrt{L(L+1) - M_L(M_L+1)}|L, M_L+1, S, M_S\rangle \\
\boldsymbol{L}_{-}|L, M_L, S, M_S\rangle &= \sqrt{L(L+1) - M_L(M_L-1)}|L, M_L-1, S, M_S\rangle \\
\boldsymbol{S}_{\mathbf{z}}|L, M_L, S, M_S\rangle &= M_S|L, M_L, S, M_S\rangle \\
\boldsymbol{S}_{+}|L, M_L, S, M_S\rangle &= \sqrt{S(S+1) - M_S(M_S+1)}|L, M_L, S+1, M_S\rangle \\
\boldsymbol{S}_{-}|L, M_L, S, M_S\rangle &= \sqrt{S(S+1) - M_S(M_S-1)}|L, M_L, S-1, M_S\rangle
\end{aligned} \tag{1.32}$$

where L, M_L, S, and M_S is are azimuthal quantum number, orbital magnetic quantum number, spin quantum number, and the z-projection of spin quantum

	$\lvert+1\rangle[+2]$	A	$\lvert-1\rangle[0]$	$\lvert-1\rangle[-2]$	B	$\lvert+1\rangle[0]$	$\lvert+1\rangle[+1]$	C	$\lvert-1\rangle[-1]$	$\lvert+1\rangle[-1]$	D	$\lvert-1\rangle[+1]$	E	$\lvert+1\rangle[-2]$	$\lvert-1\rangle[+2]$
$\lvert+1\rangle[+2]$	2λ	$\sqrt{2}\lambda$	0	0	0	0	0	0	0	0	0	0	0	0	0
A	$\sqrt{2}\lambda$	0	$-\sqrt{3}\lambda$	0	0	0	0	0	0	0	0	0	0	0	0
$\lvert-1\rangle[0]$	0	$-\sqrt{3}\lambda$	0	0	0	0	0	0	0	0	0	0	0	0	0
$\lvert-1\rangle[-2]$	0	0	0	2λ	$\sqrt{2}\lambda$	0	0	0	0	0	0	0	0	0	0
B	0	0	0	$\sqrt{2}\lambda$	0	$-\sqrt{3}\lambda$	0	0	0	0	0	0	0	0	0
$\lvert+1\rangle[0]$	0	0	0	0	$-\sqrt{3}\lambda$	0	0	0	0	0	0	0	0	0	0
$\lvert+1\rangle[+1]$	0	0	0	0	0	0	λ	$\sqrt{3}\lambda$	0	0	0	0	0	0	0
C	0	0	0	0	0	0	$\sqrt{3}\lambda$	0	$-\sqrt{3}\lambda$	0	0	0	0	0	0
$\lvert-1\rangle[-1]$	0	0	0	0	0	0	0	$-\sqrt{3}\lambda$	λ	0	0	0	0	0	0
$\lvert+1\rangle[-1]$	0	0	0	0	0	0	0	0	0	$-\lambda$	$\sqrt{2}\lambda$	0	0	0	0
D	0	0	0	0	0	0	0	0	0	$\sqrt{2}\lambda$	0	0	0	0	0
$\lvert-1\rangle[+1]$	0	0	0	0	0	0	0	0	0	0	0	$-\lambda$	$-\sqrt{2}\lambda$	0	0
E	0	0	0	0	0	0	0	0	0	0	0	$-\sqrt{2}\lambda$	0	0	0
$\lvert+1\rangle[-2]$	0	0	0	0	0	0	0	0	0	0	0	0	0	-2λ	0
$\lvert-1\rangle[+2]$	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-2λ

A $\sqrt{1/2}(|+2\rangle - |-2\rangle)[+1]$

C $\sqrt{1/2}(|+2\rangle - |-2\rangle)[0]$

E $\sqrt{1/2}(|+2\rangle - |-2\rangle)[+2]$

B $\sqrt{1/2}(|+2\rangle - |-2\rangle)[-1]$

D $\sqrt{1/2}(|+2\rangle - |-2\rangle)[-2]$

Fig. 1.3 The 15×15 matrix obtained by considering the spin-orbit coupling between the wavefunctions of octahedral FeII ion (d6) with 5T2 term with the Hamiltonian. By diagonalizing this matrix, the eigenenergies and eigenfunctions are obtained

number. As a result, we can obtain the 15 × 15 matrix based on $\langle L_i, M_{L_i}, S_i, M_{S_i}|\lambda LS|L_j, M_{L_j}, S_j, M_{S_j}\rangle$ as shown in Fig. 1.3.

By block diagonalization of this 15 × 15 matrix, we have to solve three 3 × 3 matrices and two 2 × 2 matrices, and other two eigenvalues are diagonal elements. By diagonalization, we obtain the eigenenergies ($W^{(0)}$) as shown below,

$$W^{(0)} = 3\lambda\,(g=3), W^{(0)} = \lambda\,(g=5), W^{(0)} = -2\lambda\,(g=7),$$

where g is the degree of degeneracy.

Next, we consider Zeeman splitting of these eigenfunctions.

The Zeeman energy is obtained by the following equation,

$$\text{1st order Zeeman energy: } E_{iz}^{(1)} = \mu_B \sum_j \left\langle \psi_i^{(0)}|L_z + 2S_z|\psi_j^{(0)}\right\rangle \tag{1.33}$$

$$\text{2nd order Zeeman energy: } E_{iz}^{(2)} = \mu_B^2 \sum\nolimits_j \frac{\left\langle \psi_i^{(0)}|L_z + 2S_z|\psi_j^{(0)}\right\rangle \left\langle \psi_i^{(0)}|L_z + 2S_z|\psi_j^{(0)}\right\rangle}{E_i^{(0)} - E_j^{(0)}} \tag{1.34}$$

Here, we have only to consider z component because x, y, and z are isotropic in this calculation.

1st order Zeeman splitting occurs only between the degenerated eigenfunctions, whereas 2nd order Zeeman splitting should be considered between the eigenfunctions with different energies. As a result, we obtain first and second order Zeeman energy ($W^{(1)}$ and $W^{(2)}$) as described in Fig. 1.4.

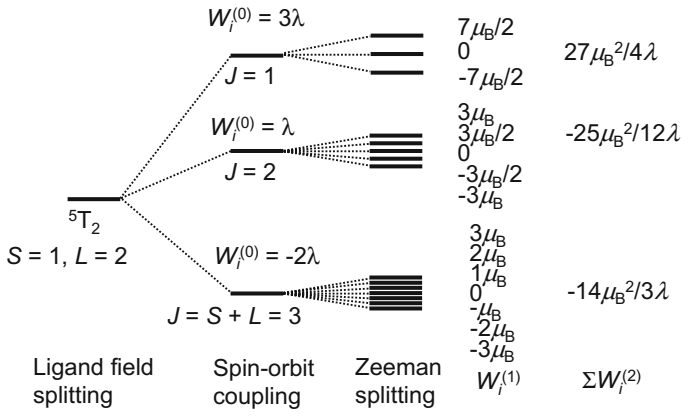


Fig. 1.4 The energy separation of octahedral FeII ion (d6) with 5T2 term due to spin-orbit coupling and Zeeman splitting. The first and second order Zeeman energies (W(1) and W(2)) are described in the *right side*

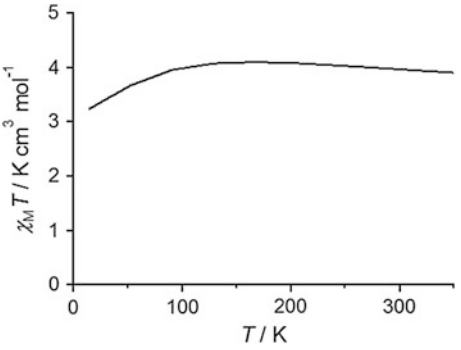


Fig. 1.5 Temperature dependence of magnetic susceptibility of octahedral Fe^{II} ion (d^6) with $^5T_2$ term when $\lambda = -100$ cm^{-1}

By substituting the obtained energy values to the Van Vleck equation, we can obtain the following magnetic susceptibility.

$$\chi = \frac{3\left[28x + \frac{28}{3} + \left(\frac{45}{2}x + \frac{25}{6}\right)\exp(-3x) + \left(\frac{49}{2}x - \frac{27}{2}\right)\exp(-5x)\right]}{8Tx[7 + 5\exp(-3x) + 3\exp(-5x)]} \tag{1.35}$$

where $x = kT/\lambda$.

The spin-orbit coupling constant of free Fe(II) ion is -100 cm^{-1} and the calculated magnetic susceptibility is shown in Fig. 1.5.

Such a calculation and the comparison with the experimental magnetic susceptibility enable us to elucidate spin-orbit coupling constant, superexchange interaction constant, zero-field splitting parameters, etc.

Since this type of approach is very useful to estimate magnetic susceptibility and magnetic anisotropy of a compound, a lot of studies about magnetic cluster molecules are reported. However, this type of approach is unable to describe magnetic phase transition behavior like molecular field theory which is described in the previous section. In 1.3 and 1.4, I have introduced basic theory for magnetic analysis of molecule based magnets. In the following section, from the view point of materials chemistry and functionalites, I will introduce molecule based magnets, especially, cyanido-bridged metal assemblies.

1.4 Cyanido-Bridged Metal Assemblies and Their Functionalities

Cyanido-bridged metal assemblies are a kind of metal complex where metal ions are bridged by cyanido group (M-CN-M'). Among cyanido-bridged metal assemblies, one of the well-known compound is Prussian blue, $Fe^{III}_4\left[Fe^{II}(CN)_6\right]_3 \cdot 7.5H_2O$, which has three-dimensional (3D) network structure in which Fe^{II} and Fe^{III} are alternatively bridged by CN ligand with –NC–Fe^{II}–CN–Fe^{III}–NC– fashion [55]. This material has been used as blue pigment for a long time. The origin of the deep

blue color in Prussian blue is metal to metal charge transfer band between Fe^{II} and Fe^{III} due to the mixed valence state of metal ions [56]. Based on this mixed valency, electrochromic properties have been extensively investigated [57, 58]. In addition, magnetic phase transition of Prussian blue was realized in 1968 with the Curie temperature (T_C) of 5.6 K [59]. In this way, Prussian blue shows interesting physical properties. Furthermore, metal ions of Fe^{II} and Fe^{III} of Prussian blue can be substituted to other metal ions and such a compound is called Prussian blue analogue. Prussian blue analogues are very attractive for designing magnetic properties and are extensively investigated from 1986 [60]. In Prussian blue analogues, mainly two types of formulas are familiar, one is $M_A^{II}[M_B^{III}(CN)_6]_{2/3} \cdot 5H_2O$ and the other is $AM_A^{II}[M_B^{III}(CN)_6]_3$, where M_A and M_B are transition metal ions, and A is alkali metal ion. Both types of Prussian blue analogues take a jungle-gym-type three-dimensional network structure, while the crystal structure is different. $M_A^{II}[M_B^{III}(CN)_6]_{2/3} \cdot 5H_2O$ takes *Fm*-3*m* space group and 1/3 of $[M_B^{III}(CN)_6]$ is vacant and water molecules occupy the vacancy (Fig. 1.6a) [61]. $AM_A^{II}[M_B^{III}(CN)_6]_3$ prefers non-centrosymmetric *F*-43*m* space group, in which there is no vacancy in the framework structure and the interstitial sites is occupied by Alkali metal ions in alternative fashion (Fig. 1.6b) [62]. Our laboratory have reported various functionalities such as humidity sensitive magnetism and humidity induced magnetic pole inversion [15], photo-induced magnetic pole inversion [9], mixed ferro-ferri magnetism [18, 19], and two compensation temperature [19]. These unique magnetic properties and functionalities are based on the following characteristics of cyanido-bridged metal assemblies. First, we can consider the magnetic interaction only among nearest neighbor metal sites because the distance between two metal centers through cyanide is sufficiently long (ca. 5 Å). Second, the magnetic interaction among nearest neighbor metal sites is strong enough to observe magnetic phase transition. Third, Prussian blue analogues can show various colors and mixed valence properties as well as original Prussian blue. These three characteristics are

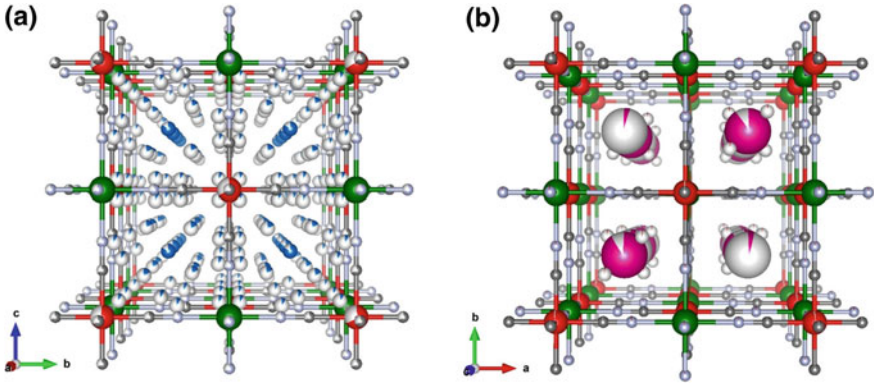


Fig. 1.6 The crystal structure of Prussian blue analogues. **a** $M_A^{II}[M_B^{III}(CN)_6]_{2/3} \cdot 5H_2O$ with vacancies and water molecules. The atomic coordinates are based on [66]. **b** $AM_A^{II}[M_B^{III}(CN)_6]_3$ with non-centrosymmetric crystal structure. The atomic coordinates are based on [67]

important to demonstrate designed magnetic properties using molecular field theory and photo-induced magnetization. Recently, in addition to $[M(CN)_6]$-based magnets, various types of cyanidometalate-based complexes are utilized for molecule-based magnet, e.g., heptacyanidometalates ($[M(CN)_7]$) and octacyanidometalates ($[M(CN)_8]$). Among them, octacyanidometalates are widely investigated for molecule-based magnet and I will describe the characteristics and functionalities of octacyanidometalate-based magnets.

1.5 Octacyanidometalate-Based Magnets

Octacyanidometalate ($[M(CN)_8]^{n-}$)-based magnets are expected to show various functionalities because octacyanmetalate-based magnets can take various crystal structures. Among octacyanidometalate, octacyanidomolybdate, octacyanidotungstate, and octcyaniobate $[M(CN)_8]^{n-}$(M = Mo, W, Nb), are mainly used as building blocks of molecular magnets for the following reasons. First, $[Mo(CN)_8]^{3-}$, $[W(CN)_8]^{3-}$, and $[Nb(CN)_8]^{4-}$ have unpaired electron, which enables strong magnetic interaction in cyanido-bridged metal assemblies. Second, diffuse 4d or 5d ortibals enhance the superexchange interaction between metal centers bridged by cyanide. Third, $[Mo(CN)_8]^{n-}$ and $[W(CN)_8]^{n-}$ ions can take multiple valence states like $[Mo(CN)_8]^{3-}$/$[Mo(CN)_8]^{4-}$ and $[W(CN)_8]^{3-}$/$[W(CN)_8]^{4-}$ [63, 64]. Fourth, $[M(CN)_8]^{n-}$ can take various coordination geometries such as square antiprism, dodecahedron, and bicapped trigonal prism (Fig. 1.7).

These properties are promising for functional magnet and high T_C magnet. In fact, our laboratory have reported a zero-dimensional cluster complex with a large spin quantum number of $S = 39/2$ [65], a one-dimensional porous chain complex having chirality [66], a two-dimensional layered complex which shows cooling rate dependent magnetism [67], nanoporous three-dimensional network complex [68], and so on (Fig. 1.8). Magnetic properties of these compounds reflect their dimensionalities. Moreover, various functional molecule-based magnets have been reported in our laboratory such as photo-reversible magnetization [69], and alcohol

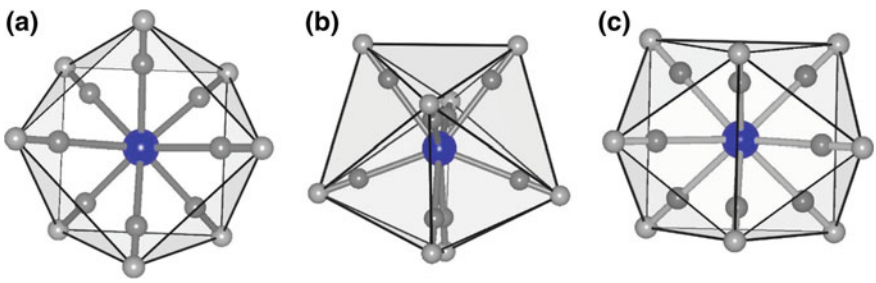


Fig. 1.7 The coordination geometries of octacyanidometalate. **a** Square-antiprism, **b** dodecahedron, and **c** bicapped trigonal prism

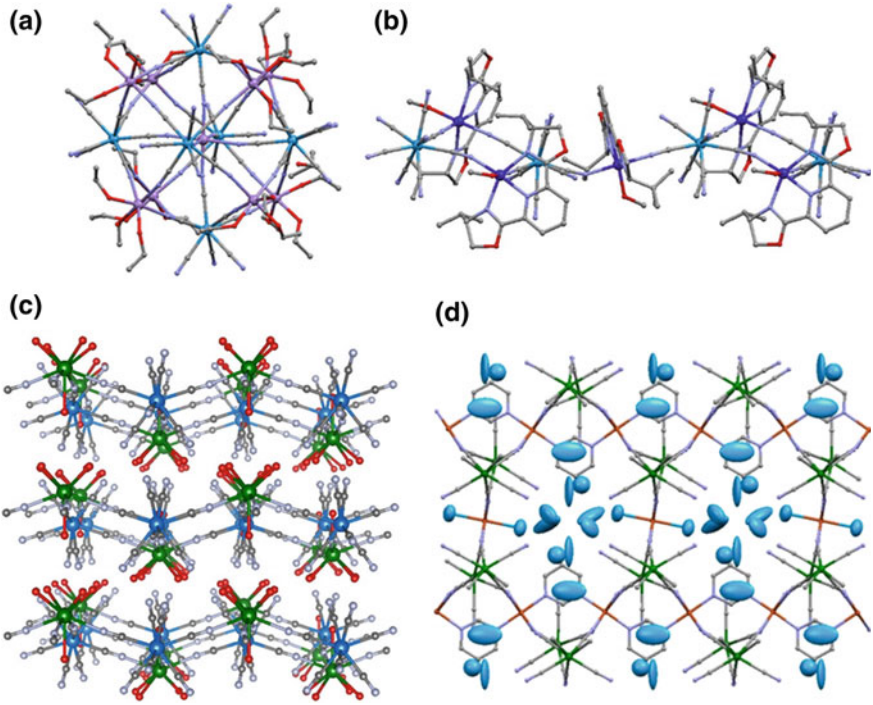


Fig. 1.8 The crystal structure of octacyanidometalate-based magnets with 0-D, 1-D, 2-D, and 3-D network structures. **a** Pentadecaanuclear Mn_9W_6 cluster (0-D), **b** 1-D Co-W-based chain, **c** 2-D Sm-W-based magnet, and **d** 3-D Cu-W-based magnet which shows alcohol vapor sensitivity. The atomic coordinates are based on [65–68], respectively

vapor sensitive magnet [68]. In addition, we have recently realized a vanadium-octacyanidoniobate magnet which shows a high Curie temperature of 210 K [70], magnetization-induced second harmonic generation (MSHG) in $[Mn^{II}(H_2O)_2][Mn^{II}(pyrazine)(H_2O)_2][Nb^{IV}(CN)_8] \cdot 4H_2O$ [71], humidity-induced magnetism in $Co_2[Nb(CN)_8] \cdot zH_2O$ [72]. These studies have shown that $[M(CN)_8]^{4-}$ is useful for constructing functionalized molecule based magnet.

1.6 Photo-Induced Magnetization in Cyanido-Bridged Metal Assemblies

Photo-induced magnetization is one of the most attractive functionalities in molecule based magnet because the control of magnetization by optical stimuli can be applied to magneto-optical device. As photo-induced magnetization of hexacyanidometalate-based assembly, $K_{0.2}Co_{1.4}[Fe(CN)_6]$ was reported [40] and our laboratory have reported photo-induced demagnetization and magnetic pole

inversion in $Fe[Cr(CN)_6]_{2/3} \cdot 5H_2O$ and $Fe_{0.40}Mn_{0.60}[Cr(CN)_6]_{2/3} \cdot 5H_2O$ [9, 73], respectively. In addition, our laboratory have reported $Rb_{0.88}Mn[Fe(CN)_6]_{0.96} \cdot 0.5H_2O$ [12], $Cu_2[Mo(CN)_8] \cdot 8H_2O$ [11], $CsCo[W(CN)_8](3\text{-cyanidopyridine})_2 \cdot H_2O$ [10], $Co_3[W(CN)_8]_2(pyrimidine)_4 \cdot 6H_2O$ [69], and so on. Photo-induced magnetization of these systems (other than $Fe[Cr(CN)_6]_{2/3} \cdot 5H_2O$ and $Fe_{0.40}Mn_{0.60}[Cr(CN)_6]_{2/3} \cdot 5H_2O$) is triggered by charge transfer phenomenon between metal ions, and the mechanisms of photo magnetization are mainly classified as charge transfer, charge transfer induced spin transition (CTIST), and charge transfer induced Jahn-Teller distortion as shown below.

1. Charge transfer

The photo-induced charge transfer phenomenon causes the change of electronic state of metal ions, which triggers photo-induced change of magnetization in three-dimensional cyanido-bridged metal assembly. For example, $Cu_2[Mo(CN)_8] \cdot 8H_2O$ before irradiation possesses the valence state of $Cu^{II}(S = 1/2)$ and $Mo^{IV}(S = 0)$, so the formula is expressed as $Cu^{II}_2\left[Mo^{IV}(CN)_8\right] \cdot 8H_2O$ After irradiating, charge transfer between $Cu^{II}(S = 1/2)$ and $Mo^{IV}(S = 0)$ occurs and the valence state changes to $Cu^{I}(S = 0)$ and $Mo^{V}(S = 1/2)$. By the irradiation of 473-nm light which correspond to the metal-to-metal charge transfer band between Cu^{II} and Mo^{IV}, the charge transfer from Mo^{IV} to Cu^{II} occurs and the electronic formula after irradiation is expressed as $Cu^{I}Cu^{II}[Mo^{V}(CN)_8] \cdot 8H_2O$. As a result, a spontaneous magnetization is generated by the magnetic interaction between the photo-produced $Mo^{V}(S = 1/2)$ and the remaining $Cu^{II}(S = 1/2)$. The photo-induced magnetization of this type of mechanism is caused by the mixed valence state. In class II mixed valence compound, the energy barrier appears between the original state and the charge separated state (Fig. 1.9) [74, 75]. This energy barrier enables the trapping of photo-induced state, resulting in the photo-induced magnetization (Fig. 1.10).

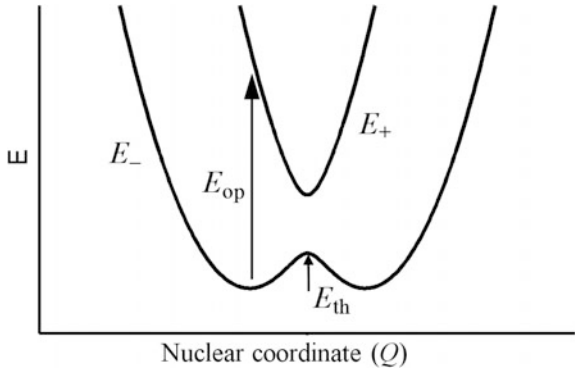


Fig. 1.9 Antisymmetrical breathing nuclear coordinate Q in a mixed-valence dinuclear compound A-B ↔ A-B with chemically equivalent A and B sites

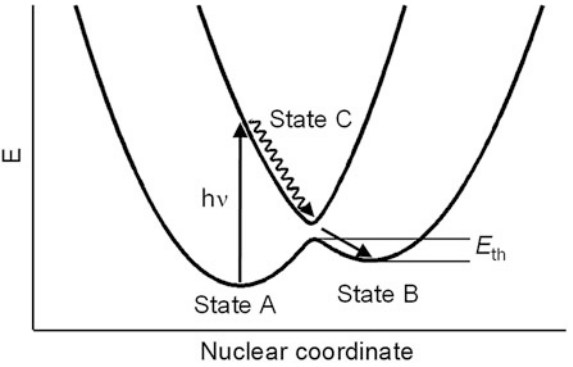


Fig. 1.10 The bistability and photo-induced change of electronic state due to the energy barrier derives from the mixed valency

2. Charge transfer induced spin transition (CTIST)

Some photomagnetic materials show photo-induced charge transfer phenomenon and the subsequent spin transition phenomenon [charge transfer induced spin transition (CTIST)]. $CsCo[W(CN)_8](3\text{-cyanidopyridine})_2 \cdot 6H_2O$ is one of a typical example of this type of photomagnet. The valence state of Co and W before irradiation is Co^{III}(low-spin (LS), $S = 0$) and $W^{IV}(S = 0)$, respectively. By irradiation, charge transfer from $W^{IV}(S = 0)$ to Co^{III}(LS, $S = 0$) occurs and the valence state changes to Co^{II}(LS, $S = 1/2$) and $W^{V}(S = 1/2)$. In this system, Co^{II} takes high-spin (HS) state rather than LS state, so spin transition takes place from Co^{II}(LS, $S = 1/2$) to Co^{II}(HS, $S = 3/2$). Magnetic interaction between Co^{II}(HS, $S = 3/2$) and $W^{V}(S = 1/2)$ generates a spontaneous magnetization. In addition to $CsCo[W(CN)_8](3\text{-cyanidopyridine})_2 \cdot H_2O$, CTIST is also observed in $Co_3[W(CN)_8]_2(\text{pyrimidine})_4 \cdot 6H_2O$ and $K_{0.2}Co_{1.4}[Fe(CN)_6]$. The origin of the bistability derives from mixed valency, and the energy of valence isomer is more stabilized by spin transition phenomenon (Fig. 1.11).

3. Charge transfer induced Jahn-Teller distortion

$Rb_{0.88}Mn[Fe(CN)_6]_{0.96} \cdot 0.5H_2O$ is representative for charge transfer induced Jahn-Teller distortion. The valence state of Mn and Fe before irradiation is

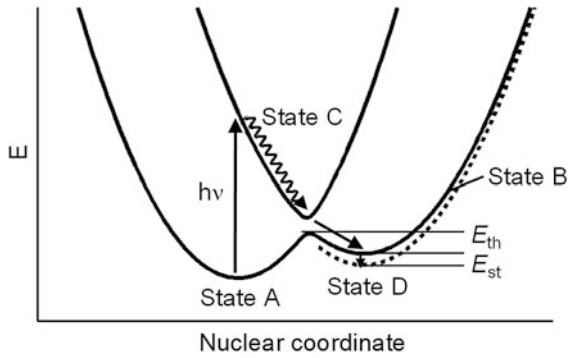


Fig. 1.11 The bistability and photo-induced change of electronic state by charge transfer induced spin transition (CTIST)

Mn^{III}(HS, $S = 2$) and Fe^{II}(LS, $S = 0$). By irradiation, charge transfer between Mn^{III}(HS, $S = 2$) and Fe^{II}(LS, $S = 0$) occurs and the valence state changes to Mn^{II}(HS, $S = 5/2$) and Fe^{III}(LS, $S = 1/2$). In this system, crystal structure changes between tetragonal to cubic because of the Jahn-Teller distortion of $Mn^{III}(d^4)$ ion. In this mechanism, the mixed valency plays a main role for the bistability.

Recently, photo-induced magnetization is also reported in photo-isomerization systems using spiro-piran (SP) in (SP)$MnCr(oxalate)_3 \cdot H_2O$ [76] and (SP)Fe[Fe(dithiooxalate)$_3$] [77]. Still, a drastic change between paramagnetism and ferromagnetism or between antiferromagnetism and ferromagnetism was reported mostly from our laboratory in cyanido-bridged metal assemblies.

1.7 Spin-Crossover Phenomenon

Spin-crossover is a phenomenon in which electronic state changes between high spin (HS) state and low spin (LS) state by external stimuli such as temperature, pressure and light [78–82]. Spin-crossover phenomenon in octahedral metal complex can occur in d^4 to d^7 metal ions, e.g., Cr^{II} (d^4), Mn^{III} (d^4), Mn^{II} (d^5), Fe^{III} (d^5), Fe^{II} (d^6), Co^{III} (d^6), Co^{II} (d^7) [78–89]. In particular, d^6 metal ions show spin-crossover phenomenon between paramagnetic high spin state ($S = 2$) and diamagnetic low spin state ($S = 0$) (Fig. 1.12), which is fascinating from the view point of application to molecular memory devices or sensors [90–94]. Before the term "Spin-crossover phenomenon" become widely used in the field of chemistry, the anomalous magnetic behavior of iron(III) complexes were reported by Cambi et al. in [Fe^{III}(dithiocarbamate)$_3$] complex in 1931 [90]. About thirty years after this report, the first iron(II) spin-crossover complex, [Fe(phen)$_2$(NCS)$_2$] (phen = 1, l0-phenanthroline) was discovered [91]. Afterwards, a lot of spin-crossover complexes have been realized and extensive studies have been executed. Spin-crossover phenomenon alters the physical properties such as color (optical absorption

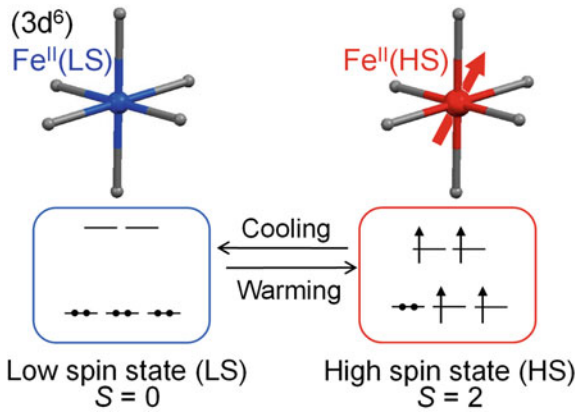


Fig. 1.12 Schematic illustration of iron(II) spin-crossover phenomenon between paramagnetic high spin state ($S = 2$) and diamagnetic low spin state ($S = 0$)

properties), crystal structure, and other spectroscopic properties. Spin-crossover properties can be controlled by various chemical modifications or physical stimuli, such as substitution of spin-crossover site to a diamagnetic ion (so called metal dilution) [95], change or desorption of solvent molecule [96], or applying pressure [97]. Furthermore, light-induced conversion from LS state to HS state was discovered and this phenomenon is called light-induced excited spin state trapping (LIESST) phenomenon [98]. Recently, the coupling effect of dielectric property and spin-crossover [99], and host-guest chemistry of a spin-crossover compound [100] were reported. In this way, many extensive researches are executed about spin-crossover phenomenon.

1.8 Light-Induced Excited Spin State Trapping (LIESST)

In 1984, Decurtins et al. have reported a conversion from LS state to HS state by light irradiation in a spin-crossover complex of $[Fe^{II}(ptz)_6](BF_4)_2$ (ptz = 1-propyl-tetrazole) [98]. This phenomenon is called Light-Induced Excited Spin State Trapping (LIESST) phenomenon, and is observed in some spin-crossover complexes [98–101]. Especially, as described in the previous section, in a Fe^{II} spin-crossover system, it is possible to change between diamagnetic LS state ($S = 0$) and paramagnetic HS state ($S = 2$). Furthermore, photo-generated HS state can be altered to the original LS state by light (reverse-LIESST phenomenon) [102], so Fe^{II} spin-crossover compounds are paid attention as a photo-reversible switching device (Fig. 1.13).

In multinuclear Fe^{II} spin-crossover complex, ferromagnetic or antiferromagnetic interaction by LIESST phenomenon has been reported [103, 104]. However, because the reported multinuclear compounds with bulky organic ligand do not have 3-dimensional network, long range magnetic ordering cannot be generated even if it is possible to observe inner molecular short range magnetic interaction.

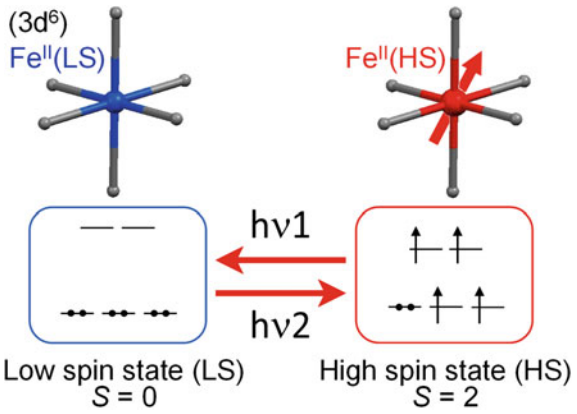


Fig. 1.13 Schematic illustration of iron(II) spin-crossover phenomenon between paramagnetic high spin state ($S = 2$) and diamagnetic low spin state ($S = 0$)

The LIESST phenomenon of 3-dimensional network spin-crossover compound based on $[M(CN)_4]^{n-}$ ($M = Ni^{II}$, Pd^{II}, Pt^{II}) or $[M(CN)_2]^{n-}$ ($M = Ag^{I}$, Au^{I}) have also been reported, [105–107] but the long range magnetic ordering was not observed since the central metal ions of these cyanidometalates are diamagnetic and the coordination number is low. In contrast, our laboratory has reported cyanido-bridged metal assemblies of $CsFe[Cr(CN)_6] \cdot 1.3H_2O$ [108] and $Fe_2[Nb(CN)_8] \cdot (3\text{-pyridylmethanol})_8 \cdot 4.6H_2O$ [109] which exhibit both spin-crossover and ferromagnetic phase transition. The coexistence of spin-crossover and ferromagnetism was observed in only these two systems and such a physical property is characteristic for a spin-crossover compound with strong magnetic interaction. In such systems, we can expect a photo-induced bulk magnetization by LIESST phenomenon, while these two compounds do not show LIESST phenomenon due to the obstruction of d-d excitation by metal-to-metal charge transfer band and the lack of cooperativeness in spin-crossover transition. To observe light-induced spin-crossover bulk magnetization, a different strategy and compound should be designed.

1.9 Ionic Conductivity

Ionic conductivity is a measure of ability to generate current by the movement of ions [110–113]. In particular, solid substances with high ionic conductivity larger than 10^{-4} S cm^{-1} are called super-ionic conductors [111]. One of the most famous examples of super-ionic conductor is α-silver iodide (α-AgI) (Fig. 1.14) [114], which is called as the first example of super-ionic conductor. α-AgI shows super-ionic conductivity of 1.9 S cm^{-1} at 473 K due to the movement of Ag^+ ions.

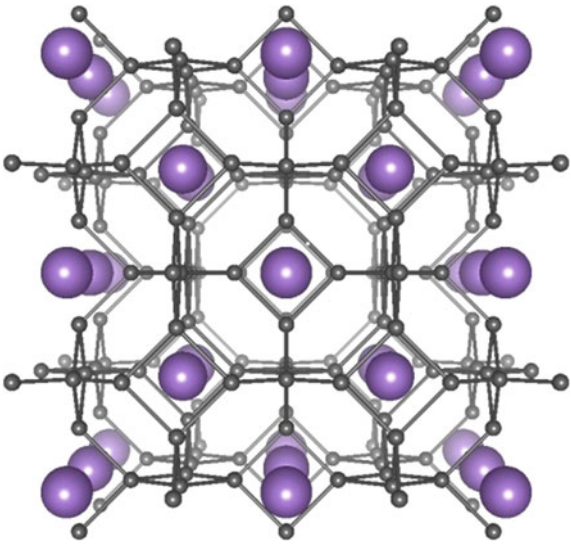


Fig. 1.14 The crystal structure of α-AgI at 353 K. The atomic coordinates are based on [120]

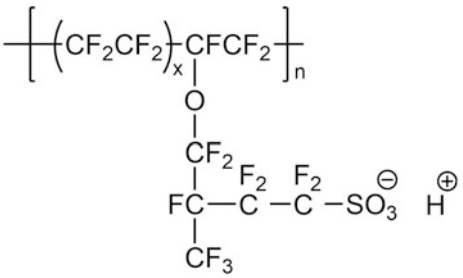


Fig. 1.15 The chemical structure of the proton conductive polymer called Nafion®

The crystal structure shown in Fig. 1.14 indicates that Ag^+ ions are disordered to many sites. Among ionic conductors, when proton is the career for ionic conductivity, the material is called proton conductor. The most famous example for proton conductor is the polymer called Nafion® (Fig. 1.15), which is utilized as the electrolyte layer of fuel cells. Nafion® is composed of sulfonated tetrafluoroethylene-based polymer and shows proton conductivity of 5.0×10^{-2} S cm^{-1} at room temperature due to the amount of sulfo-groups. Recently, ionic conductive coordination polymers are drawn attention [115–120]. In our laboratory, we observed super-ionic conductivity, and the interference between ionic conductivity and magnetic ordering in Prussian blue analogues [120], and we named this phenomenon "spin-ionic". In my doctor work, the objective is to obtain new ionic conductive magnet using octacyanidometalate-based magnet.

1.10 Objectives of This Work

In this study, I paid attention to $[M(CN)_8]$-based magnets to demonstrate novel functionalities. First, I aimed to observe photo-induced magnetization due to spin-crossover phenomenon by introducing spin-crossover sites in three-dimensional cyanido-bridged network. I have selected Fe^{II} and $[Nb^{IV}(CN)_8]$ as building blocks to achieve this goal. In addition, to achieve ionic conductive magnetic material, I have synthesized Mn^{II}-$[Nb(CN)_8]$ based bimetallic assemblies using organic ligand which is easy to be protonated. As a result, I succeeded in the demonstration of light-induced spin-crossover magnetism (Chap. 2) [121], two-step spin-crossover light-induced spin-crossover magnet (Chap. 3) [122], and super-ionic conductive magnetism (Chap. 4) [123].

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Chapter 2
Observation of Light-Induced Spin-Crossover Magnetism in a Fe-[$Nb(CN)_8$] Bimetal Assembly

Abstract In this chapter, the first example of *light-induced spin-crossover ferromagnetism* is described. This phenomenon is achieved in a three-dimensional Fe–Nb bimetallic assembly. In this compound, the spin-crossover site is introduced into cyanido-bridged metal assembly. The photo-induced magnetization is observed because of the combination of LIESST effect on spin-crossover sites, strong superexchange interaction between the magnetic centers in three-dimensional manner. The present result is the first step toward the study of the new type of photo-magnetic material.

Keywords Light-induced spin-crossover · Cyanido-bridged metal assembly · Photo-induced magnetization · Iron(II) · Octacyanidoniobate(IV)

2.1 Introduction

In a spin-crossover compound, color, electric, and magnetic properties can be altered by various external stimuli, such as temperature, pressure, and light [1–14]. Especially, the photo-induced phase transition from low-spin (LS) to high-spin (HS) states, which is recognized as light-induced excited-spin-state trapping (LIESST) effect, attracts many researchers because of the possible application to switching device [15–25]. When infinite numbers of light-induced HS sites are magnetically ordered in a three-dimensional network, bulk magnetization should be observed. However, such a bulk magnetization due to light-induced spin-crossover had not been reported yet because the reported spin-crossover compounds were usually composed of isolated molecular compounds or network compounds in which diamagnetic species are included between spin-crossover sites. In this chapter, a three-dimensional spin-crossover system using octacyanoniobate-based cyanido-bridged assembly, $Fe_2[Nb(CN)_8](4\text{-pyridinealdoxime})_8 \cdot 2H_2O$ was synthesized and realized light-induced spin-crossover bulk magnetization.

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2.2 Experiments

2.2.1 Synthesis

Synthesis of $K_4[Nb(CN)_8] \cdot 2H_2O$ [26]

The synthetic process of $K_4[Nb^{IV}(CN)_8] \cdot 2H_2O$ is mainly divided into 3 steps as shown below.

1. $Nb^{5+} + 2e^- \rightarrow Nb^{3+}$
2. $Nb^{3+} + 8KCN \rightarrow K_5[Nb^{III}(CN)_8]\downarrow + 3K^+$
3. $K_5[Nb^{III}(CN)_8] \rightarrow K_4[Nb^{IV}(CN)_8] + e^- + K^+$

All processes were carried out under dark atmosphere. Additionally, the first and second steps were performed under Ar atmosphere since Nb^{3+} is susceptible to oxidation.

1. $Nb^{5+} + 2e^- \rightarrow Nb^{3+}$

This procedure was performed under Ar atmosphere using Shrenk technique. Degassed and dehydrated methanol (12 ml) was slowly added to $NbCl_5$ (10 g). In this process, alkaline trap is equipped to trap the generated HCl gas. After all $NbCl_5$ was dissolved and the gas production stopped, methanol (28 ml) was added to the solution. Next, the resulting colorless solution was reduced electrolytically using a glassy carbon electrode with the constant current of 1 A for 6 h. Within 10 min, the solution color became dark. While the reduction, methanol was added every 1 h to maintain the amount of solution.

2. $Nb^{3+} + 8KCN \rightarrow K_5[Nb^{III}(CN)_8]\downarrow + 3K^+$

Under Ar atmosphere, the deep purple solution was slowly dropped into the 75 ml of aqueous solution of potassium cyanide (50 g) under ice-salt cooling. Stirring was continued for 1 h under ice-salt cooling and at room temperature for a further 12 h. The red colored solid of crude $K_5[Nb^{III}(CN)_8]$ was obtained by centrifugation (3500 rpm, for 10 min).

3. $K_5[Nb^{III}(CN)_8] \rightarrow K_4[Nb^{IV}(CN)_8] + e^- + K^+$

The obtained crude $K_5[Nb^{III}(CN)_8]$ was suspended to water (25 ml) in air atmosphere, and hydrogen peroxide (10%, 2 ml) was slowly added until the solution becomes yellow. The yellow-orange solution was obtained by removing the precipitate using centrifugation and methanol (600 ml) was added to the solution. The resulting yellow precipitate was filtered, recrystallized from water/methanol, and washed with methanol and diethyl ether gave yellow powder of $K_4[Nb^{IV}(CN)_8] \cdot 2H_2O$.

Synthesis of $Fe_2[Nb(CN)_8]\cdot(4\text{-pyridinealdoxime})_8 \cdot 2H_2O$

The powder sample of $Fe_2[Nb(CN)_8](4\text{-pyridinealdoxime})_8 \cdot 2H_2O$ was synthesized by reacting a mixed aqueous solution of $FeCl_2 \cdot 4H_2O$ (0.005 mol dm^{-3}) and

4-pyridinealdoxime (0.1 mol dm^{-3}) with an aqueous solution of $K_4[Nb(CN)_8] \cdot 2H_2O$ (0.05 mol dm^{-3}) under an argon atmosphere.

2.2.2 Characterization

Elemental analyses of Fe and Nb for the prepared material were conducted by HP4500 inductively coupled plasma mass spectroscopy, whereas C, H, and N were determined by standard microanalytical methods. Infrared spectra were recorded on a JASCO IRT-3000 spectrometer in the 4000–400 cm^{-1} region. XRD measurements were conducted on a Rigaku Ultima IV with Cu Kα radiation (λ = 1.5406 Å) within the range $10° \leq 2\theta \leq 70°$ using a Cu sample holder. Rietveld analyses were performed using the RIETAN-FP program [27]. The magnetic measurements were obtained from polycrystalline samples using a Quantum Design MPMS superconducting quantum interference device (SQUID) magnetometer. The UV-vis absorption spectra were measured by a Shimadzu UV-3100 spectrometer and a JASCO MSV-370 spectrometer. The optical absorption spectra before and after light irradiation were measured by a Hamamatsu Photonics PMA-12 photonic multichannel analyzer. The temperature during the optical spectra measurements was controlled by an Oxford Instruments Microstate-He. The ^{57}Fe Mössbauer spectra were measured using a Wissenschaftliche Elektronik Mössbauer spectrometer. The temperature was controlled by a Nagase Techno-Engineering PS24SS cryostat.

2.2.3 Photo-Irradiation Measurements

To investigate the magnetization changes with temperature and external magnetic fields, a diode laser with λ = 473 nm was used as the light source. The sample, which was spread on adhesive tape, was placed on the edge of an optical fiber in a SQUID. The light irradiation measurement was carried out by UV-vis absorption spectroscopy using a 473-nm diode laser light and ^{57}Fe Mössbauer spectroscopy using Xe lamp (>420 nm).

2.3 Results

2.3.1 Elemental Analysis

Elemental analysis showed that the obtained compound had the formula of $Fe_2[Nb(CN)_8] \cdot (4\text{-pyridinealdoxime})_8 \cdot 2H_2O$

Calculated: Fe, 7.8; Nb, 6.5; C, 47.2; H, 3.7; N, 23.6%.
Found: Fe, 7.8; Nb, 6.8; C, 46.7; H, 3.7; N, 23.4%.

2.3.2 XRD Patterns and Crystal Structure

The X-ray diffraction (XRD) pattern indicated that the resulting compound had a tetragonal structure with $I4_1/a$ space group [a = 20.2001(4) Å and c = 14.9565 (5) Å] (Fig. 2.1 and Table 2.1) based on Rietveld analysis using a Mn–Nb analog compound, $Mn_2[Nb(CN)_8] \cdot (4\text{-pyridinealdoxime})_8 \cdot 0.2H_2O$. The coordination geometries of the Fe and Nb sites were pseudo-octahedron and dodecahedron, respectively (Fig. 2.2a). The two axial positions of Fe were occupied by cyanide nitrogen atoms of $[Nb^{IV}(CN)_8]$, while the equatorial positions were occupied by the four nitrogen atoms of 4-pyridinealdoxime. The four equatorial CN groups of $[Nb^{IV}(CN)_8]$ were bridged to four Fe, and the other four CN groups were free. The cyano-bridged Fe-Nb moieties formed a three-dimensional cyano-bridged bimetallic framework (Fig. 2.2). Furthermore, hydrogen bond network exists between the non-bridged CN group and OH group of 4-pyridinealdoxime, between N atom and OH group of 4-pyridinealdoxime, and between OH group of 4-pyridinealdoxime and non-coordinated water molecule in the present compound (Fig. 2.3). The temperature dependence of the XRD pattern showed a continuous peak shift toward higher degrees, indicating a lattice contraction (Fig. 2.4). Between 300 K and 20 K, the lattice constant of a-axis decreased for 0.50 Å, that of c-axis decreased for 0.17 Å, and the volume contracted for 6.0%. Furthermore, the significant decrease of Fe–N bond lengths were observed (Fig. 2.5).

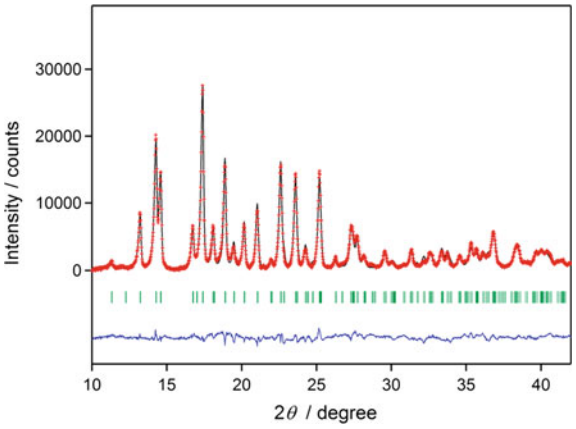


Fig. 2.1 XRD pattern of 3 at 300 K and Rietveld analysis. *Red dots*, *black line*, and *blue line* represent the observed plots, calculated pattern, and their difference, respectively. *Green bars* represent the calculated positions of the Bragg reflections. Reproduced by permission from Nature Publishing Group, a division of Macmillan Publishers Ltd: S. Ohkoshi, K. Imoto, Y. Tsunobuchi, S. Takano, H. Tokoro, ***Nature Chemistry***, 3, 564–569 (2011)

Table 2.1 Crystallographic data of $Fe_2[Nb(CN)_8] \cdot (4\text{-pyridinealdoxime})_8 \cdot 2H_2O$. Reproduced by permission from Nature Publishing Group, a division of Macmillan Publishers Ltd: S. Ohkoshi, K. Imoto, Y. Tsunobuchi, S. Takano, H. Tokoro, ***Nature Chemistry***, 3, 564–569 (2011)

Crystal system		Space group	a (Å)	c (Å)	V (Å^3)	Z	R_{wp}/R_p (%)
tetragonal		$I\,4_1/a$	20.2001(4)	14.9565(5)	6102.9(3)	4	1.98/1.52
x/a		y/b	z/c		x/a	y/b	z/c
C(l)	−0.0069	0.1449	0.5795	C(14)	0.3251	0.0206	0.5030
C(2)	0.0634	0.2559	0.5004	N(l)	−0.0025	0.0913	0.5571
C(3)	−0.0228	−0.0024	0.2849	N(2)	−0.0171	0.0331	0.3576
C(4)	−0.0222	0.1013	0.3554	N(3)	0.1125	0.0130	0.4970
C(5)	−0.0310	0.0260	0.2003	N(4)	0.0989	0.2513	0.4376
C(6)	−0.0335	0.1359	0.2729	N(5)	−0.0681	0.0986	0.0344
C(7)	−0.0368	0.0972	0.1913	N(6)	0.3581	−0.0342	0.4914
C(8)	−0.0471	0.1295	0.1013	O(l)	0	0.25	0
C(9)	0.1487	−0.0451	0.4987	O(2)	−0.0767	0.1407	−0.0407
C(10)	0.1412	0.0721	0.4949	O(3)	0.4270	−0.0244	0.4986
C(11)	0.2200	−0.0465	0.5001	Fe	0	0	0.5
C(12)	0.2133	0.0744	0.4979	Nb	0	0.25	0.625
C(13)	0.2536	0.0167	0.5017				

2.3.3 IR Spectra

In the IR spectrum, two CN stretching peaks were observed at 2130 (peak A) and 2151 cm^{-1} (peak B) whereas peaks due to 4-pyridinealdoxime were observed in the region of 534–1609 and 2650–3500 cm^{-1} (Fig. 2.6a, b). Figure 2.7c shows the temperature dependence of the IR spectra of CN stretching peaks. With decreasing temperature, the intensity of peak A decreased between 300 and 50 K and slightly increased between 50 and 20 K. As for peak B, the intensity did not change between 300 and 200 K, while between 200 and 100 K, the intensity decreased. The intensity of peak B increased between 100 and 50 K, whereas decreased again between 50 and 20 K. The peak position of A shifted to higher wavenumber between 300 and 150 K and shifted to lower wavenumber between 150 and 50 K. Between 50 and 20 K, the peak position of A shifted to higher wavenumber.

2.3.4 Magnetic Properties

The temperature (T) dependence of the molar magnetic susceptibility (χ_M) of the present compound showed a thermal transition (Fig. 2.7). The product of χ_M and T ($\chi_M T$) was 7.15 K cm^3 mol^{-1} at 290 K (high-T form), but decreased with decreasing temperature. The $\chi_M T$ value at 50 K was 1.72 K cm^3 mol^{-1} (low-T form). Half transition temperature ($T_{1/2}$), which is defined as the temperature

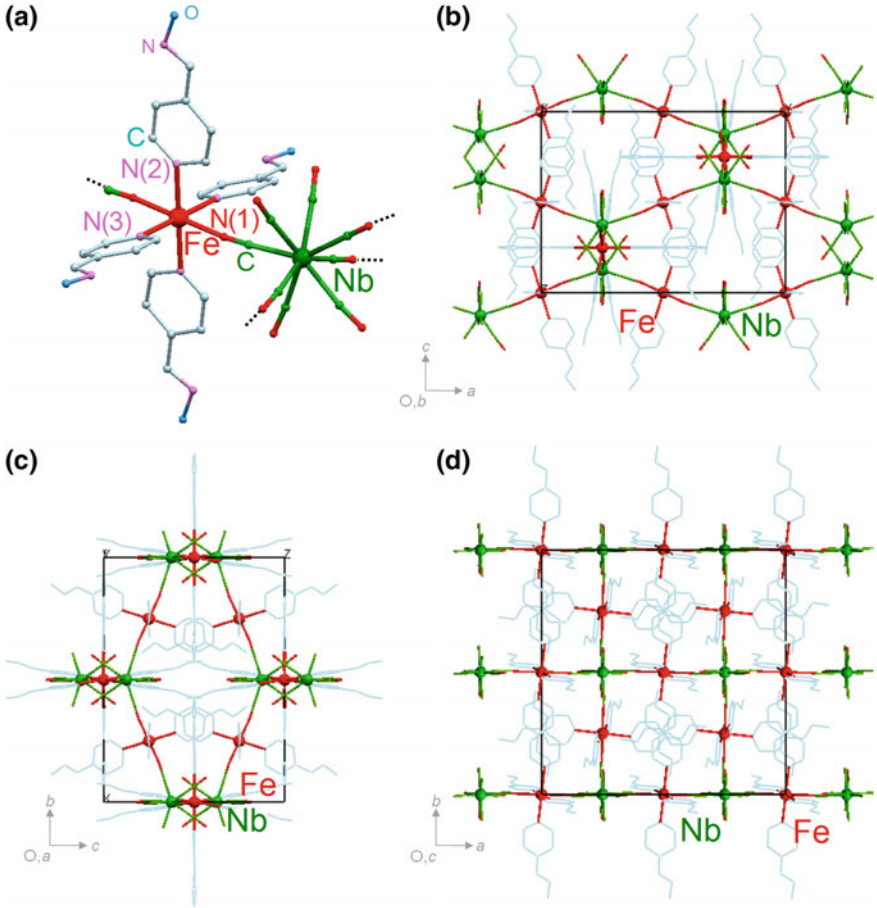


Fig. 2.2 Crystal structure of $Fe_2[Nb(CN)_8] \cdot (4\text{-pyridinealdoxime})_8 \cdot 2H_2O$. **a** Coordination geometries of Fe and Nb sites. **b** Cyano-bridged Fe–Nb 3-dimensional framework viewed along the *b*-axis, **c** *a*-axis, and **d** *c*-axis. *Red* and *green* ball-sticks denote $[FeN_6]$ and $[NbC_8]$ moieties, respectively. *Light blue* frames are 4-pyridinealdoxime molecules. Zeolitic water molecules are omitted for clarity. Reproduced by permission from Nature Publishing Group, a division of Macmillan Publishers Ltd: S. Ohkoshi, K. Imoto, Y. Tsunobuchi, S. Takano, H. Tokoro, ***Nature Chemistry***, 3, 564–569 (2011)

where a material has 50% of the population in the high-T form and 50% in the low-T form, was 130 K. Below 20 K, the $\chi_M T$ value decreased again. The $\chi_M T$ value observed in the warming process corresponded to that obtained in the cooling process at each temperature.

At very low temperature, the magnetization versus temperature plots at 10 Oe showed no spontaneous magnetization above 2 K (Fig. 2.8a). The magnetization versus external field plots showed that the coercive field was zero and the saturation magnetization value was 1.13 μ_B (Fig. 2.8b).

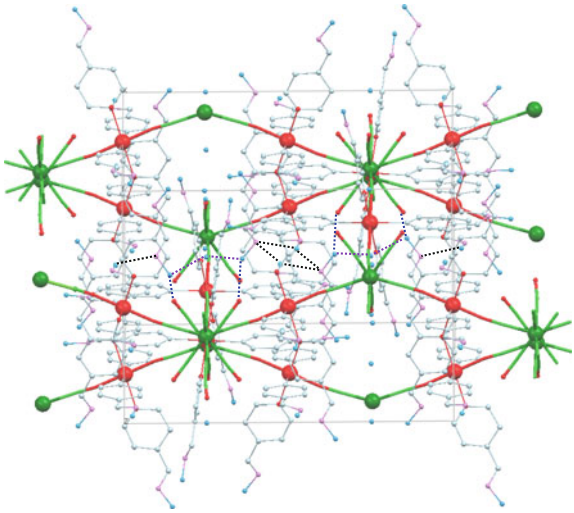


Fig. 2.3 Hydrogen-bonding network in $Fe_2[Nb(CN)_8] \cdot (4\text{-pyridinealdoxime})_8 \cdot 2H_2O$. Reproduced by permission from Nature Publishing Group, a division of Macmillan Publishers Ltd: S. Ohkoshi, K. Imoto, Y. Tsunobuchi, S. Takano, H. Tokoro, ***Nature Chemistry***, 3, 564–569 (2011)

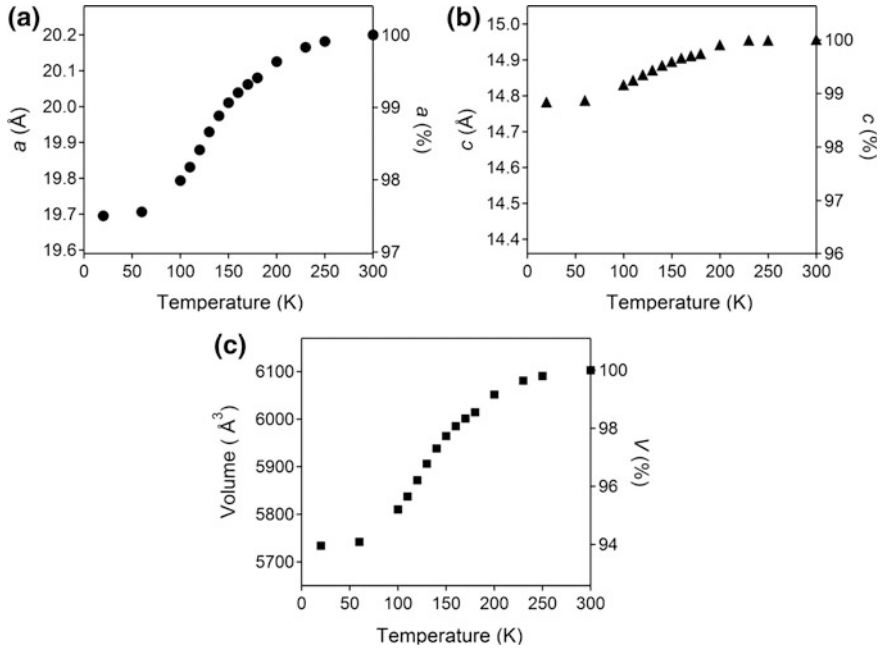


Fig. 2.4 Temperature dependence of the lattice constants and cell volume. **a** *a*-axis, **b** *c*-axis, and **c** volume. Reproduced by permission from Nature Publishing Group, a division of Macmillan Publishers Ltd: S. Ohkoshi, K. Imoto, Y. Tsunobuchi, S. Takano, H. Tokoro, ***Nature Chemistry***, 3, 564–569 (2011)

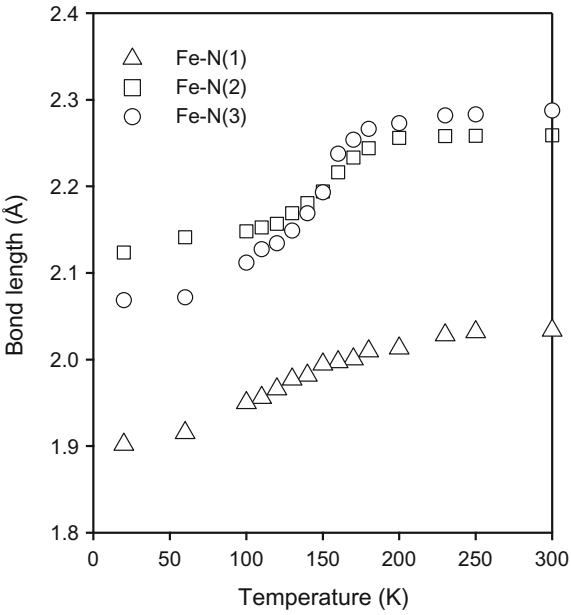


Fig. 2.5 Temperature dependence of Fe–N bond lengths. Reproduced by permission from Nature Publishing Group, a division of Macmillan Publishers Ltd: S. Ohkoshi, K. Imoto, Y. Tsunobuchi, S. Takano, H. Tokoro, ***Nature Chemistry***, 3, 564–569 (2011)

2.3.5 UV-Vis Spectra

In the variable-temperature UV-vis absorption spectra, increase of absorbance was observed as the temperature decreased and new absorption peaks appeared at 480 and 650 nm (Fig. 2.9).

2.3.6 ^{57}Fe Mössbauer Spectra

^{57}Fe Mössbauer spectra showed a doublet peak (isomer shift = 1.03 mm s^{-1}; quadrupole splitting = 1.85 mm s^{-1}) at 300 K, a doublet peak (isomer shift = 0.53 mm s^{-1}; quadrupole splitting = 0.64 mm s^{-1}) and a doublet peak (isomer shift = 1.11 mm s^{-1}; quadrupole splitting = 2.48 mm s^{-1}) at 130 K, and a doublet peak (isomer shift = 0.52 mm s^{-1}; quadrupole splitting = 0.68 mm s^{-1}) and a doublet peak (isomer shift = 1.12 mm s^{-1}; quadrupole splitting = 2.72 mm s^{-1}) at 10 K (Fig. 2.10).

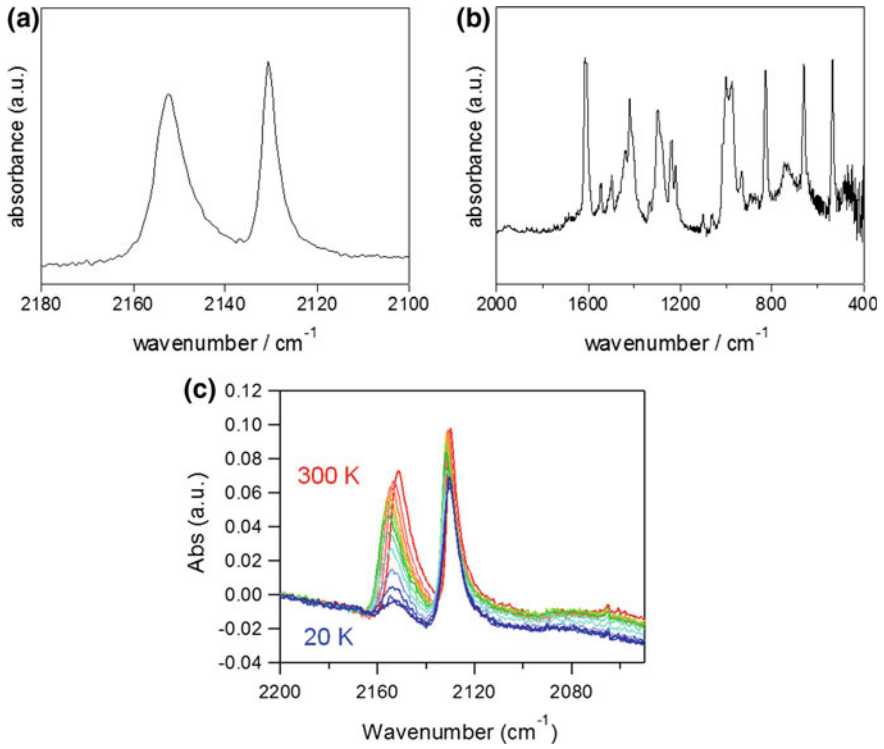


Fig. 2.6 **a** Infrared spectrum at room temperature for CN stretching region and **b** finger print region for organic ligand. **c** Temperature dependence of infrared spectrum for CN stretching region

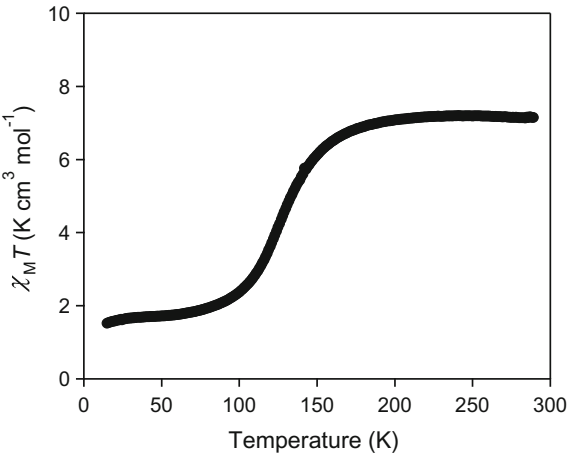


Fig. 2.7 $\chi_M T$–T plots measured under an external field of 5000 Oe. *Red* and *blue lines* are the results of fitting by molecular field theory. *Red line* is the high-T phase (Fe^{II}_{HS}: $S = 2$, Nb^{IV}: $S = ½$) with $J_{ex} = -6.9$ cm^{-1}, $g_{Fe} = 2.19$, and $g_{Nb} = 1.99$, while the *blue line* is the low-T phase (78% Fe^{II}_{LS}, 22% Fe^{II}_{HS}, and Nb^{IV}). Reproduced by permission from Nature Publishing Group, a division of Macmillan Publishers Ltd: S. Ohkoshi, K. Imoto, Y. Tsunobuchi, S. Takano, H. Tokoro, ***Nature Chemistry***, 3, 564–569 (2011)

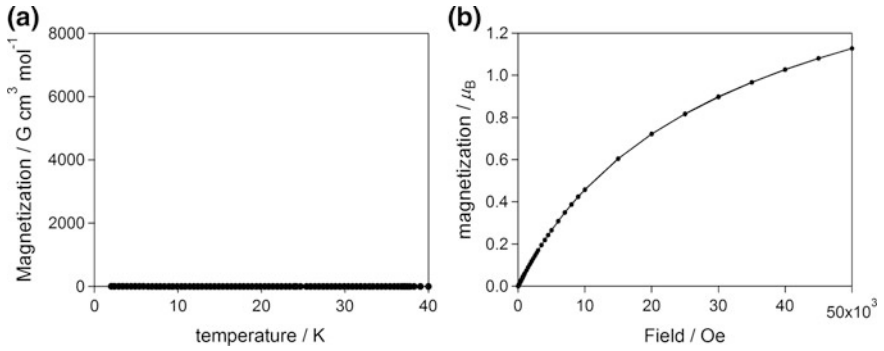


Fig. 2.8 **a** Field-cooled magnetization *curve* in an external magnetic field of 10 Oe. **b** *M-H* curve at 2 K

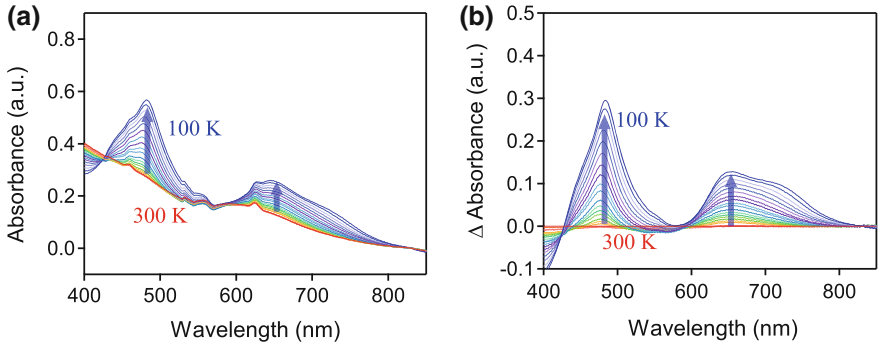


Fig. 2.9 **a** Temperature dependence of UV-vis spectra. **b** Differential UV-vis spectra relative to that at 300 K. Reproduced by permission from Nature Publishing Group, a division of Macmillan Publishers Ltd: S. Ohkoshi, K. Imoto, Y. Tsunobuchi, S. Takano, H. Tokoro, ***Nature Chemistry***, 3, 564–569 (2011)

2.3.7 Photomagnetic Measurements

The photomagnetic effect of the present compound was investigated. Because the low-*T* state possessed an absorption band due to the $^1A_1 \rightarrow {}^1T_2$ transition of Fe^{II}_{LS} at 480 nm, the low-*T* state was irradiated with a 473-nm diode laser light (17 mW cm^{-2} and 5 min) at 2 K. Consequently, a large spontaneous magnetization was observed. The magnetization versus temperature curve showed a Curie temperature (T_C) of 20 K (Fig. 2.11a). Furthermore, the magnetization versus external magnetic field plots exhibited a magnetic hysteresis loop with a coercive field (H_c) of 240 Oe at 2 K (Fig. 2.11b). The time dependence of the light-induced HS phase data showed a slow relaxation. Specifically, after irradiation, the light-induced HS phase decreased 20% after 10 h, 24% after 20 h, 26% after 30 h, and ultimately 70% of light-induced HS state remained at an asymptotic state from the fitting of decay curve (double

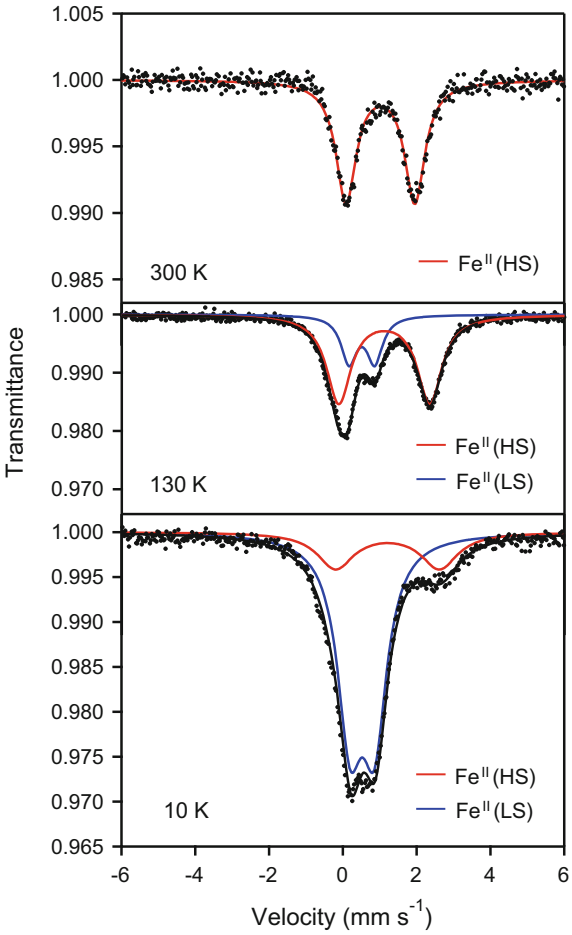


Fig. 2.10 ^{57}Fe Mössbauer spectra of $Fe_2[Nb(CN)_8] \cdot (4\text{-pyridinealdoxime})_8 \cdot 2H_2O$ at 300 K (*upper*), 130 K (*middle*) and 10 K (*lower*). Reproduced by permission from Nature Publishing Group, a division of Macmillan Publishers Ltd: S. Ohkoshi, K. Imoto, Y. Tsunobuchi, S. Takano, H. Tokoro, ***Nature Chemistry***, 3, 564–569 (2011)

exponential fitting) (Fig. 2.12). Furthermore, the temperature dependence of the light-induced HS phase plots indicated that the relaxation temperature was 55 K (Fig. 2.13). The saturation magnetization (M_s) value at 5 T was 7.5 μ_B.

2.3.8 UV-Vis Spectra Under the Light Irradiation

Upon irradiation with light at 3 K, two absorption bands at 480 and 650 nm decreased (Fig. 2.14).

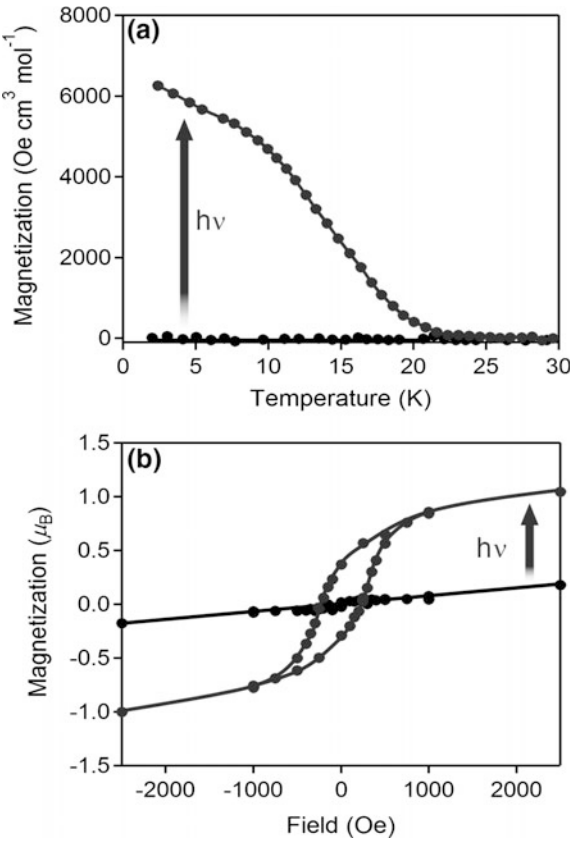


Fig. 2.11 Photo-induced magnetization caused by a light-induced spin-crossover. **a** Magnetization versus temperature curves at 100 Oe. **b** Magnetic hysteresis curves at 2 K. *Blue* and *red plots* indicate the measurement before and after irradiation with 473 nm light. Reproduced by permission from Nature Publishing Group, a division of Macmillan Publishers Ltd: S. Ohkoshi, K. Imoto, Y. Tsunobuchi, S. Takano, H. Tokoro, ***Nature Chemistry***, 3, 564–569 (2011)

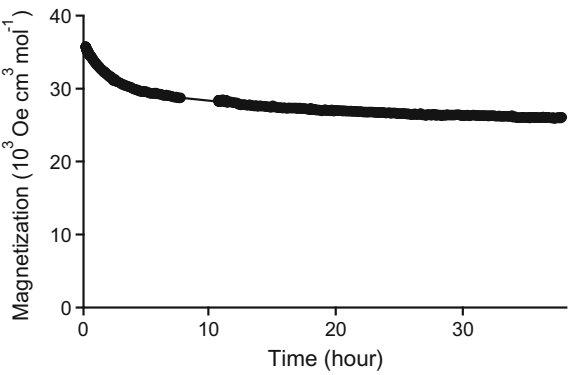


Fig. 2.12 Relaxation behavior of photo-induced state of the present compound. Reproduced by permission from Nature Publishing Group, a division of Macmillan Publishers Ltd: S. Ohkoshi, K. Imoto, Y. Tsunobuchi, S. Takano, H. Tokoro, ***Nature Chemistry***, 3, 564–569 (2011)

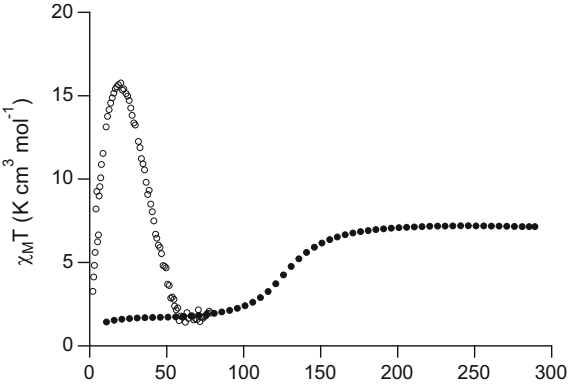


Fig. 2.13 Relaxation process of photo-induced state of the present compound measured by $\chi_M T$ versus T plot. Reproduced by permission from Nature Publishing Group, a division of Macmillan Publishers Ltd: S. Ohkoshi, K. Imoto, Y. Tsunobuchi, S. Takano, H. Tokoro, ***Nature Chemistry***, 3, 564–569 (2011)

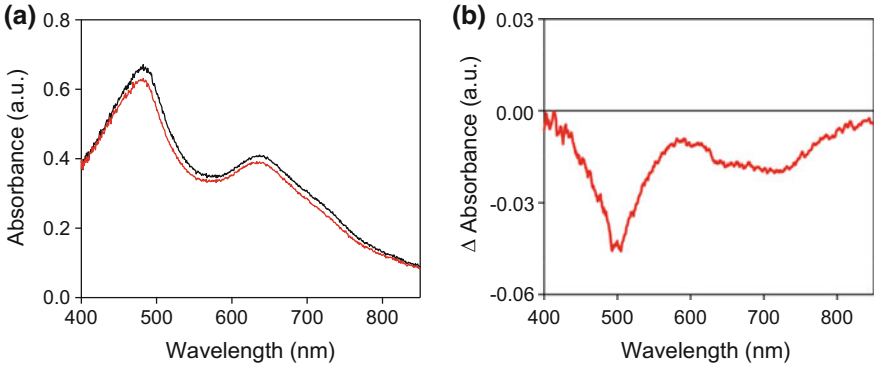


Fig. 2.14 Photo-irradiation measurement of UV-vis spectra. **a** UV-vis spectra before (*black line*) and after (*red line*) the photo-irradiation. **b** Differential UV-vis spectra before and after the photo-irradiation. Reproduced by permission from Nature Publishing Group, a division of Macmillan Publishers Ltd: S. Ohkoshi, K. Imoto, Y. Tsunobuchi, S. Takano, H. Tokoro, ***Nature Chemistry***, 3, 564–569 (2011)

2.4 Discussion

*2.4.1 Electronic States of High-*T *and Low-*T *Forms*

The IR spectra indicated that the valence state of Fe and Nb are divalent and tetravalent, respectively. The temperature dependence of $\chi_M T$ versus T plots showed a clear drop, indicating a spin transition. The temperature dependence of the UV-vis spectra showed an increase of two peaks at 480 and 650 nm, which were assigned to the $^1A_1 \rightarrow {}^1T_2$ and the $^1A_1 \rightarrow {}^1T_1$ transitions on the Fe^{II}_{LS} ($S = 0$)

site, respectively. Therefore, the transition from high-T to low-T phase in the $\chi_M T$–T plots was due to the spin-crossover from Fe^{II}_{HS} to Fe^{II}_{LS}. Furthermore, the ^{57}Fe Mössbauer spectra also supports the spin-crossover phenomenon. At 300 K, a doublet peak (isomer shift = 1.03 mm s^{-1}; quadrupole splitting = 1.85 mm s^{-1}) was observed, which was assigned to the peak of Fe^{II}_{HS}. In contrast, the Mössbauer spectrum at 10 K was composed of a doublet peak with isomer shift = 0.52 mm s^{-1} and quadrupole splitting = 0.68 mm s^{-1}, and a doublet peak with isomer shift = 1.12 mm s^{-1} and quadrupole splitting = 2.72 mm s^{-1}. The former peak is assigned to Fe^{II}_{HS} and the latter peak is assigned to Fe^{II}_{LS}. The electronic states of the high-T and low-T forms in the present spin-crossover compound were determined as follows. First, we analyzed the $\chi_M T$–T plots. From molecular field theory, χ value of bimetallic system is expressed as the following equation [28].

$$\chi = \frac{C_{Fe^{II}_{HS}} + C_{Nb^{IV}} + \frac{n_{Fe^{II}_{HS}\,Nb^{IV}} C_{Nb^{IV}} + n_{Nb^{IV}Fe^{II}_{HS}} C_{Fe^{II}_{HS}}}{T^2}}{\frac{1 - n_{Fe^{II}_{HS}\,Nb^{IV}} n_{Nb^{IV}Fe^{II}_{HS}}}{T^2}}$$

where n_{ij} are molecular-field coefficients and C_i are Curie constants. These parameters are related to the exchange coefficients (J_{ij}) by the following.

$$n_{ij} = \frac{2Z_{ij}}{\mu N (g\mu_B)^2} J_{ij}$$

$$C_i = \frac{\lambda_i N g^2 \mu_B^2 S_i (S_i + 1)}{3k}$$

where μ_B is the Bohr magneton, Z_{ij} are the numbers of the nearest-neighbor j sites, N is the total number of all types of metal ions per unit volume, and λ and μ are mole fractions.

In the high-T form, $Z^{II\,IV}_{FeHSNb}$ = 2, $Z^{IV\,II}_{NbFeHS}$ = 4, μ = 1, and λ = 2. The observed $\chi_M T$ value at 290 K was close to the estimated $\chi_M T$ value derived from the sum of Fe^{II}_{HS} (S = 2) and Nb^{IV} (S = 1/2) and considering the observed g-factors of $g_{Nb^{IV}}$ = 1.99 and $g_{Fe^{II}_{HS}}$ = 2.19. In contrast, in the low-T form, $Z^{II\,IV}_{FeHSNb}$ = 2, $Z^{IV\,II}_{NbFeHS}$ = 4x, μ = 1, and λ = 2x, where x is the rate of remaining Fe^{II}_{HS}. The least square fitting was performed for low-T form and the observed $\chi_M T$ value at 50 K of the low-T phase was close to the value where 78% of the Fe^{II}_{HS} transited to Fe^{II}_{LS} (Fig. 2.15). Additionally, in the Mössbauer spectra, the area ratio of a peak due to Fe^{II}_{HS} and a peak due to Fe^{II}_{LS} was calculated. As a result, The electronic states of the high-T and low-T forms were assigned to $(Fe^{II}_{HS})_2[Nb^{IV}(CN)_8] \cdot (4\text{-pyridinealdoxime})_8 \cdot 2H_2O$ and $(Fe^{II}_{HS})_{0.44}(Fe^{II}_{LS})_{1.56}[Nb^{IV}(CN)_8] \cdot (4\text{-pyridinealdoxime})_8 \cdot 2H_2O$, respectively.

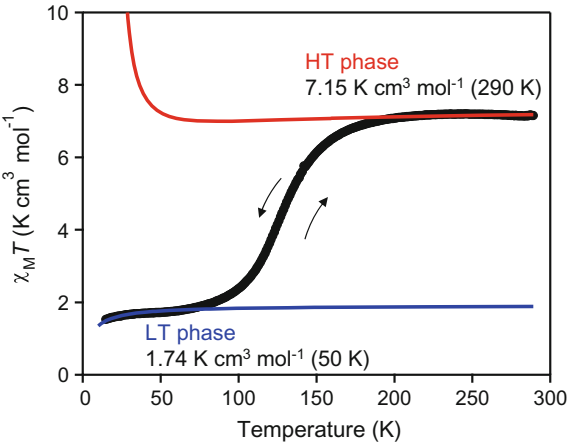


Fig. 2.15 $\chi_M T$–T plots measured under an external field of 5000 Oe. *Red* and *blue lines* are the results of fitting by molecular field theory. *Red line* is the high-T phase (Fe^{II}_{HS}: $S = 2$, Nb^{IV}: $S = ½$) with $J_{ex} = -6.9\ cm^{-1}$, $g_{Fe} = 2.19$, and $g_{Nb} = 1.99$, while the *blue line* is the low-T phase (78% Fe^{II}_{LS}, 22% Fe^{II}_{HS}, and Nb^{IV}). Reproduced by permission from Nature Publishing Group, a division of Macmillan Publishers Ltd: S. Ohkoshi, K. Imoto, Y. Tsunobuchi, S. Takano, H. Tokoro, ***Nature Chemistry***, 3, 564–569 (2011)

2.4.2 XRD Patterns and Crystal Structure

In the variable-temperature XRD measurements, the lattice constants decreased and compression of the Fe–N bond lengths, 2.03 (300 K) → 1.90 Å (20 K) on Fe–N (1), 2.26 → 2.12 Å on Fe–N(2), and 2.29 → 2.07 Å on Fe–N(3), was observed (Figs. 2.4 and 2.5). In typical Fe^{II} spin-crossover complex, the Fe-N bond length is about 2.1 Å for Fe^{II}_{HS} and 1.9 Å for Fe^{II}_{LS}. Therefore this change of crystal structure supports the $Fe^{II}_{HS} \rightarrow Fe^{II}_{LS}$ transition.

2.4.3 Magnetic Properties of Low-T Form

At very low temperature, the magnetization versus temperature plots at 10 Oe showed no spontaneous magnetization above 2 K and the magnetization versus external field plots showed that the coercive field was zero, indicating that the present compound is paramagnetic above 2 K.

2.4.4 Photomagnetic Properties

The photomagnetic effect of the present compound was investigated. Because the low-T phase possessed an absorption band due to the $^1A_1 \rightarrow {}^1T_2$ transition of Fe^{II}_{LS} at 480 nm, the low-T phase was irradiated with a 473-nm diode laser light (17 mW cm^{-2} and 5 min) at 2 K. Consequently, a large spontaneous magnetization was observed. The magnetization versus temperature curve showed a Curie temperature (T_C) of 20 K, while the magnetization versus external magnetic field plots exhibited a magnetic hysteresis loop with a coercive field (H_c) of 240 Oe at 2 K. The saturation magnetization (M_s) at 5 T was 7.5 μ_B, which was close to the expected M_s value of 7.8 μ_B due to ferrimagnetic coupling between Nb^{IV} (S = 1/2) and the photo-produced Fe^{II}_{HS} (S = 2) with the observed g-factors of g^{IV}_{Nb} = 1.99 and g^{II}_{FeHS} = 2.19.

The time dependence of the magnetization value after the light irradiation under 3 T showed that the photo-induced phase slowly relaxes and ultimately 70% of the initial magnetization remained. Specifically, after irradiation, the light-induced HS phase decreased 20% after 10 h, 24% after 20 h, 26% after 30 h, and ultimately 70% of light-induced HS state remained at an asymptotic state from the double exponential fitting. Furthermore, the temperature dependence of the light-induced HS phase plots indicated that the relaxation temperature was 55 K.

The observed photomagnetism can be explained by the following scheme. Before irradiation, paramagnetic Nb^{IV} and diamagnetic Fe^{II}_{LS} of the low-T phase were connected by a CN ligand in an alternating fashion, which showed paramagnetism. Irradiation with 473-nm light caused the light-induced excited-spin-state trapping (LIESST) effect on the Fe^{II} site (Fig. 2.16); the $^1A_1$ state on Fe^{II}_{LS} transited to the excited state ($^1T_2$) due to the light irradiation and then partially proceeded to the metastable state ($^5T_2$) through the triplet state ($^3T_1$). In the photo-induced metastable high-T phase, the magnetic spins on the photo-produced Fe^{II}_{HS} (S = 2) and neighboring Nb^{IV} (S = 1/2) interacted antiferromagnetically by a strong superexchange interaction through the CN ligand, resulting in bulk magnetization (Fig. 2.15). Based on molecular field theory, the values of the superexchange interaction constant ($J_{ex,ij}$) between the i site and the nearest neighbour j site in $(Fe^{II}_{HS})_2[Nb^{IV}(CN)_8] \cdot (4\text{-pyridinealdoxime})_8 \cdot 2H_2O$ (the photo-induced phase) are related to the T_C value via the following equation [29]:

$$T_C = \frac{\left|J_{ex,Fe^{II}_{HS}\,Nb^{IV}}\right|\left\{Z_{Nb^{IV}Fe^{II}_{HS}} Z_{Fe^{II}_{HS}\,Nb^{IV}} S_{Fe^{II}_{HS}}\left(S_{Fe^{II}_{HS}}+1\right) S_{Nb^{IV}}\left(S_{Nb^{IV}}+1\right)\right\}^{1/2}}{3k_B},$$

where S_i is the spin quantum number ($S_{Nb^{IV}} = 1/2$ and $S_{Fe^{II}_{HS}} = 2$), Z_{ij} is the number of the nearest neighbour j sites around i site, and k_B is the Boltzmann constant. In the case of $(Fe^{II}_{HS})_2[Nb^{IV}(CN)_8] \cdot (4\text{-pyridinealdoxime})_8 \cdot 2H_2O$, T_C = 20 K, and Z_{ij} were $Z_{Nb^{IV}Fe^{II}_{HS}} = 4$ and $Z_{Fe^{II}HSNb^{IV}} = 2$. Consequently, $J_{ex,Fe^{II}_{HS}\,Nb^{IV}}$ was estimated to be −6.9 cm^{-1}, which is quite a large J_{ex} value.

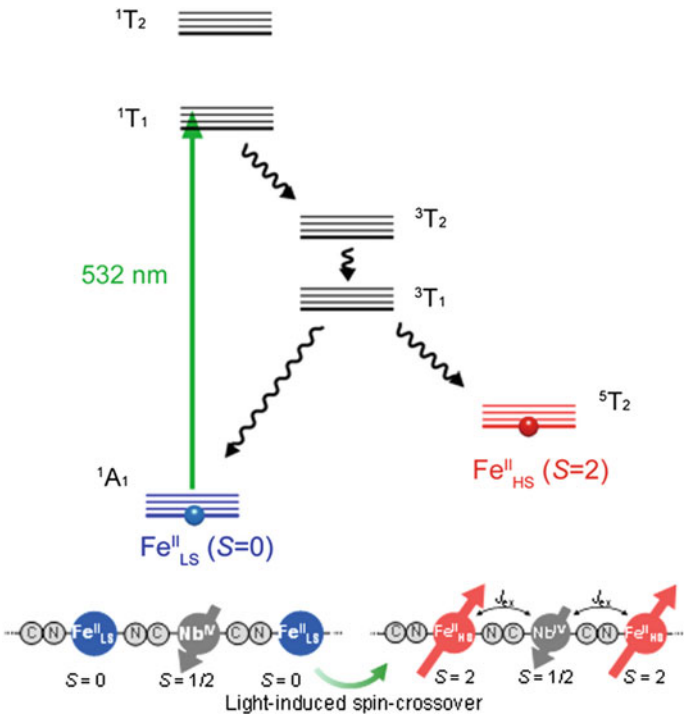


Fig. 2.16 Mechanism of photo-induced spin-crossover

2.5 Conclusion

In this chapter, we report the first example of a *light-induced spin-crossover ferromagnet* composed of a three-dimensional Fe–Nb bimetallic assembly. In this compound, the spin-crossover site is introduced in a metal-organic framework. Photo-induced magnetization was observed because the LIESST effect occurred in a spin-crossover complex with strong superexchange interaction between the magnetic centers, and three-dimensional network frame. The present result may be the first step toward the study of this new type of photo magnetic material.

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Chapter 3
Two-Step Spin-Crossover and Photo-Induced Spin-Crossover Ferromagnetism in $Fe^{II}_2[Nb^{IV}(CN)_8]$ (4-Methylpyridine)$_8 \cdot 2H_2O$

Abstract In Chap. 2, photo-induced spin-crossover magnetism is demonstrated in iron-ocatacyanoniobate-based three-dimensional spin-crossover compound, $Fe^{II}_2[Nb^{IV}(CN)_8](4\text{-pyridinealdoxime})_8 \cdot 2H_2O$. In this chapter, utilizing an advantage of spin-crossover phenomenon that can be controlled by ligand substitution, a cyanido-bridged metal assembly was synthesized using Fe^{II}, $[Nb^{IV}(CN)_8]$, and 4-methylpyridine and obtained $Fe^{II}_2[Nb^{IV}(CN)_8](4\text{-methylpyridine})_8 \cdot 2H_2O$. As a result, I succeeded in observing coexistence of two step spin-crossover behavior and light-induced spin-crossover magnetization.

Keywords Two-step spin-crossover · Cyanido-bridged metal assembly · Photo-induced magnetization · Iron(II) · Octacyanidoniobate(IV)

3.1 Introduction

In Chap. 2, the photo-induced spin-crossover magnetism is demonstrated in iron-ocatacyanoniobate-based three-dimensional cyanido-bridged spin-crossover compound, $Fe^{II}_2[Nb^{IV}(CN)_8](4\text{-pyridinealdoxime})_8 \cdot 2H_2O$. Spin-crossover phenomenon has been widely investigated as a excellent way to switch physical properties of a material, because spin-crossover behavior can be tuned by various physical and chemical perturbations, e.g., temperature, pressure, light, or solvent molecules. [1–23] These tunabilities have enabled realization of many kinds of spin-crossover compounds with abrupt, gradual, incomplete, hysteretic, or multi-step spin transitions. Especially, multi-step spin-crossover phenomenon [21–23] is fascinating since the switching between more than three different states is possible, which has a capability of multiple-valued information storage. In a spin-crossover compound with octahedral coordination geometry, the ligand field strength between t_{2g} orbitals and $e_{g.}$ orbitals and cooperativity between spin-crossover sites play an important role in the properties of spin-crossover behavior. From this point of view, ligand substitution is one of a useful method to control spin-crossover property. In this chapter, by introducing 4-methylpyridine as an organic ligand, coexistent system of two step

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spin-crossover behavior and light-induced spin-crossover magnetization is realized in an iron-ocatacyanoniobate-based system. The synthesis, structural analysis, detailed investigations of magnetic properties, thermodynamical analysis, and photomagnetic studies are described.

3.2 Experiments

3.2.1 Synthesis

The preparation of $K_4[Nb(CN)_8] \cdot 2H_2O$

$K_4[Nb(CN)_8] \cdot 2H_2O$ was synthesized as described in Chap. 2.

The preparation of $Fe_2[Nb(CN)_8](4\text{-methylpyridine})_8 \cdot 2H_2O$

The target sample of $Fe_2[Nb(CN)_8](4\text{-methylpyridine})_8 \cdot 2H_2O$ was prepared as both powder form and single crystal.

The preparation of powder: The aqueous solution of $K_4[Nb(CN)_8] \cdot 2H_2O$ (0.05 mol dm^{-3}) is added to the mixed aqueous solution of $FeCl_2 \cdot 4H_2O$ (0.05 mol dm^{-3}), ascorbic acid, and 4-methylpyridine (0.1 mol dm^{-3}). After stirring for 1 h, the resulting purple powder is filtered, washed twice with 2 ml of distilled water, and dried in air.

The preparation of single crystal: The mixed aqueous solution of $FeCl_2 \cdot 4H_2O$ (0.005 mol dm^{-3}), ascorbic acid, and 4-methylpyridine (0.1 mol dm^{-3}) is put in a 100 ml vial, and the aqueous solution of $K_4[Nb(CN)_8] \cdot 2H_2O$ (0.05 mol dm^{-3}) is quietly added to the vial. After 1 day, purple block shaped crystal is formed at the wall surface of the vial.

3.2.2 Characterization

Elemental analyses of Fe and Nb were conducted by Agilent-7700 inductively coupled plasma mass spectroscopy and those of C, H, and N were performed by standard microanalytical methods in the microanalytical laboratory at the department of chemistry, school of science the University of Tokyo. X-ray structural analysis was performed on a Rigaku RAXIS RAPID imaging plate area detector with graphite monochromated Mo Kα radiation. The structure was solved by a direct method and refined by a full-matrix least-squares technique using SHELXL-97 [24]. All non-hydrogen atoms were refined anisotropically. The hydrogen atoms of 4-methylpyridine were refined using the riding model. The calculation was performed using the Crystal Structure crystallographic software package. Infrared spectra were recorded on a JASCO IRT-3000 spectrometer in the 4000–400 cm^{-1} region. XRD measurements were conducted on a Rigaku Ultima IV with Cu Kα radiation (λ = 1.5406 Å) within the range of $10° \leq 2\theta \leq 70°$ using a Cu sample

holder. Rietveld analyses were performed using the RIETAN-FP program [25]. The magnetic measurements were obtained from polycrystalline samples using a Quantum Design MPMS superconducting quantum interference device (SQUID) magnetometer. The UV-vis absorption spectra were measured by a Shimadzu UV-3100 spectrometer. The temperature during the optical spectra measurements was controlled by an Oxford Instruments Microstate-He.

3.2.3 Photo-Irradiation Measurements

To investigate the magnetization changes with temperature and external magnetic fields, a diode laser with $\lambda = 532$ nm was used as the light source. The sample, which was spread on adhesive tape, was placed on the edge of an optical fiber in a SQUID. The light irradiation measurement was carried out by UV-vis absorption spectroscopy using a 532-nm diode laser light.

3.3 Results

3.3.1 Elemental Analysis

Elemental analysis showed that the obtained compound had the formula of $Fe_2[Nb(CN)_8](4\text{-methylpyridine})_8 \cdot 2H_2O$.

Calculated: Fe, 9.4; Nb, 7.8; C, 56.3; H, 5.1; N, 18.8%.
Found: Fe, 9.7; Nb, 8.1; C, 56.1; H, 5.0; N, 18.7%.

3.3.2 Single Crystal X-Ray Analysis at Various Temperatures

The single crystal X-ray analysis indicated that the resulting compound had an orthorhombic crystal system and *Fddd* space group with $a = 13.9704(3)$ Å, $b = 26.3002(6)$ Å, and $c = 31.7577(7)$ Å (Figs. 3.1, 3.2 and Table 3.1). The iron site was coordinated by two cyanide nitrogen atoms of $[Nb^{IV}(CN)_8]$ in axial position and the equatorial positions were occupied by the four nitrogen atoms of 4-methylpyridine. The Nb site is coordinated by eight cyano groups in dodecahedral geometry and four equatorial CN groups of $[Nb^{IV}(CN)_8]$ were bridged to Fe, and the other four CN groups were not bridged. The extended network structure of Fe-NC-Nb fragment formed a three-dimensional cyano-bridged bimetallic framework (Fig. 3.2). In view of the weak interaction, the CH-N interaction between methyl group of 4-methylpyridine and cyanide-nitrogen atoms is suggested by the

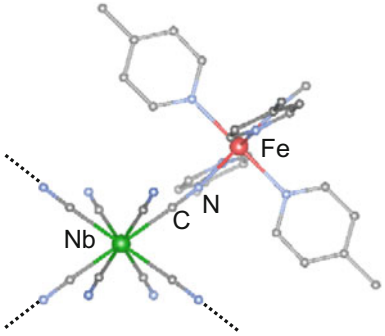


Fig. 3.1 Coordination geometries of Fe and Nb sites at 296 K. Reproduced by permission from the Chemical Society of Japan: K. Imoto, S. Ohkoshi, ***Chem. Lett***. 45, 359–361 (2016)

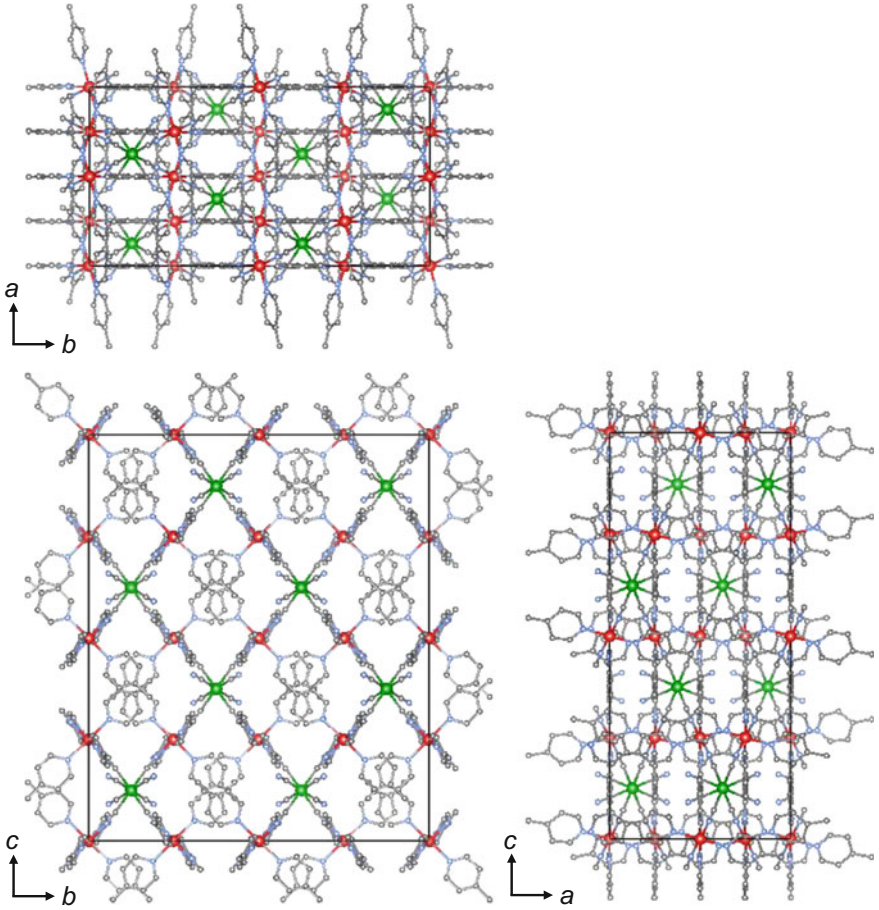


Fig. 3.2 The crystal structure viewed from each axis

Table 3.1 Crystallographic data of [Fe(4-methylpyridine)$_4$]$_2$[Nb(CN)$_8$] · 2H$_2$O at 296 K. Reproduced by permission from the Chemical Society of Japan: K. Imoto, S. Ohkoshi, ***Chem. Lett***. 45, 359–361 (2016)

Formula	$C_{56}H_{56}Fe_2N_{16}Nb$
Formula weight	1193.80
Crystal system	Orthorhombic
Space group	*Fddd*
a/Å	13.9704(3)
b/Å	26.3002(6)
c/Å	31.7577(7)
V/Å^3	11,668.6(3)
Z	8
Calculated density/g cm^{-3}	1.359
Temperature/K	296
R_1	0.0589
wR_2	0.1528
GOF on F_2	1.130
Absorption coefficient μ/mm^{-1}	0.738
Number of measured reflections	45,393
Number of refined parameters	172

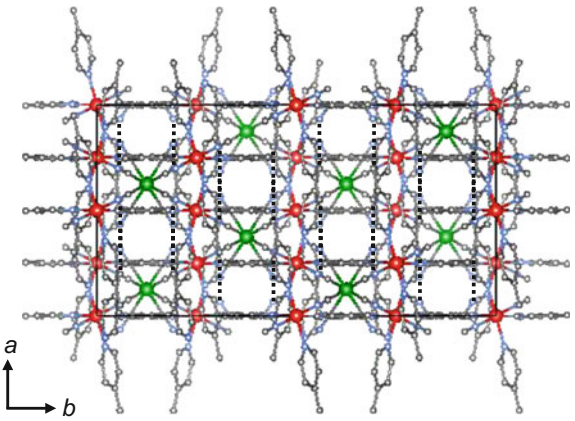


Fig. 3.3 Illustration of interaction between methyl carbon atom and non-bridged cyano-nitrogen atom whose distance is 3.6 Å

carbon-nitrogen distance of 3.6 Å (Fig. 3.3). The Fe–N bond lengths are 2.085 Å (Fe–N(1)), 2.248 Å (Fe–N(2)), and 2.267 Å (Fe–N(3)), which are typical values for an Fe^{II}_{HS} complex (Table 3.2).

3.3.3 IR Spectra

Figure 3.4 shows the temperature dependence of the IR spectra of CN stretching peaks. At 300 K, two CN stretching peaks were observed at 2152 (peak A) and

Table 3.2 Selected bond distances (Å)

Fe1	–	N1	2.084(3)	C1	–	N1	1.149(5)
Fe1	–	N1$^{(i)}$	2.084(3)	C2	–	N2	1.142(6)
Fe1	–	N2	2.248(3)				
Fe1	–	N2$^{(i)}$	2.248(3)				
Fe1	–	N3	2.267(4)				
Fe1	–	N3$^{(i)}$	2.267(4)				
Nb1	–	C1	2.230(3)				
Nb1	–	C1$^{(ii)}$	2.263(3)				
Nb1	–	C1$^{(iii)}$	2.237(3)				
Nb1	–	C1$^{(iv)}$	2.263(3)				
Nb1	–	C2	2.252(4)				
Nb1	–	C2$^{(ii)}$	2.259(3)				
Nb1	–	C2$^{(iii)}$	2.256(3)				
Nb1	–	C2$^{(iv)}$	2.267(3)				

(i) $1/2 - x, -1/2 + y, -z$, (ii) $+x, 1/4 - y, 1/4 - z$
(iii) $1/4 - x, + y, 1/4 - z$, (iv) $1/4 - x, 1/4 - y, +z$

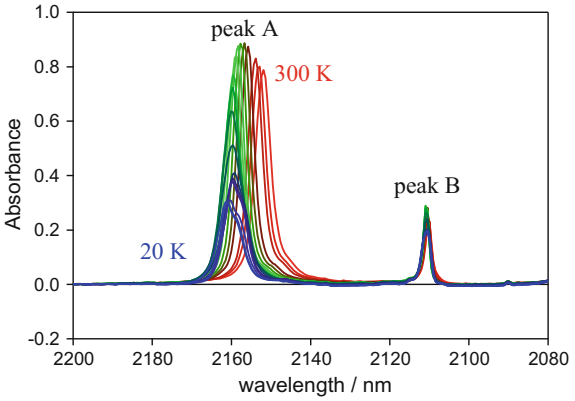


Fig. 3.4 Temperature dependence of infra-red spectrum in CN stretching region with every 20 K

2110 cm^{-1} (peak B). With decreasing temperature, between 300 and 200 K, the peak position of peak A shifted to higher wavenumber and a slight increase of intensity was observed.

However, below 200 K, the intensity of peak A turned into decrease and the most abrupt decrease in intensity occurred between 120 and 90 K. In addition, below 200 K, the shift of peak position gradually stopped and the peak position shifted to lower frequency between 100 and 80 K. Below 80 K, the peak splitted to two components and the slight increase of peak frequency and the decrease in intensity was observed. The change of peak B is not as large as that of peak A, while the tendency was almost the same as peak A. The higher-wavenumber shift and slight increase of intensity occurred between 300 and 200 K, and between 200

and 110 K, the peak height and peak position was almost maintained. The peak height decrease and the broading of peak was observed between 110 and 90 K. Below 90 K, no notable change was observed.

3.3.4 Magnetic Properties

The temperature (T) dependence of the molar magnetic susceptibility (χ_M) of the present compound showed that the product of χ_M and T ($\chi_M T$) was 7.73 K cm^3 mol^{-1} at 300 K (Fig. 3.5). As decreasing temperature, the $\chi_M T$ value decreased in a two-step fashion with the transition temperatures of $T_{1/2\downarrow}$ = 105 K and 92 K. After the first spin transition, the $\chi_M T$ value exhibited a plateau around 100 K with the $\chi_M T$ value of 4.19 K cm^3 mol^{-1} (at 100 K) The second spin transition exhibited a thermal hysteresis of 2 K and the $\chi_M T$ value reached 1.72 K cm^3 mol^{-1} at 50 K (LT phase).

The magnetization vs external field plots showed that the magnetization value was 0.67 μ_B at 50 kOe (Fig. 3.6).

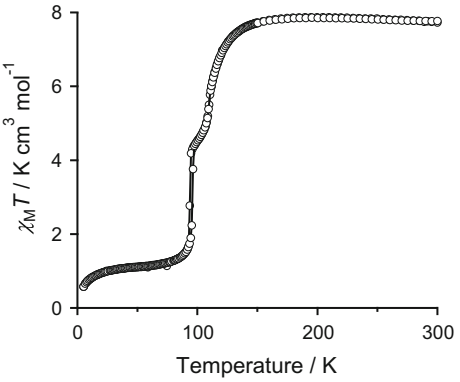


Fig. 3.5 The temperature (T) dependence of the product of molar magnetic susceptibility (χ_M) and T ($\chi_M T$). Reproduced by permission from the Chemical Society of Japan: K. Imoto, S. Ohkoshi, ***Chem. Lett***. 45, 359–361 (2016)

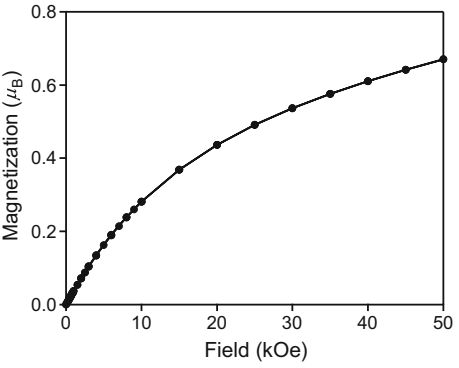


Fig. 3.6 Magnetization versus external magnetic field curve at 2 K

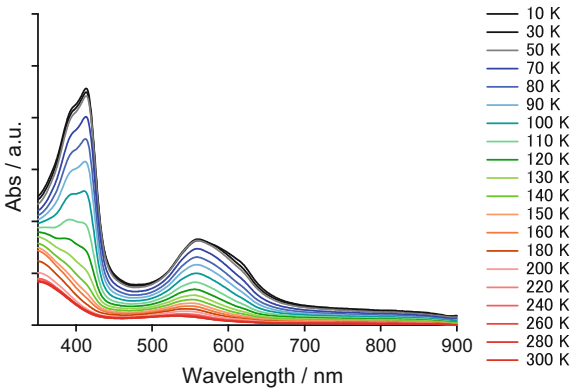


Fig. 3.7 Temperature dependence of UV-vis spectrum. Reproduced by permission from the Chemical Society of Japan: K. Imoto, S. Ohkoshi, ***Chem. Lett***. 45, 359–361 (2016)

3.3.5 Temperature Dependence of UV-Vis Spectra

In the variable-temperature UV-vis absorption spectra, optical absorptions at 400 nm, 415 nm, 560 nm, and 610 nm appeared as the temperature decreased (Fig. 3.7) and the changing temperature corresponded to that of $\chi_M T$ value.

3.3.6 Photomagnetic Measurements

The photomagnetic effect of the present compound was investigated. Because the low-T state possessed an absorption band at 560 nm, the low-T phase was irradiated with a 532-nm diode laser light (13 mW cm^{-2} and 5 min) at 2 K. Consequently, a large spontaneous magnetization was observed. The magnetization versus temperature curve showed a Curie temperature (T_C) of 15 K (Fig. 3.8a), while the magnetization versus external magnetic field plots exhibited a magnetic hysteresis loop with a coercive field (H_c) of 2300 Oe at 2 K (Fig. 3.8b).

The time dependence of the light-induced HS phase data showed a slow relaxation. After irradiation, 90% of light-induced HS state remained after 1 day (Fig. 3.9). Furthermore, the temperature dependence of the $\chi_M T$ value of light-induced HS phase plots indicated that the relaxation temperature was 55 K (Fig. 3.10). The saturation magnetization (M_s) value at 5 T was 7.5 μ_B.

3.3.7 UV-Vis Spectra Under the Light Irradiation at 3 K

By the irradiation of 532 nm light, absorption bands around 400, 415, 560, and 610 nm decreased (Fig. 3.11). The spectrum recovered to the original one after thermal annealing to 100 K.

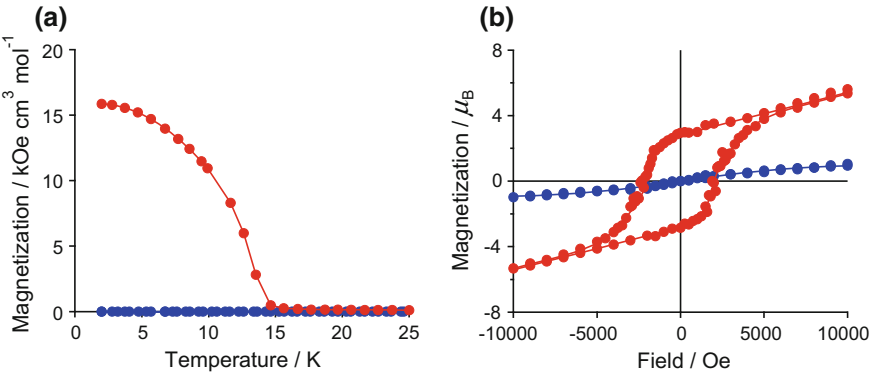


Fig. 3.8 Photomagnetic properties of the present compound. **a** Magnetization versus temperature curve at 100 Oe, and **b** Magnetic hysteresis curve at 2 K. *Blue plot* represent before irradiation and the *red plot* is after irradiation. Reproduced by permission from the Chemical Society of Japan: K. Imoto, S. Ohkoshi, ***Chem. Lett***. 45, 359–361 (2016)

Fig. 3.9 Time dependence of the light-induced state at 1 T

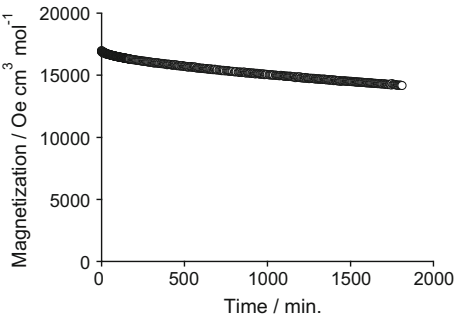
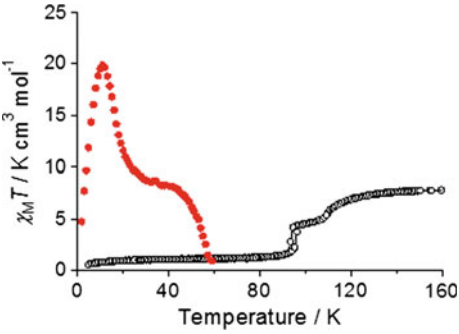


Fig. 3.10 Temperature dependence of the light-induced state at 1 T. Reproduced by permission from the Chemical Society of Japan: K. Imoto, S. Ohkoshi, ***Chem. Lett***. 45, 359–361 (2016)



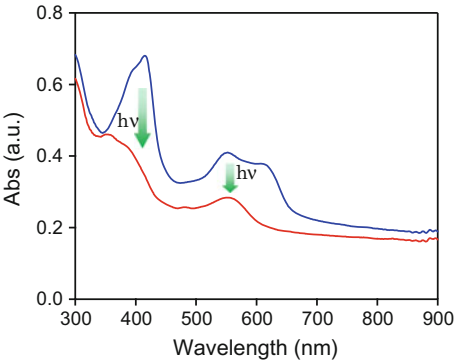


Fig. 3.11 Photoirradiation measurement of UV-vis spectrum before (*blue line*) and after (*red line*) irradiation of 532-nm light. Reproduced by permission from the Chemical Society of Japan: K. Imoto, S. Ohkoshi, ***Chem. Lett***. 45, 359–361 (2016)

3.4 Discussion

3.4.1 The Assignment of Spin-Transition in the Present System

The IR spectra indicated that the valence state of Fe and Nb are divalent and tetravalent, respectively. The temperature dependence of $\chi_M T$ versus T plots showed a clear drop, indicating a spin transition. The temperature dependence of the UV-vis spectra showed an increase of absorptions at 400, 415, 560, and 610 nm as the temperature decreased (Fig. 3.7), which were assigned to the $^1A_1 \rightarrow {}^1E$, $^1A_1 \rightarrow {}^1B_2$, $^1A_1 \rightarrow {}^1A_2$, and $^1A_1 \rightarrow {}^1E$ d-d transitions on the Fe_{LS}^{II} ($S = 0$) with D_{4h} symmetry, respectively. Therefore, the transition from high-T to low-T phase in the $\chi_M T$–T plots was due to the spin-crossover from Fe_{HS}^{II} to Fe_{LS}^{II}. Furthermore, the temperature dependence of single crystal X-ray diffraction showed a clear decrease of Fe–N bond lengths below 120 K, supporting the spin-crossover.

3.4.2 The Analysis of Magnetic Susceptibility and the Electronic State

The electronic states of the HT phase and LT phase in the present spin-crossover compound were determined by the analysis of the magnetic susceptibility. First, molecular field approach is applied as shown below,

$$\chi = \frac{C_{Fe^{II}HS} + C_{Nb^{IV}} + \frac{n_{Fe^{II}HS\ Nb^{IV}} C_{Nb^{IV}} + n_{Nb^{IV}Fe^{II}HS} C_{Fe^{II}HS}}{T^2}}{\frac{1 - n_{Fe^{II}HS\ Nb^{IV}} n_{Nb^{IV}Fe^{II}HS}}{T^2}}$$

where n_{ij} are molecular-field coefficients and C_i are Curie constants. These parameters are related to the exchange coefficients (J_{ij}) by the following.

$$n_{ij} = \frac{2Z_{ij}}{\mu N (g\mu_B)^2} J_{ij}$$

$$C_i = \frac{\lambda_i N g^2 \mu_B^2 S_i (S_i + 1)}{3k}$$

where μ_B is the Bohr magneton, Z_{ij} are the numbers of the nearest-neighbor j sites, N is the total number of all types of metal ions per unit volume, and λ and μ are mole fractions.

In the HT phase, $Z^{II\ IV}_{FeHSNb} = 2$, $Z^{IV\ II}_{NbFeHS} = 4$, $\mu = 1$, and $\lambda = 2$. The observed $\chi_M T$ value at 290 K was close to the estimated $\chi_M T$ value derived from the sum of Fe^{II}_{HS} ($S = 2$) and Nb^{IV} ($S = 1/2$) and considering the observed g-factors of $g_{Nb^{IV}} = 1.99$ and $g_{Fe^{II}HS} = 2.4$. In contrast, in the low-T form, $Z^{II\ IV}_{FeHSNb} = 2$, $Z^{IV\ II}_{NbFeHS} = 4x$, $\mu = 1$, and $\lambda = 2x$, where x is the rate of remaining Fe^{II}_{HS}. The least square fitting was performed for low-T form and the observed $\chi_M T$ value at 50 K of the low-T phase was close to the value where 90% of the Fe^{II}_{HS} transited to Fe^{II}_{HS} (Fig. 3.12). As shown in Fig. 3.12, in the magnetic susceptibility of high temperature region, the gradient of experimental susceptibility and calculated one do not agree, which indicate some other contribution has to be considered. Based on the knowledge of magnetic susceptibility of $^5T_2$ term in octahedral geometry as shown in Chap. 1, the consideration of spin-orbit coupling is inevitable for the analysis of magnetic susceptibility for the present compound. In this work, I have analyzed the magnetic susceptibility by combining the Van Vleck approach and molecular field approach as shown below.

(i) Van Vleck approach

First, the magnetic susceptibility of $^5T_2$ term with axial ligand field is calculated based on the previous paper.[28]

In this model, the basic process of calculation is the same as introduced in Chap. 1, while the axial ligand field (Δ) and spin-orbit coupling are simultaneously considered.

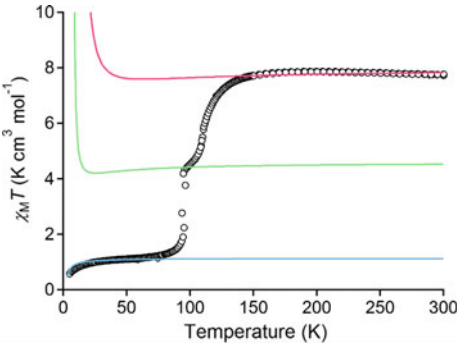


Fig. 3.12 The analysis of $\chi_M T$ versus T plot using molecular field theory. *Red*, *green*, and *blue curves* correspond to the magnetic susceptibility with $Fe^{II}_{HS}(100\%)$, $Fe^{II}_{HS}(55\%)$ and $Fe^{II}_{LS}(45\%)$, and $Fe^{II}_{HS}(10\%)$ and $Fe^{II}_{LS}(90\%)$

The Hamiltonian is

$$H = \kappa\lambda LS + \Delta(L_z^2 - \frac{1}{3}),$$

where κ is orbital reduction factor.

Since the wave functions of $^5T_2$ term is described as the following shown as

$|m_L\rangle[m_S]$,
$|1\rangle[+2], \quad |1\rangle[+1], \quad |1\rangle[0], \quad |1\rangle[-1], \quad |1\rangle[-2],$
$-|1\rangle[+2], \quad |-1\rangle[+1], \quad |-1\rangle[0], \quad |-1\rangle[-1], \quad |-1\rangle[-2]$
$\sqrt{1/2}(|+2\rangle - |-2\rangle)[+2], \sqrt{1/2}(|+2\rangle - |-2\rangle)[+1],$
$\sqrt{1/2}(|+2\rangle - |-2\rangle)[0], \sqrt{1/2}(|+2\rangle - |-2\rangle)[-1], \sqrt{1/2}(|+2\rangle - |-2\rangle)[-2]$

we obtain the 15 × 15 matrix shown in Fig. 3.13.

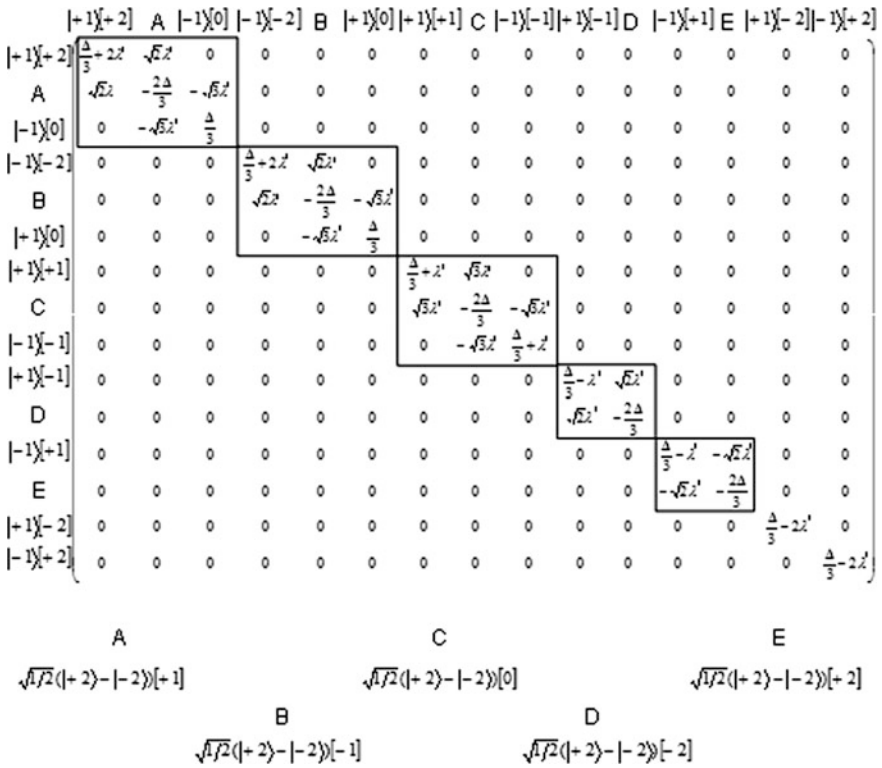


Fig. 3.13 The 15 × 15 matrix obtained by considering the spin-orbit coupling between the wavefunctions of octahedral Fe^{II} ion (d^6) with $^5T_2$ term with the Hamiltonian $H = \kappa\lambda LS + \Delta(L_z^2 - 1/3)$ ($\lambda' = \kappa\lambda$). By diagonalizing this matrix, the eigenenergies and eigenfunctions are obtained

By block diagonalization of this 15 × 15 matrix, we have only to solve three 3 × 3 matrices and two 2 × 2 matrices, and other two eigenvalues are diagonal elements without non-diagonal components. Since it is difficult to obtain the analytical solution of eigenenergies ($W^{(0)}$), I numerically calculated the engenenergies and eigenfunctions using C-language.

Since the Zeeman energies of z direction and x (y) direction are different, the following equation is needed to be solved.

1st order Zeeman energy:

$$E_{iz}^{(1)} = \mu_B \sum_j \left\langle \psi_i^{(0)} | kL_z + 2S_z | \psi_j^{(0)} \right\rangle$$

$$E_{ix}^{(1)} = \mu_B \sum_j \left\langle \psi_i^{(0)} | kL_x + 2S_x | \psi_j^{(0)} \right\rangle$$

2nd order Zeeman energy:

$$E_{iz}^{(2)} = \mu_B^2 \sum\nolimits_j \frac{\left\langle \psi_i^{(0)} | kL_z + 2S_z | \psi_j^{(0)} \right\rangle \left\langle \psi_i^{(0)} | kL_z + 2S_z | \psi_j^{(0)} \right\rangle}{E_i^{(0)} - E_j^{(0)}}.$$

$$E_{ix}^{(2)} = \mu_B^2 \sum\nolimits_j \frac{\left\langle \psi_i^{(0)} | kL_x + 2S_x | \psi_j^{(0)} \right\rangle \left\langle \psi_i^{(0)} | kL_x + 2S_x | \psi_j^{(0)} \right\rangle}{E_i^{(0)} - E_j^{(0)}},$$

where κ is orbital reduction factor.

By substituting the obtained energy values to the Van Vleck equation, we are able to obtain the results. First, I will show the results of the calculation which is shown in the reference paper.[26]

The results perfectly agree with the experimental value described in the paper, indicating the reliability of the present calculation (Fig. 3.14).

Next, I will show the procedure to combine this Van Vleck approach and molecular field approach.

(ii) Combination of Van Vleck approach and molecular field approach

To combine the Van Vleck approach and molecular field approach, the Curie constant of molecular field approach should be replaced to the equation of magnetic

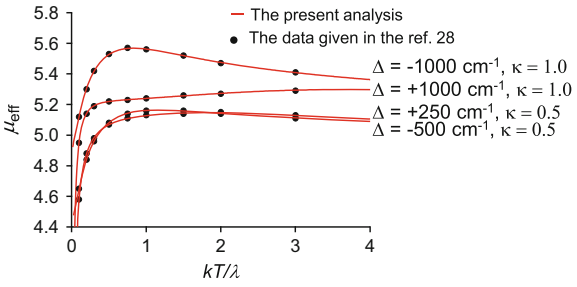


Fig. 3.14 Temperature dependence of magnetic susceptibility of octahedral Fe^{II} ion (d^6) with $^5T_2$ term in axial field. The comparison of the present analysis with the data given in [26].

susceptibility based on Van Vleck approach. Such a type of analysis is sometimes performed to analyze magnetic susceptibility of metal cluster complexes, in which magnetic interaction and/or spin-obrit coupling is considered with single metal cluster based on Van Vleck equation, and then the obtained susceptibility is replaced to the Curie constant of molecular field theory to estimate the weak magnetic interaction between clusters.[27] In such case, molecular field theory of unary metal system is sufficient, while in the case of $(Fe^{II})_2[Nb^{IV}(CN)_8](4\text{-methylpyridine})_8 \cdot 2H_2O$, the molecular field theory of ternary metal system (Fe^{II}_{HS}, Fe^{II}_{LS}, and Nb^{IV}) is needed. Based on the Ohkoshi's papar,[28] the magnetic susceptibility of $(Fe_{HS,x}Fe_{LS,1-x})_2[Nb(CN)_8](4\text{-methylpyridine})_8 \cdot 2H_2O$ is expressed as the following equation,

$$\chi = \frac{(C_{Fe^{II}_{HS}} + C_{Fe^{II}_{LS}} + C_{Nb})T^2 + (n_{Fe^{II}_{HS}Nb}n_{NbFe^{II}_{LS}} + n_{Fe^{II}_{LS}Nb}n_{NbFe^{II}_{HS}} - n_{Fe^{II}_{HS}Nb}n_{NbFe^{II}_{HS}} - n_{Fe^{II}_{LS}Nb}n_{NbFe^{II}_{LS}})C_{Fe^{II}_{HS}}C_{Fe^{II}_{LS}}C_{Nb}}{T^3 - (C_{Fe^{II}_{LS}}C_{Nb}n_{Fe^{II}_{LS}Nb}n_{NbFe^{II}_{LS}} + C_{Fe^{II}_{HS}}C_{Nb}n_{Fe^{II}_{HS}Nb}n_{NbFe^{II}_{HS}})T} + \frac{\left\{(n_{Fe^{II}_{HS}Nb} + n_{NbFe^{II}_{HS}})C_{Fe^{II}_{HS}}C_{Nb} + (n_{Fe^{II}_{LS}Nb} + n_{Nb})C_{Fe^{II}_{LS}}C_{Nb}\right\}T}{T^3 - (C_{Fe^{II}_{LS}}C_{Nb}n_{Fe^{II}_{LS}Nb}n_{NbFe^{II}_{LS}} + C_{Fe^{II}_{HS}}C_{Nb}n_{Fe^{II}_{HS}Nb}n_{NbFe^{II}_{HS}})T}$$

where n_{ij} is the molecular field coefficients and C_i is the Curie constants given by the following equations.

$$n_{Fe^{II}_{HS}Nb} = \frac{Z_{Fe^{II}_{HS}Nb}}{N_A\mu_B^2 g_{Fe^{II}_{HS}} g_{Nb}} J_{Fe^{II}_{HS}Nb},\ n_{Fe^{II}_{LS}Nb} = \frac{Z_{Fe^{II}_{LS}Nb}}{N_A\mu_B^2 g_{Fe^{II}_{LS}} g_{Nb}} J_{Fe^{II}_{LS}Nb}$$

$$n_{NbFe^{II}_{HS}} = \frac{Z_{NbFe^{II}_{HS}}}{2xN_A\mu_B^2 g_{Fe^{II}_{HS}} g_{Nb}} J_{Fe^{II}_{HS}Nb},$$

$$n_{NbFe^{II}_{LS}} = \frac{Z_{NbFe^{II}_{LS}}}{2(1-x)N_A\mu_B^2 g_{Fe^{II}_{LS}} g_{Nb}} J_{Fe^{II}_{LS}Nb},$$

$$C_{Nb} = \frac{N_A g_{Nb}^2 \mu_B^2 S_{Nb}(S_{Nb}+1)}{3k_B},$$

$$C_{Fe^{II}_{LS}} = \frac{2(1-x)N_A g_{Fe^{II}_{LS}}^2 \mu_B^2 S_{Fe^{II}_{LS}}(S_{Fe^{II}_{LS}}+1)}{3k_B} = 0$$

In addition, we use the magnetic susceptibility obtained from the Van Vleck approach described in (i) to the Curie constant of Fe^{II}_{HS}, $C_{Fe^{II}_{HS}}$.

As a result, the matching of the high temperature region dramatically improved with the axial splitting parameter (Δ) of Δ = 200 cm^{-1} and the orbital reduction factor (κ) of κ = 0.89. In the low temperature region, the best agreement of experimental magnetic susceptibility and the calculated one is obtained with x = 0.12 by using Δ and κ obtained from the analysis of high temperature region. Thus, the electronic states of the high temperature state and low temperature state are assigned to $(Fe^{II}_{HS})_2[Nb^{IV}(CN)_8](4\text{-methylpyridine})_8 \cdot 2H_2O$ and $(Fe^{II}_{HS})_{0.24}(Fe^{II}_{LS})_{1.76}[Nb^{IV}(CN)_8](4\text{-methylpyridine})_8 \cdot 2H_2O$, respectively (Fig. 3.15). Moreover, at middle

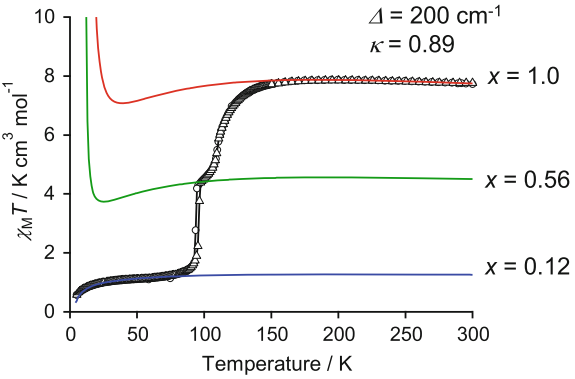


Fig. 3.15 The analysis of $\chi_M T$ versus T plot by combining Van Vleck approach and molecular field theory. *Red*, *green*, and *blue curves* correspond to the magnetic susceptibility with Fe_{HS}^{II}(100%), Fe_{HS}^{II}(56%) and Fe_{LS}^{II}(44%), and Fe_{HS}^{II}(12%) and Fe_{LS}^{II}(88%)

temperature range, the $\chi_M T$ value corresponds to the value with 44% of Fe_{LS}^{II}, which is the half value of 88%. This indicates that plateau comes at the high spin fraction with the half value of the final transition ratio.

3.4.3 Temperature Dependence of Crystal Structure

As shown in the above discussion, in the variable-temperature single crystal X-ray analysis, the lattice constants decreased and compression of the Fe–N bond lengths, 2.085 (300 K) → 1.944 Å (80 K) on Fe–N(1), 2.248 → 2.071 Å on Fe–N(2), and 2.267 → 2.089 Å on Fe–N(3), was observed (Fig. 3.16a). In typical Fe^{II} spin-crossover complex, the Fe–N bond length is about 2.1 Å for Fe_{HS}^{II} and 1.9 Å for Fe_{LS}^{II}. Therefore this change of crystal structure supports the Fe_{HS}^{II} → Fe_{LS}^{II} transition. In addition, the Nb-C distances remain almost constant value between 80 K and 300 K, indicating that the electronic state of Nb ion is not changed (Fig. 3.16b). According to this change the lattice parameters also change drastically below 110 K (Fig. 3.17).

3.4.4 Thermodynamical Properties

The thermodynamical analysis of the $\chi_M T$-T curve was performed using advanced Slichter Drickamer's model. The Slichter Drickamer's model is often used to analyze the one step spin crossover behavior and the Gibbs free energy is described as the following.

$$G = x\Delta H + \gamma x(1-x) - T[x\Delta S - R\{x\,ln\,x + (1-x)\,ln\,(1-x)\}],$$

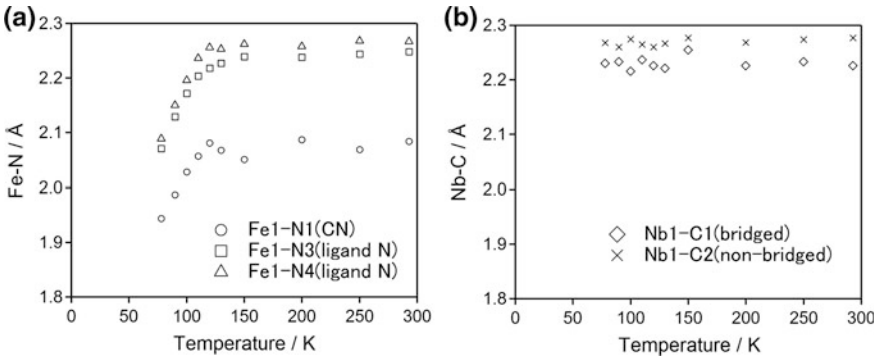


Fig. 3.16 The temperature dependence of **a** Fe–N and **b** Nb–C bond lengths obtained by the temperature variation measurements of single crystal X-ray diffraction

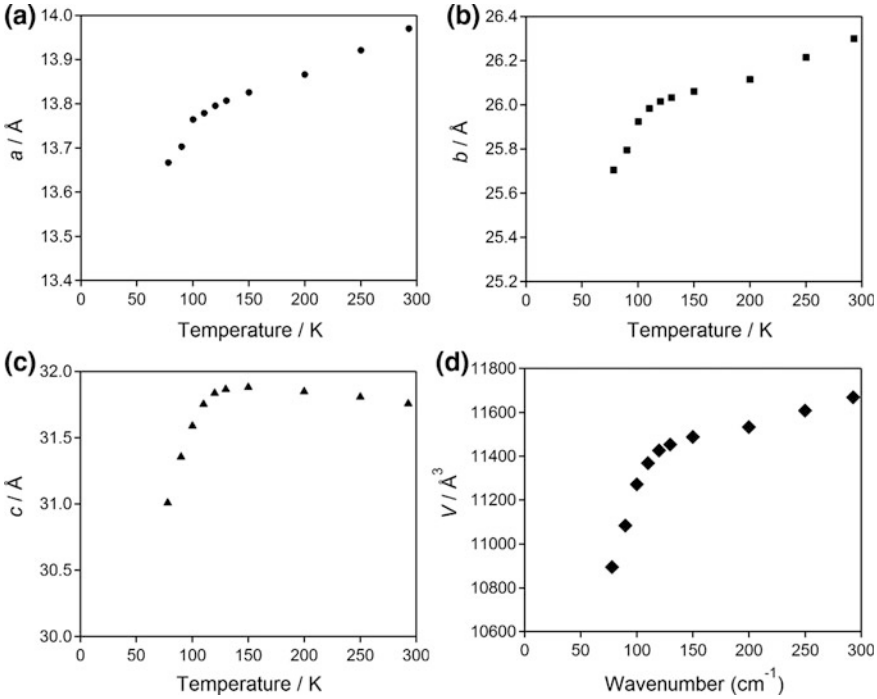


Fig. 3.17 The temperature dependence of lattice constants obtained by the temperature variation measurements of single crystal X-ray diffraction. **a** *a*-axis, **b** *b*-axis, **c** *c*-axis, and **b** cell volume

where x is the fraction of LS state, ΔH is transition enthalpy from HS state to LS state, γ is interaction parameter between HS and LS states, ΔS is transition entropy from HS state to LS state, and R is gas constant. The schematic illustration of Slichter Drickamer's model is shown in Fig. 3.18.

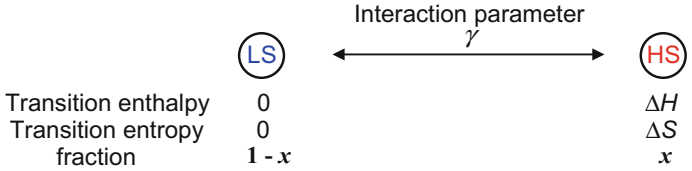


Fig. 3.18 The formula and schematic illustration of Slichter Drickamer's model. In this model, two states of LS and HS are considered, where x is the fraction of LS state, ΔH is transition enthalpy from HS state to LS state, γ is interaction parameter between HS and LS states, ΔS is transition entropy from HS state to LS state, and R is gas constant

In the present case, the analysis of two step spin-crossover phenomenon is needed, and three states of HS-HS, HS-LS, and LS-LS are considered with the transition entropy and transition enthalpy as shown in Fig. 3.19. As a result, the following equation is obtained,

$$G = x\Delta H + y\left(\frac{\Delta H}{2} + W\right) + \gamma\{2x(1-x-y) + y(1-x-y) + xy\} - T\left[\left(\frac{y}{2} + x\right)\Delta S - R\{x\,ln\,x + y\,ln\,y + (1-x-y)\,ln\,(1-x-y)\}\right]$$

Here, x is the fraction of HS-HS state, y is the fraction of LS-HS state, ΔH is transition enthalpy from HS-HS state to LS-LS state, $\Delta H/2 + W$ is the transition enthalpy from LS-HS state to LS-LS state, γ is interaction parameter, ΔS is transition entropy from HS state to LS state, and R is gas constant.

Based on this model, the magnetic behavior of the present compound is analyzed. As a result, a good agreement of HS fraction and the $\chi_M T$ versus T plot is obtained with the parameters of ΔH = 8.3 kJ, ΔS = 80 J K^{-1} mol^{-1}, γ = 1.8 kJ mol^{-1}, W = −350 J mol^{-1} (Fig. 3.20). Figure 3.21 shows the high spin fraction dependence of Gibbs free energy curve at each temperature with the above parameters. At high temperature, the minimum of the Gibbs free energy curve is

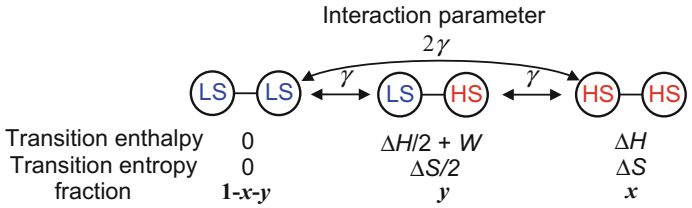


Fig. 3.19 The formula and schematic illustration of advanced Slichter Drickamer's model. In this model, three states of LS-LS, LS-HS, and HS-HS are considered, where x is the fraction of HS-HS state, y is the fraction of LS-HS state, ΔH is transition enthalpy from HS-HS state to LS-LS state, $\Delta H/2 + W$ is the transition enthalpy from LS-HS state to LS-LS state, γ is interaction parameter, ΔS is transition entropy from HS state to LS state, and R is gas constant

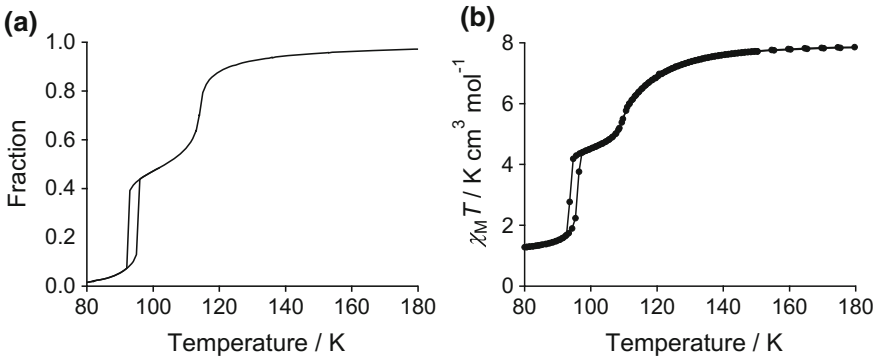


Fig. 3.20 **a** The HS fraction vs temperature plot obtained by the analysis using advanced Slichter-Drickamer's model with the parameters of ΔH = 8.3 kJ, ΔS = 80 J K^{-1} mol^{-1}, γ = 1.8 kJ mol^{-1}, W = −350 J mol^{-1}. **b** The observed $\chi_M T$ versus T plot

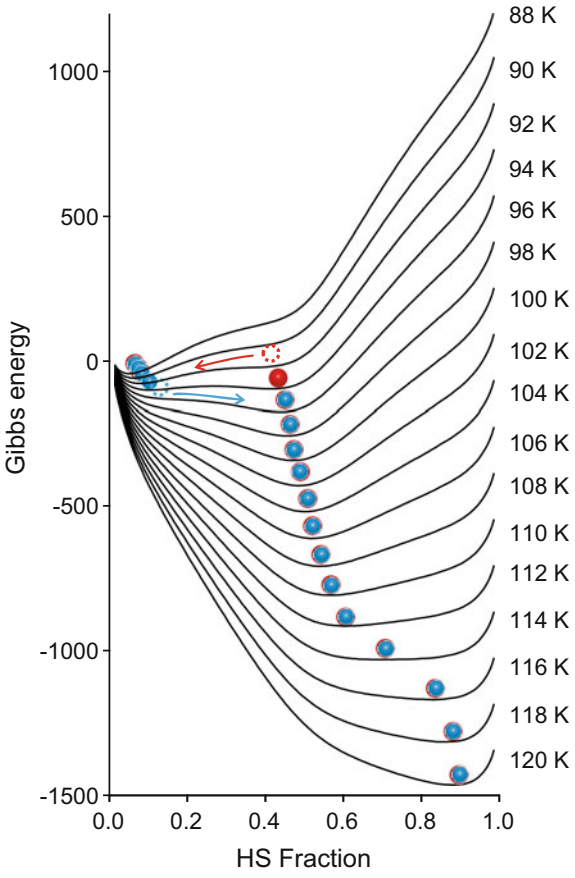


Fig. 3.21 The high spin fraction dependence of Gibbs free energy curve at each temperature with the parameters of ΔH = 8.3 kJ, ΔS = 80 J K^{-1} mol^{-1}, γ = 1.8 kJ mol^{-1}, W = −350 J mol^{-1}. *Red* and *blue* balls represent the equilibrium states with decreasing temperature and increasing temperature, respectively

located near the high spin fraction of 1. As decreasing temperature, the minimum shifts to the high spin fraction of about 0.5. As further decreasing temperature, the minimum disappears at 94 K and the sample transits to low temperature state with almost no high spin. With increasing temperature, the minimum of low temperature state disappears at 96 K and transits to middle temperature phase, and then returns to the original high temperature phase around 115 K.

These results indicate the present transition behavior can be explained by the advanced Slichter Drickamer's model.

3.4.5 Magnetic Properties of Low Temperature State

At very low temperature, the magnetization vs temperature plots at 10 Oe showed no spontaneous magnetization above 2 K and the magnetization vs external field plots showed that the coercive field was zero, indicating that the present compound is paramagnetic above 2 K.

3.4.6 Photomagnetic Properties

As shown in 3.3.6, The irradiation with a 532-nm diode laser light (13 mW cm^{-2} and 5 min) at 2 K produced a large spontaneous magnetization. The magnetization versus temperature curve showed a Curie temperature (T_C) of 15 K (Fig. 3.8a), while the magnetization versus external magnetic field plots exhibited a magnetic hysteresis loop with a coercive field (H_c) of 2300 Oe at 2 K (Fig. 3.8b). The time dependence of the light-induced HS phase data showed a slow relaxation. After irradiation, 90% of light-induced HS state remained after 1 day (Fig. 3.9) Furthermore, the temperature dependence of the light-induced HS phase plots indicated that the relaxation temperature was 55 K (Fig. 3.10). The saturation magnetization (M_s) value at 5 T was 7.5 $\mu_{B,}$ indicating that an antiferromagnetic interaction operates between photogenerated Fe^{II}_{HS} and Nb^{IV}.

The mechanism of the photo-magnetization can be understood by the result of the photo-irradiation measurement of the UV-vis spectra. By the irradiation of 532 nm light, absorption bands around 400 nm, 415 nm, 560 nm, and 610 nm decreased (Fig. 3.11), which are assigned to the $^1A_1 \rightarrow {}^1E$, $^1A_1 \rightarrow {}^1B_2$, $^1A_1 \rightarrow {}^1A_2$, and $^1A_1 \rightarrow {}^1E$ d-d transitions on the Fe^{II}_{LS} ($S = 0$) with D_{4h} symmetry. This strongly suggests the occurrence of light-induced spin-crossover phenomenon from LS state to HS state.

Therefore, the observed photomagnetism can be explained by the following scheme. Before irradiation, paramagnetic Nb^{IV} and diamagnetic Fe^{II}_{LS} of the low-T phase were connected by a CN ligand in an alternating fashion, which showed paramagnetism. Irradiation with 532-nm light caused the light-induced excited-spin-state trapping (LIESST) effect on the Fe^{II} site (Fig. 3.22); the $^1A_1$

state on Fe^{II}_{LS} transited to the excited state due to the light irradiation and then partially proceeded to the metastable quintet state through the triplet state. By the observed saturation magnetization value, it is indicated that 100% of Fe^{II} sites transit to HS state and antiferromagnetic interaction works between photo-generated Fe^{II}_{HS} and Nb^{IV} by a strong superexchange interaction through the CN ligand, resulting in bulk magnetization. Based on molecular field theory, the values of the superexchange interaction constant ($J_{ex,ij}$) between the i site and the nearest neighbour j site in $(Fe^{II}_{HS})_2[Nb^{IV}(CN)_8](4\text{-pyridinealdoxime})_8 \cdot 2H_2O$ (the photo-induced phase) are related to the T_C value via the following equation,

$$T_C = \frac{\left|J_{ex,Fe^{II}HSNb^{IV}}\right| \left\{ Z_{Nb^{IV}Fe^{II}HS} Z_{Fe^{II}HS\ Nb^{IV}} S_{Fe^{II}HS} \left(S_{Fe^{II}HS} + 1\right) S_{Nb^{IV}} \left(S_{Nb^{IV}} + 1\right) \right\}^{1/2}}{3k_B},$$

where S_i is the spin quantum number ($S_{Nb^{IV}} = 1/2$ and $S_{Fe^{II}HS} = 2$), Z_{ij} is the number of the nearest neighbour j sites around i site, and k_B is the Boltzmann constant. In the case of $(Fe^{II}_{HS})_2[Nb^{IV}(CN)_8](4\text{-pyridinealdoxime})_8 \cdot 2H_2O$, T_C = 15 K, and Z_{ij} were $Z_{Nb^{IV}Fe^{II}HS} = 4$ and $Z_{Fe^{II}HSNb^{IV}} = 2$. Consequently, $J_{ex,Fe^{II}HS\ Nb^{IV}}$ was estimated to be −5.2 cm^{-1}.

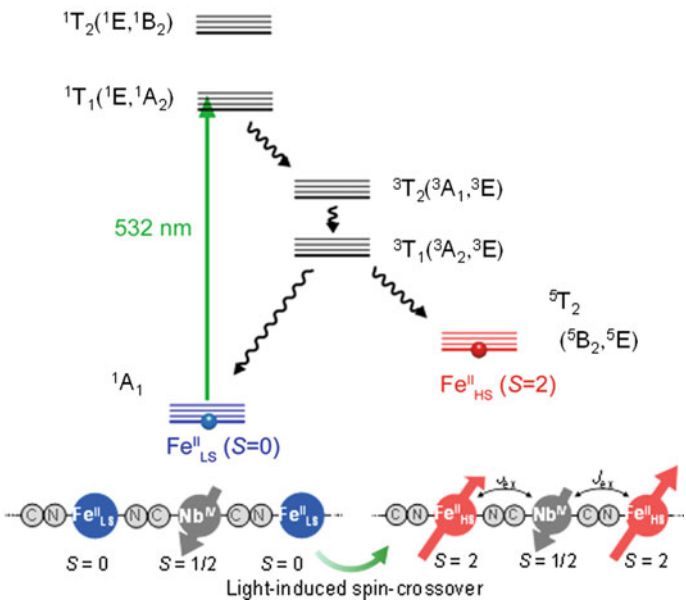


Fig. 3.22 The mechanism of spin-crossover and photo-induced spin-crossover magnetism

3.5 Conclusion

In this chapter, a two-step spin-crossover phenomenon and light-induced spin-crossover magnetization is demonstrated in a three-dimensional Fe–Nb based magnet, $Fe_2^{II}[Nb^{IV}(CN)_8](4\text{-methylpyridine})_8 \cdot 2H_2O$ with an orthorhombic crystal structure. The bulk magnetic properties of the material are switched by light irradiation in a two-step spin transition material. The present results demonstrate that new type of photo-magnetic materials can be created by using light-induced spin-crossover mechanism. The present results were obtained by ligand modification of three-dimensional Fe–Nb based magnet. Further investigations of ligand exchange, solvent exposure, etc., will lead to the realization of other novel functionalities.

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Chapter 4
Observation of Coexistence of Super-Ionic Conductivity and Metamagnetism in $Mn_3[Nb^{IV}(CN)_8]_2$(4-Aminopyridine)$_{10}$ (4-Aminopyridinium)$_2 \cdot 12H_2O$

Abstract In this chapter, to obtain new ionic conductive magnetic material using octacyanometalate-based magnet, the incorporation of 4-aminopyridine as organic ligand is performed due to the capability of forming hydrogen bond. The reaction of Mn^{II}, $[Nb^{IV}(CN)_8]$, and 4-aminopyridine yielded yellow crystal formulated as $Mn^{II}_3[Nb^{IV}(CN)_8]_2$(4-aminopyridine)$_{10}$(4-aminopyridinium)$_2 \cdot 12H_2O$. This compound possesses non-coordinated pyridinium cation and 1-dimensional infinite hydrogen-bonding networks constructed with ligand water molecules, non-coordinated water molecules, and 4-aminopyridinium cations. Magnetic measurements revealed that the present compound shows metamagnetism with a Neel temperature of 9 K. The impedance measurement showed that the present compound exhibits super-ionic conductivity ($>10^{-4}$ S cm^{-1}) which is caused by proton conduction based on the Grotthuss mechanism. In the present material, 4-aminopyridinium cation is incorporated into the flexible octacyanidometallate-based magnetic framework and plays an important role in the formation of the hydrogen-bonding network, which leads to the observation of super-ionic conductivity.

Keywords Ionic conductivity · Cyanido-bridged metal assembly · Hydrogen bond network · Metamagnetism

4.1 Introduction

Ionic conductivity is one of a promising electric functionality of a material because an ionic conductive material has a possibility to be applied to fuel cell [1–11]. Especially, solid state substance which shows ionic conductivity higher than 10^{-4} S cm^{-1} is called "super-ionic conductor" [2]. Conventionally, most of ionic conductive materials are based on inorganic compounds, ceramics or polymer based materials. The most famous examples are alfa-phase silver iodide (α-AgI) and polymer called Nafion® which is a poly-(sulfonated tetrafluoroethylene). The

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former example exhibit super-ionic conductivity of 0.1 S cm^{-1} at 473 K due to Ag^+ ion conductivity. This material is known as the first super-ionic conductor. The latter example shows super-ionic conductivity of 10^{-2} S cm^{-1} at room temperature due to proton conduction. Protons are released and carried through the amount of sulfo groups ($SO_3^-H^+$), resulting in high ionic conductivity. This material is applied as electrolyte layer in fuel cell. Recently, ionic conductive coordination polymers are drawn attention [5–11]. Our laboratory has reported super-ionic conductivity in cobalt-hexacyanochromate and vanadium-hexacyanochromate based Prussian blue analogues and observed an interference effect between ionic conductivity and magnetic phase transition in vanadium-hexacyanochromate based Prussian blue analogue [11]. The objective of this chapter is to obtain new ionic conductive magnetic material using octacyanometalate-based magnet. To achieve this target using proton conductivity, the formation of infinite hydrogen bonding network is one of an important factor. The incorporation of organic ligand which is capable of forming hydrogen bond is one of a good way to assist the formation of hydrogen bonding network. Along this line, a cyanido-bridged metal assembly is synthesized using Mn^{II}, $[Nb^{IV}(CN)_8]$, and 4-aminopyridine and the physical properties of the obtained compound is investigated.

4.2 Experiments

4.2.1 Synthesis

Synthesis of $K_4[Nb(CN)_8] \cdot 2H_2O$

$K_4[Nb(CN)_8] \cdot 2H_2O$ was synthesized as described in Chap. 2.

Synthesis of $Mn_3[Nb(CN)_8]_2(4\text{-aminopyridine})_{10}(4\text{-aminopyridinium})_2 \cdot 12H_2O$

The target sample of $Mn_3[Nb(CN)_8]_2(4\text{-aminopyridine})_{10}(4\text{-aminopyridine})_2 \cdot 12H_2O$ was prepared using slow diffusion method. Aqueous solution of $K_4[Nb(CN)_8] \cdot 2H_2O$ (0.05 mol dm^{-3}) is added to the mixed aqueous solution of $MnCl_2 \cdot 4H_2O$ (0.005 mol dm^{-3}) and 4-aminopyridine (0.1 mol dm^{-3}). After stirring for 1 h, the resulting purple powder was filtered, washed twice with distilled water, and dried under the exposure of air.

4.2.2 Characterization

Elemental analyses of Mn and Nb for the prepared material were conducted by Agilent-7700 inductively coupled plasma mass spectroscopy. C, H, and N elemental analyses were performed by standard microanalytical methods in the microanalytical laboratory at the department of chemistry, school of science the University of Tokyo.

Infrared spectra were recorded on a JASCO IRT-3000 spectrometer in the 4000–400 cm^{-1} region. X-ray single crystal structural analysis was measured by a Rigaku RAXIS RAPID imaging plate area detector with graphite monochromated Mo Kα radiation. The crystal structure was solved by a direct method and refined by a full-matrix least-squares technique using SHELXL-97 [12]. Anisotropic refinement was applied to all non-hydrogen atoms. The hydrogen atoms of 4-aminopyridine, 4-aminopyridinium on aromatic ring, and ligand water molecules were refined using the riding model, while those of non-coordinated water molecules and amino group of 4-aminopyridinium were not included in the refinement. The calculation was performed using the CrystalStructure crystallographic software package. Crystallographic data have been deposited with Cambridge Crystallographic Data Centre as supplementary publication No. CCDC-916575. XRD measurements were conducted on a Rigaku Ultima IV with Cu Kα radiation (λ = 1.5406 Å) within the range of $10° \leq 2\theta \leq 70°$. The magnetic measurements were obtained from grinded single crystals using a Quantum Design MPMS superconducting quantum interference device (SQUID) magnetometer.

4.2.3 Impedance Measurement

Impedance measurements were carried out with an Agilent 4294A precision impedance analyzer from the 40 Hz–110 MHz frequency range using a four-terminal pair method. The powdered sample was compressed into a pellet with a 7-mm diameter and 0.10-mm thickness using a sample holder between two steel electrodes for the impedance measurement.

4.3 Results

4.3.1 Elemental Analysis

Elemental analysis showed that the obtained compound had the formula of $Mn_3[Nb(CN)_8]_2$(4-aminopyridine)$_{10}$(4-aminopyridinium)$_2 \cdot 12H_2O$.

Calculated: Mn, 7.8; Nb, 8.8; C, 43.2; H, 4.7; N, 26.5%;
Found: Mn, 7.7; Nb, 8.7; C, 42.7; H, 4.7; N, 26.4%.

4.3.2 Single Crystal X-ray Analysis

The single crystal X-ray analysis indicated that the resulting compound had a monoclinic *C2/c* space group with a = 19.1492(5), b = 17.7054(4), c = 29.3099 (7), and β = 103.255(1)° (Fig. 4.1 and Table 4.1).

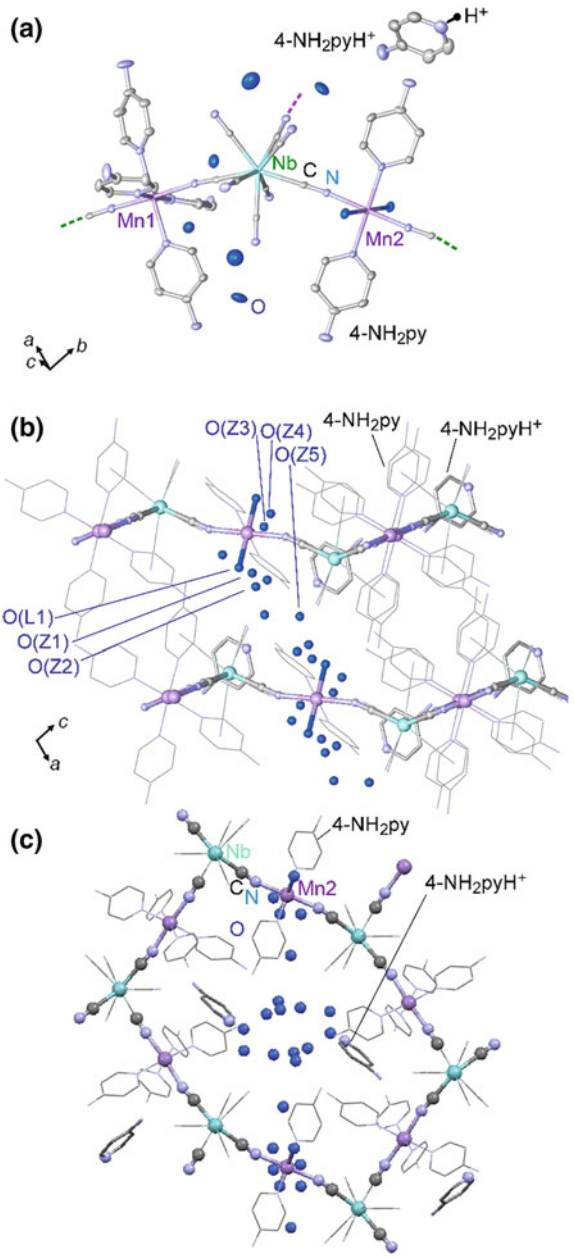


Fig. 4.1 Crystal structure of $Mn_3[Nb(CN)_8]_2$ (4-aminopyridine)$_{10}$ (4-aminopyridine)$_2 \cdot 12H_2O$. **a** Coordination environment, **b** crystal structure viewed from *b*-axis, and **c** the crystal structure viewed from the orthogonal direction to b-axis. Reproduced by permission from American Chemical Society: K. Imoto, K. Nakagawa, H. Miyahara, S. Ohkoshi, ***Cryst. Growth Des***., 13, 4673–4677 (2013)

There were two pseudo-octahedral Mn sites (Mn1, Mn2) and one Nb site in the crystal structure. A $[Nb^{IV}(CN)_8]^{4-}$ anion, a $[Mn^{II}(4\text{-aminopyridine})_4]^{2+}$ cation, half of $[Mn^{II}(4\text{-aminopyridine})_2(H_2O)_2]^{2+}$ cation, five non-coordinated (zeolitic) water molecules, and a non-coordinated 4-aminopyridinium cation are the component of

Table 4.1 Crystallographic and refinement data of $Mn^{II}_3[Nb^{IV}(CN)_8]_2$(4-aminopyridine)$_{10}$(4-aminopyridinium)$_2 \cdot 12H_2O$

Formula	$C_{76}H_{98}Mn_3N_{40}Nb_2O_{12}$
Formula weight	2114.50
Crystal system	Monoclinic
Space group	*C*2/*c*
a/Å	19.1498(6)
b/Å	17.7070(4)
c/Å	29.3088(7)
β/degrees	103.2520(10)
V/Å^3	9673.5(4)
Z	4
Calculated density/g cm^{-3}	1.452
Temperature/K	90(1)
R_1	0.0446
wR_2	0.1124
GOF on F_2	1.152
Absorption coefficient μ/cm^{-1}	6.88
Number of measured reflections	46,867
Number of independent reflections	11,058
Number of refined parameters	614

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the asymmetric unit (Fig. 4.1a). The coordination geometry of the Nb site is square-antiprism. The three CN groups of $[Nb(CN)_8]$ bridge two Mn1 sites and one Mn2 site, whereas the other five CN groups are not bridged. The terminal CN groups form hydrogen-bonds with oxygen atoms of zeolitic water molecules towards three directions. The bond distances between Nb and cyanide carbon atoms range from 2.23 to 2.27 Å. The Nb-N-C angles are between 177.0° and 179.3°. The Mn1 site is coordinated by two cyanide nitrogen atoms in apical positions and four pyridyl nitrogen atoms of 4-aminopyridine in equatorial positions, while the Mn2 site is coordinated by two cyanide nitrogen atoms, two pyridyl nitrogen atoms of 4-aminopyridine, and two oxygen atoms of ligand water molecules. The bond distances between Mn1 and cyanide nitrogen atoms are 2.18 and 2.20 Å, whereas those between Mn1 and pyridyl nitrogen atoms of 4-aminopyridine range from 2.27 to 2.31 Å. The Mn2-N(cyanide), Mn2-N(pyridyl), and Mn2-O distances are 2.23, 2.22, and 2.26 Å, respectively. The Mn-N-C angles are 168.7° and 175.9° for Mn1, and 160.2° for Mn2 (Tables 4.2 and 4.3). The Nb-CN-Mn cyano-bridged framework forms a 2D coordination network structure parallel to (1 0 −1) crystallographic plane with an interlayer distance of 8.8 Å (Fig. 4.1b). The 2D layer is

Table 4.2 Selected bond distances (Å)

Mn1	–	N1	2.203(3)	C1	–	N1	1.150(4)
Mn1	–	N3[(i)]	2.178(3)	C2	–	N2	1.153(4)
Mn1	–	N9	2.288(3)	C3	–	N3	1.155(4)
Mn1	–	N11	2.265(3)	C4	–	N4	1.155(4)
Mn1	–	N13	2.314(3)	C5	–	N5	1.158(4)
Mn1	–	N15	2.298(3)	C6	–	N6	1.153(4)
				C7	–	N7	1.155(4)
Mn2	–	N2	2.228(2)	C8	–	N8	1.154(5)
Mn2	–	N17	2.223(3)				
Mn2	–	O(L1)	2.265(3)				
Nb1	–	C1	2.230(3)				
Nb1	–	C2	2.263(3)				
Nb1	–	C3	2.237(3)				
Nb1	–	C4	2.263(3)				
Nb1	–	C5	2.252(4)				
Nb1	–	C6	2.259(3)				
Nb1	–	C7	2.256(3)				
Nb1	–	C8	2.267(3)				

(i) $1/2 - x, -1/2 + y, 1/2 - z$

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constructed with cyano-bridged Mn_6Nb_6 dodecanuclear hexagonal grids, and two non-coordinated 4-aminopyridinium cations are placed inside the hexagonal grid (Fig. 4.1c). The non-coordinated water molecules are located inside and between the layers. In the crystal, 1D infinite hydrogen-bonding networks are constructed with O(water)–O(water), O(water)–N(4-aminopyridinium), and O(water)–N(terminal CN group) with the distance between 2.7 and 3.2 Å along [1 −1 0] and [−1 1 0] crystallographic directions, which penetrate the 2D network structure (Table 4.4 and Fig. 4.2).

4.3.3 Powder X-ray Diffraction Pattern

The powder X-ray diffraction (XRD) pattern of the grinded single crystal corresponded to the simulated X-ray diffraction pattern obtained from the crystal structure of single crystal, indicating that the grinded single crystal and single crystal is isostructural (Fig. 4.3).

Table 4.3 Selected bond angles (deg.)

C1	–	Nb1	–	C2	143.38(10)						
C1	–	Nb1	–	C3	105.75(10)						
C1	–	Nb1	–	C4	142.75(10)						
C1	–	Nb1	–	C5	72.48(10)						
C1	–	Nb1	–	C6	79.43(9)						
C1	–	Nb1	–	C7	70.24(10)						
C1	–	Nb1	–	C8	83.52(11)						
C2	–	Nb1	–	C3	80.87(9)						
C2	–	Nb1	–	C4	73.78(10)						
C2	–	Nb1	–	C5	75.85(10)						
C2	–	Nb1	–	C6	75.10(10)						
C2	–	Nb1	–	C7	143.89(10)						
C2	–	Nb1	–	C8	111.71(10)						
C3	–	Nb1	–	C4	76.97(11)						
C3	–	Nb1	–	C5	71.86(10)						
C3	–	Nb1	–	C6	143.01(10)						
C3	–	Nb1	–	C7	74.77(10)						
C3	–	Nb1	–	C8	145.38(10)						
C4	–	Nb1	–	C5	139.21(10)						
C4	–	Nb1	–	C6	121.14(10)						
C4	–	Nb1	–	C7	75.02(10)						
C4	–	Nb1	–	C8	76.18(11)						
C5	–	Nb1	–	C6	75.23(10)						
C5	–	Nb1	–	C7	119.61(10)						
C5	–	Nb1	–	C8	141.42(10)						
C6	–	Nb1	–	C7	138.14(10)						
C6	–	Nb1	–	C8	70.91(10)						
C7	–	Nb1	–	C8	77.47(10)						
N1	–	Mn1	–	$N3^{(i)}$	174.68(10)	O(L1)	–	Mn2	–	$O(L1)^{(ii)}$	180.00(11)
N1	–	Mn1	–	N9	90.61(9)	O(L1)	–	Mn2	–	N2	95.41(9)
N1	–	Mn1	–	N11	88.74(10)	O(L1)	–	Mn2	–	$N2^{(ii)}$	84.59(9)
N1	–	Mn1	–	N13	86.73(9)	O(L1)	–	Mn2		N17	91.49(9)
N1	–	Mn1	–	N15	88.47(9)	O(L1)	–	Mn2	–	$N17^{(ii)}$	88.51(9)
N3	–	Mn1	–	$N9^{(i)}$	92.25(9)	$O(L1)^{(ii)}$	–	Mn2	–	N2	84.59(9)
N3	–	Mn1	–	$N11^{(i)}$	95.41(10)	$O(L1)^{(ii)}$	–	Mn2	–	$N2^{(ii)}$	95.41(9)
N3	–	Mn1	–	$N13^{(i)}$	89.10(10)	$O(L1)^{(ii)}$	–	Mn2	–	N17	88.51(9)
N3	–	Mn1	–	$N15^{(i)}$	88.51(9)	$O(L1)^{(ii)}$	–	Mn2	–	$N17^{(ii)}$	91.49(9)

Table 4.4 Hydrogen bond distances (Å)

O(L1)	–	O(Z5)	2.88	O(Z4)	–	$N8^{(ii)}$	2.88
O(L1)	–	O(Z4)	2.81	O(Z5)	–	$N8^{(ii)}$	2.93
O(Z4)	–	O(Z3_2)	3.10	$O(Z5)^{(ii)}$	–	$N4^{(ii)}$	2.96
O(Z3)	–	O(Z2)	2.70	$O(Z1)^{(iii)}$	–	$N4^{(ii)}$	2.91
O(Z3)	–	$O(Z2)^{(iii)}$	2.80	$O(Z2)^{(ii)}$	–	$N6^{(ii)}$	2.87
O(Z2)	–	O(Z1)	2.86	O(Z3_2)	–	$N16^{(iv)}$	2.97
O(Z5_2)	–	$O(Z1)^{(i)}$	3.21	O(Z1)	–	N18	2.91
O(Z5)	–	$N19^{(i)}$	2.80		–		

(i) $x + 1/2$, $y + 1/2$, z; (ii) $x - 1/2$, $y + 1/2$, z; (iii) $-x$, $-y$, $-z$; (iv) $x + 1/2$, $-y + 1/2$, $-z + 1/2$

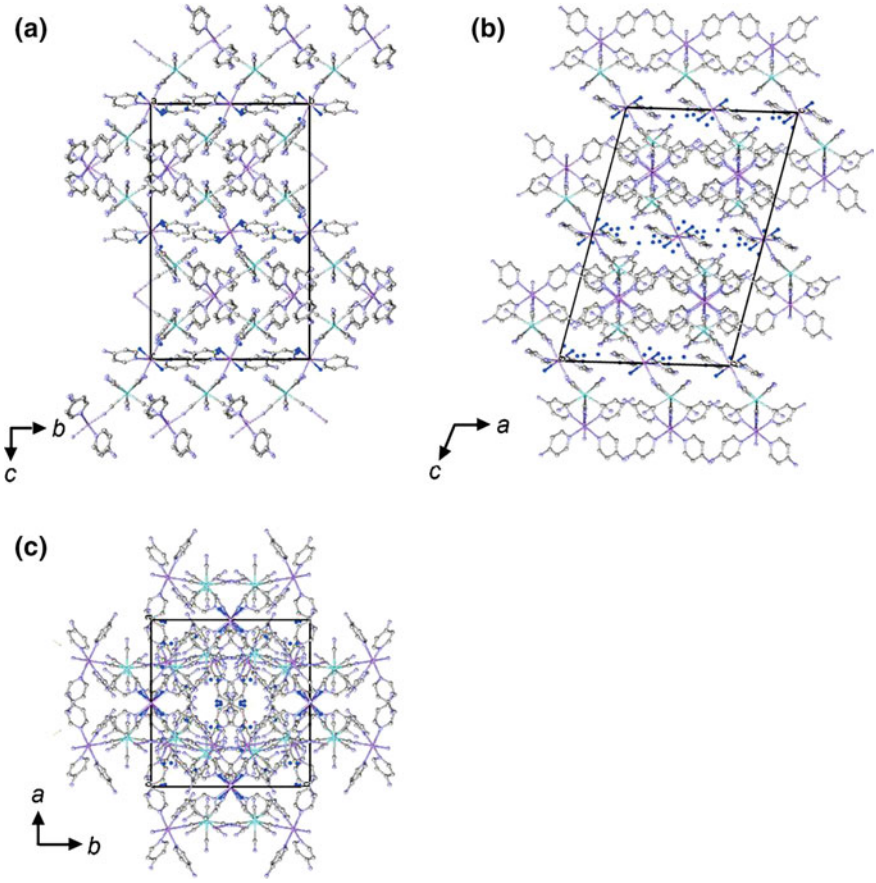


Fig. 4.2 Crystal structure viewed from the **a** *a*-axis, **b** *b*-axis, and **c** *c*-axis. *Red*, *green*, *gray*, *cyan*, *light blue*, and *blue balls* represent Mn, Nb, C, protonated pyridinium N, other N, and O atoms, respectively. H atoms are omitted for clarity. Reproduced by permission from American Chemical Society: K. Imoto, K. Nakagawa, H. Miyahara, S. Ohkoshi, ***Cryst. Growth Des***., 13, 4673–4677 (2013)

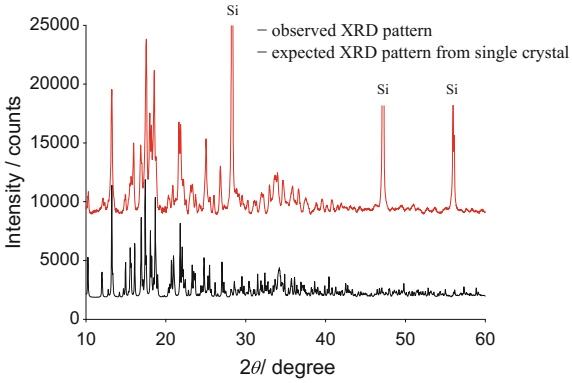


Fig. 4.3 XRD pattern and the simulated XRD pattern. Reproduced by permission from American Chemical Society: K. Imoto, K. Nakagawa, H. Miyahara, S. Ohkoshi, ***Cryst. Growth Des***., 13, 4673–4677 (2013)

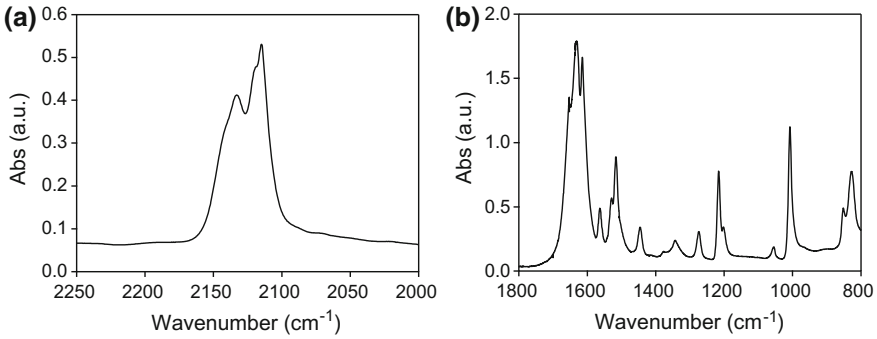


Fig. 4.4 Infrared spectrum of $Mn_3[Nb(CN)_8]_2(4\text{-aminopyridine})_{10}$ $(4\text{-aminopyridine})_2 \cdot 12H_2O$. **a** CN stretching region and **b** the vibrational modes of organic ligand. Reproduced by permission from American Chemical Society: K. Imoto, K. Nakagawa, H. Miyahara, S. Ohkoshi, ***Cryst. Growth Des***., 13, 4673-4677 (2013)

4.3.4 IR Spectra

The infrared (IR) spectrum showed CN stretching peaks at 2144, 2133, 2120, and 2114 cm^{-1} (Fig. 4.4).

4.3.5 Magnetic Properties

The χ_M^{-1} versus temperature plot under the external magnetic field of 5000 Oe exhibited a linear behavior between 150 K and 300 K region (Fig. 4.5). The linear

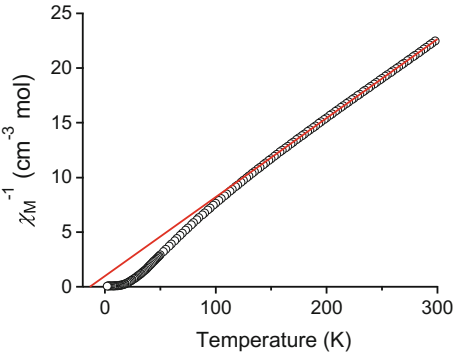


Fig. 4.5 Inverse of magnetic susceptibility (χ_M^{-1}) versus temperature curve under 5000 Oe. *Red line* shows the linear fitting between 150 and 300 K. Reproduced by permission from American Chemical Society: K. Imoto, K. Nakagawa, H. Miyahara, S. Ohkoshi, ***Cryst. Growth Des***., 13, 4673–4677 (2013)

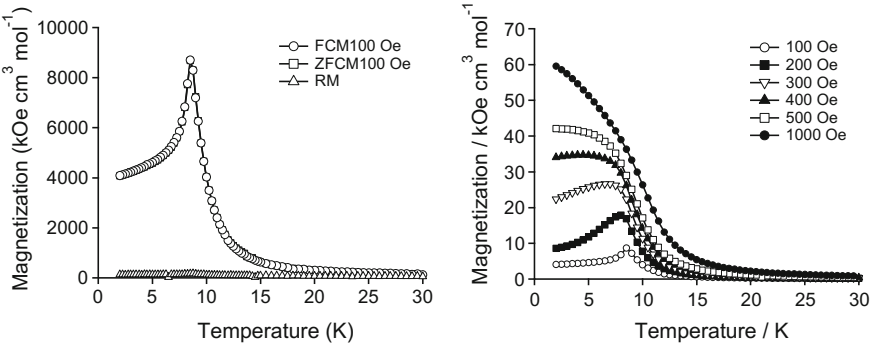


Fig. 4.6 Magnetic properties of $Mn_3[Nb(CN)_8]_2$(4-aminopyridine)$_{10}$ (4-aminopyridine)$_2$ · $12H_2O$. **a** Field-cooled magnetization, remnant magnetization, and zero-field-cooled magnetization. **b** Field cooled magnetization curve at various external magnetic field. Reproduced by permission from American Chemical Society: K. Imoto, K. Nakagawa, H. Miyahara, S. Ohkoshi, ***Cryst. Growth Des***., 13, 4673–4677 (2013)

fitting of this region indicated that the Weiss temperature is −10 K. In the field cooled magnetization (FCM) curve of the present compound at 100 Oe, the magnetization value increases below 20 K, but suddenly decreases below 9 K (Fig. 4.6a). The magnetization versus temperature curve in various external magnetic fields exhibits a sudden decrease in magnetization under the external magnetic fields from 100 to 400 Oe. However, the magnetization value monotonously increases under the external magnetic fields above 500 Oe (Fig. 4.6b). The initial magnetization curve shows an anomalous behavior around the magnetic field value of 200 Oe (Fig. 4.7a) and the magnetization value at 5 T was 12.3 μ_B (Fig. 4.7b).

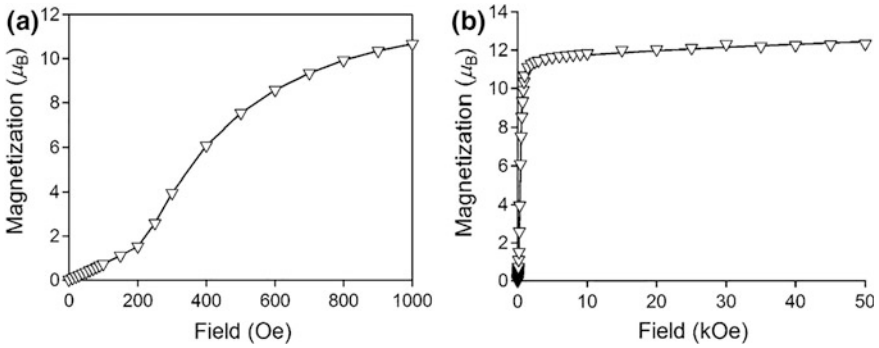


Fig. 4.7 Magnetic properties of $Mn_3[Nb(CN)_8]_2(4\text{-aminopyridine})_{10}(4\text{-aminopyridine})_2 \cdot 12H_2O$. **a** Magnetization vs external magnetic field curve below 1000 Oe. **b** Initial magnetization curve at 2 K. Reproduced by permission from American Chemical Society: K. Imoto, K. Nakagawa, H. Miyahara, S. Ohkoshi, ***Cryst. Growth Des***., 13, 4673–4677 (2013)

4.3.6 Impedance Measurements

The ionic conductivity (σ) of this compound is measured by impedance technique. Figure 4.8 shows the real (Z') and imaginary (Z'') parts of the complex-plane impedance data from 295 to 322 K. In the present experiment, the Cole-Cole plot was well fitted by a single half-circle, indicating that the observed conductivity is attributed to the material itself. The circular arc fitting indicates that the σ value is 1.0×10^{-4} S cm^{-1} at 295 K at 100% RH. As the temperature increases, the σ value increases to 4.6×10^{-4} S cm^{-1} at 322 K (Fig. 4.8a). The activation energy

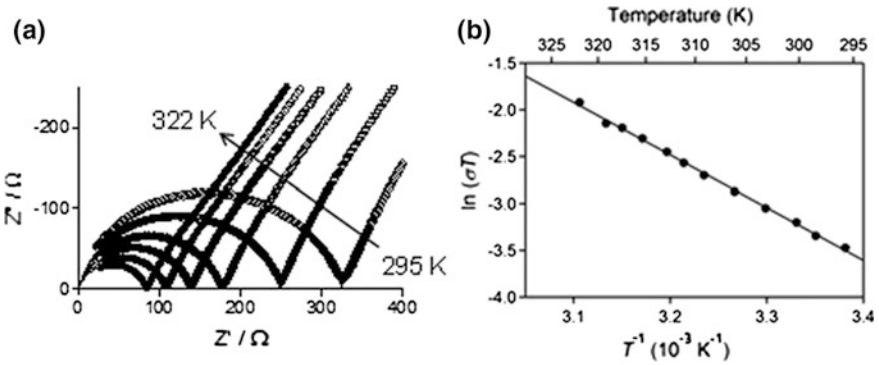


Fig. 4.8 **a** Cole-Cole impedance plot of $Mn_3[Nb(CN)_8]_2(4\text{-aminopyridine})_{10}(4\text{-aminopyridine})_2 \cdot 12H_2O$ from 295 to 322 K. **b** $\ln(\sigma T)$ versus T^{-1} plot and the result of linear fitting. Reproduced by permission from American Chemical Society: K. Imoto, K. Nakagawa, H. Miyahara, S. Ohkoshi, ***Cryst. Growth Des***., 13, 4673–4677 (2013)

Table 4.5 Humidity dependence of ionic conductivity of $Mn_3[Nb(CN)_8]_2(4\text{-aminopyridine})_{10}(4\text{-aminopyridine})_2 \cdot 12H_2O$ at 298 K

Humidity (% RH)	Conductivity (S cm^{-1})
100	1.2×10^{-4}
80	9.7×10^{-8}
62	4.4×10^{-8}
23	4.2×10^{-10}

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(E_a) of the ionic conductivity is understood by the Grotthuss mechanism, which takes into account the hopping of an ion (in the present case, proton from the Lewis acid sites), which obeys the following equation, where k is the Boltzmann constant.

$$\ln(\sigma T) = \exp\left(-\frac{E_a}{kT}\right)$$

In Fig. 4.8b, the ln (σT) value varies linearly with T^{-1}, indicating that the E_a value of ionic conductivity is 0.48 eV. The thermal stability of the sample for impedance measurement was confirmed by the reproducibility of the data at 295 K after cooling. The σ value clearly depends on the humidity from 1.0×10^{-4} S cm^{-1} (100% RH) to 4.2×10^{-10} S cm^{-1} (23% RH) at 300 K, indicating that the high conductivity is attributed to proton conduction (Table 4.5). The observed σ value of 4.6×10^{-4} S cm^{-1} is sufficient for a super-ionic conductor ($\sigma > 10^{-4}$ S cm^{-1}).

4.4 Discussion

4.4.1 The Valence State, the Form of Organic Ligand, and the Crystal Structure

The consideration about the valence states, the form of organic ligand, and the crystal structure are performed by the results of single crystal X-ray analysis, stoichiometric aspect and IR spectrum. First, the result of the single crystal X-ray analysis indicated the presence of non-coordinated organic ligand molecule with pyridyl-N-protonated form. In addition, we can assume the valence state of Mn ions by the fact that the bond lengths around the Mn sites are between 2.18 to 2.31 Å. Compared to Mn^{II}, the bond distance of Mn^{III} is about 1.9–2.1 Å and the difference in distance of axial and equatorial positions should be larger due to a stronger electropositivity and a Jahn-Teller effect. This indicates that the Mn ions are divalent.

In addition, stoichiometric consideration also matches with the above assignment. The unit cell contains twelve Mn atoms, eight $[Nb(CN)_8]$ moieties, and eight

4-aminopyridinium molecules. When we consider that the charge of Mn is +2 and that of $[Nb(CN)_8]$ is −4, the total charge in the unit cell is (+2) × 12 + (−4) × 8 + (+1) × 8 = 0, which satisfies the charge neutrality.

The result of the IR spectrum supports the discussion. The IR spectrum showed CN stretching peaks at 2144, 2133, 2120, and 2114 cm^{-1}, which is slightly higher than the CN stretching peaks of $K_4[Nb^{IV}(CN)_8] \cdot 2H_2O$, indicating that the valence state of Nb^{IV} is not changed and the existence of bridged CN groups. In addition the peak around 1630 cm^{-1} is assigned to the (pyridyl)N-H^+ bending mode, indicating that the existence of pyridyl-N-protonated form of 4-aminopyridine. In fact, the IR spectrum of 4-aminopyridinium chloride shows the peak around 1630 cm^{-1}, while this peak is almost absent in the IR spectrum of 4-aminopyridine (Fig. 4.9).

In the present synthesis, 4-aminopyridinium cation is incorporated in the crystal structure although 4-aminopyridinium cation is not intentionally introduced as a reagent. The pKa of 4-aminopyridine is 9.4 [13], indicating that 4-aminopyridine exists as both 4-aminopyridine and 4-aminopyridinium in aqueous medium. The stabilization of 4-aminopyridium cation due to the resonant structure as shown in Fig. 4.10 is one of a reason for a high pKa value. Such a high pKa value may promote the incorporation of 4-aminopyridinium cation into the crystal structure.

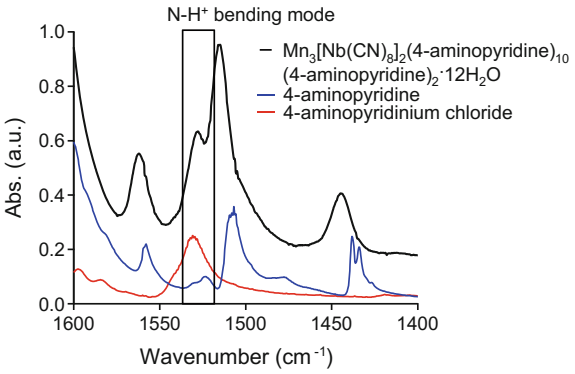


Fig. 4.9 Infrared spectra of $Mn_3[Nb(CN)_8]_2(4\text{-aminopyridine})_{10}(4\text{-aminopyridine})_2 \cdot 12H_2O$, 4-aminopyridine, and 4-aminopyridinium chloride. Reproduced by permission from American Chemical Society: K. Imoto, K. Nakagawa, H. Miyahara, S. Ohkoshi, ***Cryst. Growth Des.***, 13, 4673–4677 (2013)

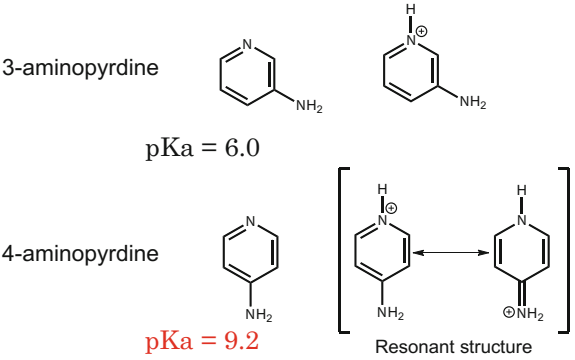


Fig. 4.10 pKa values and chemical structure of 3-aminopyridine, 4-aminopyridine, and their protonated form

In fact, the reaction of Mn^{II}, $[Nb^{IV}(CN)_8]$, and 3-aminopyridine gives a different crystal structure, which does not contain protonated organic ligand [14]. The pKa values of 3-aminopyridine is 6.3, which is significantly lower than that of 4-aminopyridine due to the absence of resonant structure in protonated form [13]. This experimental fact supports that the pKa value of organic ligand affects the composition of the product.

4.4.2 Magnetic Properties

Magnetic interaction between metal centers

In the χ_M^{-1} versus temperature plot, the Weiss temperature of −10 K obtained by the linear fitting between 150 and 300 K region suggests the antiferromagnetic interaction between Mn^{II} (S = 5/2) and Nb^{IV} (S = 1/2). In addition, the magnetization vs external magnetic field curve at 2 K showed a magnetization value of 12.3 μ_B at 5 T. This value agrees with the calculated saturation magnetization value of 13.0 μ_B for the ferrimagnetic ordering between Mn^{II} (S = 5/2) and Nb^{IV} (S = 1/2) for the given formula. These results suggests that an antiferromagnetic interaction works between Mn^{II} (S = 5/2) and Nb^{IV} (S = 1/2).

Magnetic ordering

The magnetic behavior of the present compound is explained by “metamagnetism”. Before going to the detailed discussion, let me explain “metamagnetism” here. In an antiferromagnetic material, magnetic moments of each moiety are aligned in an antiparallel manner, resulting in cancelling the total magnetic moment. However, when a very strong magnetic field (which is stronger enough than antiferromagnetic interaction) is applied, spin flip transition occurs and the magnetic moments of each moiety order in a parallel fashion. When the spin-flip transition occurs in an observable magnetic field, the material is called a “metamagnet”. Based on this knowledge, the present magnetic behavior is discussed from here. In the field cooled magnetization (FCM) curve of the present compound at 100 Oe, the magnetization value increases below 20 K, but suddenly decreases below 9 K, which is typical behavior for an antiferromagnetic material. However, though the magnetization versus temperature curve in various external magnetic fields exhibits a sudden decrease in magnetization under the external magnetic fields from 100 to 400 Oe, the magnetization value monotonously increases under the external magnetic fields above 500 Oe. This indicates that above 500 Oe, the sample is no longer an antiferromagnetic material. In addition, the initial magnetization curve shows an anomalous behavior around the magnetic field value of 200 Oe, indicating the occurrence of a spin-flip transition. These observations indicate that the present material is a metamagnet.

Based on these discussions, the magnetic ordering of the present compound is explained as follows. In the FCM curve, as the temperature decreases in a low

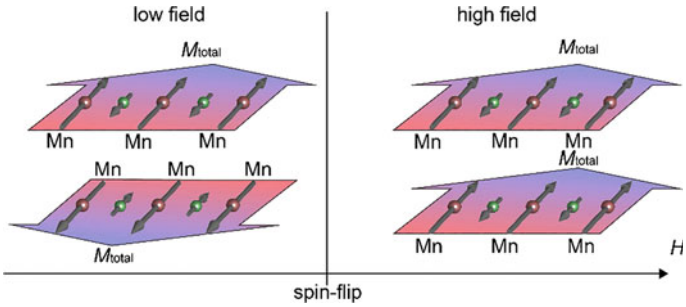


Fig. 4.11 Spin configuration at low (*left*) and high external magnetic field (*right*). Reproduced by permission from American Chemical Society: K. Imoto, K. Nakagawa, H. Miyahara, S. Ohkoshi, ***Cryst. Growth Des***., 13, 4673–4677 (2013)

external magnetic field (below 400 Oe), the magnetic ordering of Mn and Nb sites begins below 20 K, while the magnetic moments in each layer, which have ferrimagnetic ordering between Mn^{II}(S = 5/2) and Nb^{IV}(S = 1/2), are antiparallel below 9 K, resulting in exhibiting antiferromagnetism (Fig. 4.11). In contrast, at a high external magnetic field (above 500 Oe), magnetic moments of the layers order in a parallel manner described with the thick arrows in Fig. 4.11. In the magnetization vs external magnetic field curve, as increasing magnetic field, the increase of magnetization is gradual in low external magnetic field region (below 200 Oe) due to the antiferromagnetic interlayer interaction, while the increase of magnetization become fast in the high external region because the magnetic moments of the layers starts to order in a parallel manner, resulting in a spin-flip transition.

4.4.3 Conducting Properties

The real (Z') and imaginary (Z'') parts of the complex-plane impedance data from 295 K to 322 K is shown in Fig. 4.8a. The circular arc fitting indicates that the σ value is 1.0×10^{-4} S cm^{-1} at 295 K at 100% RH. As the temperature increases, the σ value increases to 4.6×10^{-4} S cm^{-1} at 322 K. In Fig. 4.8b, the ln (σT) value varies linearly with T^{-1}, indicating that the E_a value of ionic conductivity is 0.44 eV. The σ value clearly depends on the humidity from 1.0×10^{-4} S cm^{-1} (100% RH) to 4.2×10^{-10} S cm^{-1} (23% RH), indicating that the high conductivity is attributed to proton conduction. The observed σ value of 4.6×10^{-4} S cm^{-1} is sufficient for a super-ionic conductor ($\sigma > 10^{-4}$ S cm^{-1}).

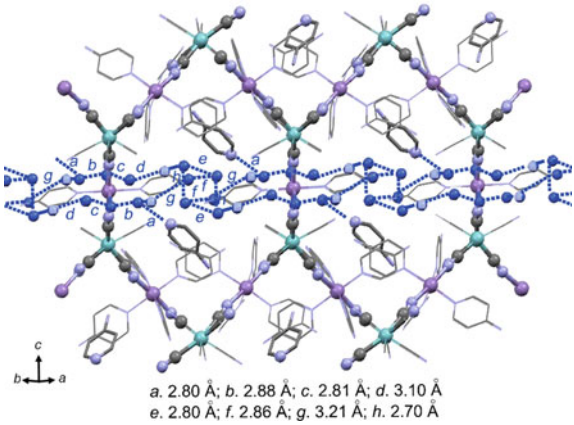


Fig. 4.12 Hydrogen bonding network of $Mn_3[Nb(CN)_8]_2$(4-aminopyridine)$_{10}$(4-aminopyridine)$_2 \cdot 12H_2O$. Reproduced by permission from American Chemical Society: K. Imoto, K. Nakagawa, H. Miyahara, S. Ohkoshi, ***Cryst. Growth Des***., 13, 4673–4677 (2013)

4.4.4 The Mechanism for Super-Ionic Conductivity

As described in 4.3.2, the present compound possesses 1D infinite hydrogen-bonding networks constructed with ligand water molecules, non-coordinated water molecules, and 4-aminopyridinium cations along [1 −1 0] and [−1 1 0] crystallographic directions. Therefore, the mechanism of the proton conduction is explained by the Grotthuss mechanism as follows. The Lewis acidity of Mn ion accelerates the deprotonation from the ligand water molecules, and then the released proton propagates through hydrogen-bonding network of ligand water molecules, non-coordinated water molecules, and 4-aminopyridinium cation (Fig. 4.12).

4.5 Conclusion

In this chapter, the coexistence of super-ionic conductivity and metamagnetism is demonstrated in $Mn^{II}_3[Nb^{IV}(CN)_8]_2$(4-aminopyridine)$_{10}$(4-aminopyridinium)$_2 \cdot 12H_2O$. The super-ionic conductivity is caused by proton conduction based on the Grotthuss mechanism. In the present material, the 4-aminopyridinium cation is incorporated into the flexible octacyano-bridged magnetic framework and plays an important role in the formation of the hydrogen-bonding network, which leads to the observation of super-ionic conductivity.

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Chapter 5
Concluding Remarks

Abstract In this work, I investigated to observe novel functionalities in octacyanometalate-based magnet and successfully realized i) light-induced spin-crossover magnetization, ii) phtoomagnetism in a two-step spin-transition material, and iii) the coexistence of super-ionic conductivity and metamagnetism. In spin-crossover phenomenon, the conversion of electronic state occurs on single ion, so partial metal substitution is possible. By incorporating other functional metal ions, multi-functional photo-magnetic materials, such as high Curie temperature photo-magnetic materials, are expected to be obtained. In addition, super-ionic conductivity and photo-induced change of ionic conductivity is also a fascinating functionality. The present study opens up the new possibilities for novel functionalities.

Keywords Molecule-based magnet · Photo-induced magnetization · Spin-crossover · Ionic conductivity · cyanido-bridged metal assemblies

In Chap. 1, general introduction and advantages of molecule-based magnets are explained with principal method of theoretical analysis of magnetic properties, examples of functional molecule-based magnets, and detailed description of spin-crossover phenomenon, light-induced spin-crossover phenomenon, photo-induced magnetization, and ionic-conductivity. Furthermore, observation of coupling effect between spin-crossover phenomenon and magnetic phase transition and demonstration of ionic conductive magnetic material in octacyanidometalate-based magnets are described as the objectives of this thesis.

In Chap. 2, the first example of *light-induced spin-crossover ferromagnetism* is described. This phenomenon is achieved in a three-dimensional Fe−Nb bimetallic assembly using 4-pyridinealdoxime as organic ligand. The structural analysis showed that the present compound possesses tetragonal crystal structure with $I4_1/a$ space group and Fe and Nb are connected in three-dimensional fashion through bridging CN. A spin-crossover phenomenon between Fe^{II}_{HS} ($S = 2$) and Fe^{II}_{LS} ($S = 0$) is observed around 130 K. By irradiating 473-nm light, a large spontaneous

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magnetization with a Curie temperature (T_C) of 20 K and a coercive field (H_c) of 240 Oe is observed. In this compound, the spin-crossover site is introduced into cyanido-bridged metal assembly. The photo-induced magnetization is observed because of the combination of LIESST effect on spin-crossover sites and strong superexchange interaction between the magnetic centers in three-dimensional manner. The present result is the first step toward the study of the new type of photo-magnetic material.

In Chap. 3, I demonstrated two-step spin transition and light induced bulk magnetization in $[Fe^{II}(4\text{-methylpyridine})_4]_2[Nb^{IV}(CN)_8] \cdot 2H_2O$. The compound showed an abrupt two step spin-crossover transition and was paramagnetic at low temperature. By irradianting 532-nm laser light, a large spontaneous magnetization was observed. The present observation proves the usefulness of light-induced spin-crossover magnetization for realizing novel type of photo-magnet.

In Chap. 4, I have succeeded in observing the coexistence of super-ionic conductivity and metamagnetism in two-dimensional $Mn_3[Nb(CN)_8]_2(4\text{-aminopyridine})_{10}(4\text{-aminopyridinium})_2 \cdot 12H_2O$. The compound showed a metamagnetism, which is explained by the strong intralayer ferrimagnetic ordering of Mn and Nb through cyanide ligand and the weak interlayer antiferromagnetic interaction. The super-ionic conductivity is caused by proton conduction based on the Grotthuss mechanism. In this compound, the 4-aminopyridinium cation is incorporated into the flexible octacyano-bridged magnetic framework and plays an important role in the formation of the hydrogen-bonding network, which leads to the observation of super-ionic conductivity.

In this thesis, I have investigated to observe novel functionalities in octacyanometalate-based magnet and successfully realized (i) light-induced spin-crossover magnetization, (ii) photomagnetism in a two-step spin-transition material, and (iii) the coexistence of super-ionic conductivity and metamagnetism. In spin-crossover phenomenon, the conversion of electronic state occurs on single ion, so partial metal substitution is possible. By incorporating other functional metal ions, multi-functional photo-magnetic materials, such as high Curie temperature photo-magnetic materials, are expected to be obtained. In addition, super-ionic conductivity and photo-induced change of ionic conductivity is also a fascinating functionality. Such a new functionality is expected by combining photo-induced magnetization demonstrated in (i) and super-ionic conductivity observed in (iii). The present study opens up the new possibilities for novel functionalities.

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